

Phenix Overview and General Reference (Part 4 of 5)

20 documentation pages

Generated from local Phenix documentation

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PredictModel: Predict structures of chains in a sequence file

predict_model.html

Author(s)

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Tutorial video

A tutorial video for AlphaFold prediction is available on the Phenix YouTube channel and covers the following topics:

- basic overview of AlphaFold model prediction
- how to run PredictModel with the Phenix AlphaFold server using the Phenix GUI
- how to interpret the output

Purpose

Predict model can be used to generate predicted models from a sequence file. One predicted model is generated for each unique sequence. It does not make predictions of a complex.

You can choose whether to use a multiple sequence alignment (MSA), whether go use templates from the PDB, and whether to supply your own template to be used as a guide in prediction.

Sequence file

The sequence file that you supply specifies what is going to be predicted and how many copies of each chain are present in the structure (one for every copy in the sequence file). All models that are created and input will be associated with one or more of the chains in your sequence file based on sequence identity (normally every model should match a chain in the sequence file exactly).

Model prediction

Predict model normally uses the Phenix server notebook to carry out AlphaFold prediction of one chain at a time.

You can specify what inputs the AlphaFold prediction should use. These always include a sequence file, but it can include an optional multiple sequence alignment file, optional templates, keywords for model prediction such as the number of models to generate, random seed, and whether to use multiple sequence alignment.

When prediction is being carried out, the Phenix GUI waits for the predicted models to appear in the working directory, then it writes out the resulting models.

Using the Phenix server for model prediction

Prediction is fully automated with the Phenix server through the Phenix GUI, so all you have to do is let it run. The Phenix GUI will use the Phenix server to carry out the prediction and put it in the working directory.

MSA calculation vs model prediction

The calculation of multiple sequence alignments (MSAs) is a separate step from model prediction. The Phenix GUI on your computer sends a request to the mmseqs2 server, which creates an MSA and sends it back to your computer. Then your computer uploads the MSA to the Phenix server, which uses it in an AlphaFold prediction.

If you want, you can supply your own MSAs. The key requirement is that the sequence of the first entry in your MSA must exactly match the sequence to be predicted.

You can also skip the use of MSAs. This can be useful if you supply a template and you want AlphaFold to rebuild your template instead of doing a new prediction.

Number of models

AlphaFold can carry out multiple predictions for a sequence. You can specify how many of these to carry out. The PredictModel tool will choose the one with the highest value of pLDDT (predicted local difference distance test, Mariani et al., 2013). Normally the default (5 models) is a reasonable compromise between optimal prediction and the length of time needed to carry out the prediction. Note that the Phenix server will stop carrying out predictions if it appears unlikely that further predictions will significantly improve the pLDDT (the estimate is based on the variability of pLDDT values so far.)

Using templates from the PDB

You can request that templates from the PDB be used in prediction. If you use this feature, models will be predicted both with and without the templates, and the model with the highest pLDDT will be saved.

Using supplied templates

You can supply your own templates. As for templates from the PDB, if you use this feature, models will be predicted both with and without the templates, and the model with the highest pLDDT will be saved.

Predicting very long sequences in chunks and reassembling

Sequences longer than about 1500 residues take quite a long time to run. By default if your sequence is more than 1500 residues it will be split into overlapping chunks of length 600 with an overlap of 200. These chunks will be run individually, then reassembled at the end. To improve the contacts between domains in different chunks, a contact map is created and domains that are in contact but in different chunks are re-predicted together. During assembly the domain pairs are used to adjust the positions of domains, and also positions of domains are adjusted to minimize overlaps. The repositioned domains are then used as a template to morph the full reassembled prediction. The result of this procedure is to assemble a full-length prediction from overlapping chunk predictions and to distort this assembled prediction to minimize overlaps and make domain-domain contacts present between chunks.

Note that if you have a sequence longer than the threshold for breaking into chunks then you can only include one sequence in a PredictModel run.

Precalculated MSAs and models save time if you rerun with the same sequence

When you run a prediction on the Phenix server, it saves the MSA and the predicted model. Then if you run the same request again (same number of AlphaFold models, same sequence, same choice of including templates from the PDB, no supplied template), the server will return the result from the original run. You can turn this off with the precalculated results and precalculated MSA keywords if you want. Note that the server

does not save your sequence or parameters. It just makes a hash string from them so that if the same request is made again it can detect it. The results are simply saved in a file that has the hash string as a name.

Precalculated MSAs and models allow you go get your result even if disconnected

If you get disconnected from the server during a prediction job, you can just wait for the job to finish, then submit a new request with the same parameters. As the server saves the predictions, it will return the result to you right away (if it is finished).

Using AlphaFold to improve a model you already have

As AlphaFold often produces models with quite good geometry, you can use it as a procedure for geometry optimization. You supply your working model and your sequence, and you turn off the use of MSAs and the use of templates from the PDB. Of course AlphaFold does not know about your density map, so it could move the model away from the density. Normally you would use refinement to help with this.

Using AlphaFold to build parts of a model that are missing

If you want AlphaFold to try and build the missing parts of a chain that you have partially built, supply your working partial model and your sequence, and allow the use of MSAs (and optionally, the use of templates from the PDB). Your working model will be used as a template for the part of the structure that you have already built. Note that you have to have the correct sequence for your partially-built model.

Notes on using the Phenix server from the Phenix GUI for model prediction

The Phenix GUI can automatically send a request to the Phenix server for an AlphaFold prediction. This requires an internet connection, the Phenix server to be up, and a token that is supplied with Phenix.

If the Phenix GUI successfully communicates with the server, it will wait for the server to carry out the prediction, showing the status of the prediction while it is being carried out. The status will usually start out as waiting in the queue for a place on the GPU. If this status lasts more than a few seconds, an estimated waiting time (before starting processing) is displayed in the Phenix GUI. The waiting time is very approximate. It is based on how long previous jobs waited to start and how many jobs are ahead of yours. If the currently-running jobs happen to take a very short or long time, the estimate could be off in either direction. If you submit more than one job at a time, the first job you submit will get higher priority and may move nearer the front of the queue than the others.

Once your job starts, the status in the Phenix GUI will change to reflect that it is running. There won't be an estimated run time shown, but most of the time jobs with a few hundred residues take 10-20 min, and longer sequences take more or less proportionally longer. If you include templates from the PDB, it takes longer to run. If you specify that the number of AlphaFold models to use is more than the default (5), it will also take longer.

If your prediction is successful, the Phenix GUI will list the estimated confidence (pLDDT) and the output data files. The output files include the AlphaFold model (named something like PredictAndBuild_10_ALPHAFOLD_A.pdb), the multiple sequence alignment used (PredictAndBuild_10_ALPHAFOLD_A_MSA.a3m), and the Predicted Aligned Error (PAE) file (PredictAndBuild_10_ALPHAFOLD_A_PAE.jsn).

Privacy of your data when using the Phenix server

When you run the Phenix server to get an AlphaFold prediction, your data are uploaded by the Phenix GUI to the Phenix server, the prediction is carried out, and then the results are downloaded.

Normally your sequence is also sent to the mmseqs2 MSA server to obtain a multiple sequence alignment. You can also supply your own MSA and then the mmseqs2 server is not used. The MSA and the hashed sequence are sent to the Phenix server and are saved unless you specify `allow_precalculated_msa_file=False`.

When the job is completed and Phenix downloads the data, all data is deleted if you specify `allow_precalculated_result_file=False`. If you specify `allow_precalculated_result_file=True` (default), a hashed version of the inputs (sequence, number of models, etc) is saved along with the result (the AlphaFold model). In this way it is not possible to see the inputs, but if you run the exact same job again, the server will see that the hash of the inputs matches and it will return the model directly.

Here are some details on the privacy of your data when using the Phenix server:

- Accessible only from Phenix

- Password-protected (password supplied by Phenix)

- Server provides Phenix with a passkey when data are uploaded. That passkey allows download of the data. All this is done within Phenix.

- Phenix sends your IP address to the server. This is used as a unique identifier in the queueing system. The IP address is deleted when the data are downloaded.

- Server is accessible in two modes (user can choose, default is Mode 1 (save hashed inputs and results)).

- A. Mode 1: hashed inputs and results saved on server
 - a. in this mode if anyone submits the identical job (identical sequence, identical parameters) later, they get back the result model from this job
 - b. result model does not contain any information about the user
 - c. inputs and log files are deleted after data are downloaded
 - d. results removed from server after 1 year
- B. Mode 2: nothing saved. Input and results deleted from server after data are downloaded

- Server can only carry out pre-specified commands (i.e., model prediction), and only allows json text to be uploaded.

- Data is accessible to the Phenix developers while running. The developers will only access this data to debug the system.

The output predicted model

The predicted model produced by AlphaFold with Phenix will consist of a single chain with exactly the sequence that is supplied. It will be in the standard PDB (version 3.2) format. It will contain all (non-hydrogen) atoms for all the residues. The coordinates of the atoms have the usual meaning (locations of atoms in 3D space), but the orientation and position of the chain are arbitrary. The B-value field (normally atomic displacement parameters) will contain the values of pLDDT (confidence), with one value per residue. This means that some processing of this model is necessary before use (this is normally done by `phenix.process_predicted_model`).

The pLDDT confidence measure

AlphaFold provides residue-level estimates of model accuracy in the form of predicted values of the LDDT (Local Distance Difference Test, Mariani et al., 2013). These predicted values are referred to as pLDDT values (predicted Local Distance Difference Test). The LDDT measure is a number from zero to one that reflects the similarity of CA positions in two structures.

The LDDT is not a simple measure of accuracy, but rather a composite that is based on the CA distances between all pairs of CA atoms in one model and how similar these inter-atomic distances are to those in the other model. It is a local test, where only inter-atomic distances that are less than 15 Å are considered.

The LDDT is a composite that is generated from the fraction of CA-CA distances that are very accurate (less than 0.5 Å off), the fraction that are quite accurate (less than 1 Å off), moderately accurate (less than 2 Å off), and somewhat accurate (less than 4 Å off). If all CA-CA distances are within 0.5 Å, the resulting score will be 1, if all are more than 4 Å off, the score is zero. Note that the LDDT is related to RMSD, but not in a simple way.

The pLDDT values (predicted Local Distance Difference Test) provided by AlphaFold are estimates of the LDDT values. The pLDDT values are essentially unbiased (just as likely to be too low as too high), and reasonably accurate. The correlation between pLDDT values and actual LDDT values calculated using AlphaFold models and models in the PDB is about 0.7 - 0.75, which means that the values of pLDDT you get are useful indicators of model quality but also that sometimes AlphaFold will have high confidence in an incorrect prediction or low confidence in a correct prediction.

As the pLDDT metric is an estimate of local error, it does not tell you that much about long-range distortions or domain movements between the predicted and actual structure. The PAE matrix does have some information about these features because it is not restricted to short distances.

The output MSA file

The MSA file that was used in AlphaFold prediction is available if the Phenix server is used. This MSA file can be used in subsequent predictions (if the sequence is identical). Normally there is no need to supply an MSA file because the MSA is saved by the Phenix server and it will just be re-used if the same sequence is supplied. However if you want to edit this MSA file before running an AlphaFold prediction, you can do that.

The output PAE file

AlphaFold provides a Predicted Aligned Error matrix in the form of a PAE file. The error that it is describing reflects the estimated error in CA positions in the predicted model compared to the (unknown) true structure.

Suppose we had both the prediction and the true structure. Then suppose we superpose the predicted model on the true structure, basing the superposition on the residues near residue *i*. Using that superposition, we can now look at the location of the CA atom of residue *j* in the predicted model and in the true structure. We could in particular note the distance between the predicted and actual position of the CA atom of residue *j*, when the structures are superposed using residues near residue *i*. We could do this for every residue pair (*i,j*), and the result would be the Aligned Error matrix.

The Predicted Aligned Error matrix is an estimate of the Aligned Error matrix. It is generated by AlphaFold. Basically, AlphaFold predicts the model and also it generates an error estimate in the prediction.

The PAE matrix is useful for identifying domains within a predicted chain that are accurately predicted, along with connecting regions that are more uncertain. In the Phenix GUI the PAE matrix is plotted with dark green indicating low PAE values (relationships between pairs of residues that are predicted to be accurate). A domain that is accurately predicted can be seen in such a plot as a solid green square along the diagonal. If your plot has just one big green square, it is probably all quite accurate. If it looks instead like a few squares

along the diagonal, probably the individual domains are accurately predicted, but their relationship is not known.

Relationship between the pLDDT values and the PAE matrix

The PAE matrix is related to the pLDDT values, but not in a simple way.

The pLDDT value for a residue describes the expected errors in the distances between that CA atom and nearby CA atoms (compared to the true structure) on a scale of 0 to 1, where 1 means all CA-CA distances are expected to match within 0.5 Å.

The value of the PAE matrix for a pair of residues is the estimated error in Å in the position of the CA atom for the second residue if the model is superposed on the true structure using CA atoms near the first CA atom.

Both measures tell you something about the uncertainties in your model, but neither tells you exactly how well the model would superpose on the true structure.

Rough guide as to expected RMSD for a residue if you have the pLDDT

Here is a table that gives you an idea of the RMSD of CA atoms in an AlphaFold prediction to a superposed deposited model (see Terwilliger et al., 2023):

pLDDT Median prediction error % with error over 2 Å

pLDDT > 90	0.6 Å	10%
pLDDT 80-90	1.1 Å	22%
pLDDT 70-80	1.5 Å	33%
pLDDT < 70	3.5 Å	77%

This table shows that high-confidence residues (pLDDT >90) are generally incredibly accurate, but a few (10%) still are off by more than 2 Å. Low-confidence residues (pLDDT < 70) are usually pretty far off (median of 3.5 Å) but still almost a quarter are better than 2 Å.

Examples

Standard run of predict_model

Running predict_model is easy. From the command-line you can type:

```
phenix.predict_and_build jobname=myjob seq_file=seq.dat \
  prediction_server=PhenixServer \
  stop_after_predict=True
```

Common questions

How long will it take?

A typical protein chain with 200 residues will take about 5-10 minutes to run, once it has started on the Phenix server.

Timing is more or less proportional to chain length. If you specify that more than 5 models are to be built, it will typically take longer.

If your chain has already been predicted for you or for someone else, you normally should get a result (a copy of that prediction) in about 30 seconds or less.

If the Phenix server is full (typically 6 jobs can run at once), you can use the Server button on the Phenix GUI to see the expected wait time to start a job.

What to do next after PredictModel

Normally the next step after PredictModel is running ProcessPredictedModel to trim off low-confidence residues, identify compact domains, and convert pLDDT values to pseudo-B values. If you have cryo-EM data the next step after ProcessPredictedModel is usually docking the trimmed model into your map with DockInMap or EMPlacement or automated docking and rebuilding with PredictAndBuild. If you have X-ray data, the next step after ProcessPredictedModel is usually molecular replacement with Phaser or automated molecular replacement and rebuilding with PredictAndBuild.

Note that a model produced by PredictModel is not yet aligned with your density map for cryo-EM or for your crystal unit cell in crystallography. This means you must carry out some step for placing and orienting your PredictModel model before carrying out refinement or any steps that involve comparing your model to data.

Server problems

The most common problem in running Predict model is that the Phenix server is not working as expected. Normally the first thing to try is just let the program retry (it will do this for a while normally). If that does not work, you can take the files in the packaged .tgz file (listed in the GUI output), use them to get your own prediction with any server, and put the resulting predicted models in the place specified in the GUI or program output.

Specific limitations and problems:

Literature

Accelerating crystal structure determination with iterative AlphaFold prediction. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *Acta Cryst D* 79, 234-244 (2023). Improved AlphaFold modeling with implicit experimental information. T.C. Terwilliger, B.K. Poon, P.V. Afonine, C.J. Schlicksup, T.I. Croll, C. Millán, J.S. Richardson, R.J. Read, and P.D. Adams. *Nature Methods* 19, 1376-1382 (2022). AlphaFold predictions: great hypotheses but no match for experiment. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *BiorXiv* (2023). Making protein folding accessible to all. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, and M. Steinegger. *Nature Methods* 19, 679–682 (2022). Highly accurate protein structure prediction with AlphaFold. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S.A.A. Kohl, A.J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A.W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. *Nature* 596, 583-589 (2021). MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. M. Steinegger, and J. Söding. *Nature biotechnology* 35, 1026-1028 (2017). IDDT: a local superposition-free score for comparing protein structures and models using distance difference tests. V. Mariani, M. Biasini, A. Barbato, and T.. Schwede. *Bioinformatics* 29, 2722-2728 (2013).

- Accelerating crystal structure determination with iterative AlphaFold prediction. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *Acta Cryst D* 79, 234-244 (2023).

- Improved AlphaFold modeling with implicit experimental information. T.C. Terwilliger, B.K. Poon, P.V. Afonine, C.J. Schlicksup, T.I. Croll, C. Millán, J.S. Richardson, R.J. Read, and P.D. Adams. *Nature Methods* 19, 1376-1382 (2022).
- AlphaFold predictions: great hypotheses but no match for experiment. T.C. Terwilliger, P.V. Afonine, Liebschner, D., T.I. Croll, McCoy, AJ, Oeffner, RD., Williams, CJ, Poon, BK, J.S. Richardson, R.J. Read, and P.D. Adams. *BiorXiv* (2023).
- Making protein folding accessible to all. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, and M. Steinegger. *Nature Methods* 19, 679–682 (2022).
- Highly accurate protein structure prediction with AlphaFold. J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Žídek, A. Potapenko, A. Bridgland, C. Meyer, S.A.A. Kohl, A.J. Ballard, A. Cowie, B. Romera-Paredes, S. Nikolov, R. Jain, J. Adler, T. Back, S. Petersen, D. Reiman, E. Clancy, M. Zielinski, M. Steinegger, M. Pacholska, T. Berghammer, S. Bodenstein, D. Silver, O. Vinyals, A.W. Senior, K. Kavukcuoglu, P. Kohli, and D. Hassabis. *Nature* 596, 583-589 (2021).
- MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. M. Steinegger, and J. Söding. *Nature biotechnology* 35, 1026-1028 (2017).
- IDDT: a local superposition-free score for comparing protein structures and models using distance difference tests. V. Mariani, M. Biasini, A. Barbato, and T.. Schwede. *Bioinformatics* 29, 2722-2728 (2013).

Additional information

List of all available keywords (same as predict_and_build keywords)

jobname = " Name of this job. Used to create a directory that will hold all the working files for the run. Normally set automatically by the GUI to match the directory where the job is run: directory: PredictAndBuild_30, jobname: PredictAndBuild_30. When run on command line you can specify any 4-or-more character jobname if you want. carry_on_directory = None Name of carry-on directory (stores intermediate and final results). You can continue with a previous run by specifying this directory from a previous run. To rerun or carry on from the job in /Users/Documents/PredictAndBuild_30, you specify carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn. Then if you also specify carry_on=True, files will be read from this directory and copied to a new carry_on_directory inside the working directory. Normally set automatically by the GUI in the directory containing your job: directory: PredictAndBuild_30, jobname: PredictAndBuild_30 carry_on_directory: PredictAndBuild_30_CarryOn. NOTE: Cannot be the the current directory. NOTE: if you specify a CarryOn directory you must specify carry_on=True if you want it to be used. upload_download_directory = /Users/terwill/Downloads/ Directory to use for uploading and downloading files from Colab. Normally use your standard Downloads folder. alphafold_model_suffix = AlphaFold_cycle_1.pdb Name of AlphaFold model is <jobname>_<alphafold_model_suffix> when coming from Colab server. pae_suffix = PAE_cycle_1.jsn Name of PAE file is <jobname>_<pae_suffix> when coming from Colab server. carry_on = False Carry on with a job from where it left off. Read in all files that were created instead of making new ones. To use, run the job you want exactly as you did the first time except set carry_on=True and specify carry_on_directory=xxx where xxx is the carry_on_directory from the run you want to continue. If your previous run was /Users/Documents/PredictAndBuild_30, specify: carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn. NOTE: if you specify a CarryOn directory you must also specify carry_on=True (False is the default) if you want this to be used. prediction_server = *PhenixServer Colab ManualPrediction LocalServer Choose where prediction

calculations will be done. Normally use the Phenix server (PhenixServer). As a backup you can use Colab (you will need a Google account.) Use ManualPrediction if you are going to do the prediction off-line and copy the result to the location specified by PredictAndBuild. LocalServer is used internally only.
`run_directly = False` Run directly. Used internally only
`max_len_at_server = 1500` Maximum length of sequence at server. Used internally only
`dummy_run = False` Write dummy file. Used internally only
`run_server_test = False` Run test of server and stop (does nothing else and does not require any inputs)
`copy_of_final_model = None` Copy final model here. Used internally and by predict_model. (Also copy map files to same directory). If None, copy to working directory. Use `copy_of_final_model=null` to not write these files at all.
`copy_of_msa = None` Copy final msa here (only if there is just one). Used internally
`copy_of_pae = None` Copy final pae here (only if there is just one). Used internally
`predicted_model_prefix = None` Predicted models will have this prefix. Used by predict_model.
`job_title = None` Job title in PHENIX GUI, not used on command line
`predict_and_build_max_rmsd_to_predict_together = 2` Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
`template_file = None` Template model to be used in prediction.
`get_msa_with_mmseqs2 = True` Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply an MSA or to run the mmseqs2 server from the Phenix server or Colab.
`combine_rebuilt_and_autobuild = True` Combine rebuilt and autobuild models and use if better Rfree (X-ray only)....

- `jobname = "` Name of this job. Used to create a directory that will hold all the working files for the run. Normally set automatically by the GUI to match the directory where the job is run: directory: PredictAndBuild_30, jobname: PredictAndBuild_30. When run on command line you can specify any 4-or-more character jobname if you want.
- `carry_on_directory = None` Name of carry-on directory (stores intermediate and final results). You can continue with a previous run by specifying this directory from a previous run. To rerun or carry on from the job in /Users/Documents/PredictAndBuild_30, you specify `carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn`. Then if you also specify `carry_on=True`, files will be read from this directory and copied to a new `carry_on_directory` inside the working directory. Normally set automatically by the GUI in the directory containing your job: directory: PredictAndBuild_30, jobname: PredictAndBuild_30 `carry_on_directory: PredictAndBuild_30_CarryOn`. NOTE: Cannot be the the current directory. NOTE: if you specify a CarryOn directory you must specify `carry_on=True` if you want it to be used.
- `upload_download_directory = /Users/terwill/Downloads/` Directory to use for uploading and downloading files from Colab. Normally use your standard Downloads folder.
- `alphafold_model_suffix = AlphaFold_cycle_1.pdb` Name of AlphaFold model is <jobname>_<alphafold_model_suffix> when coming from Colab server.
- `pae_suffix = PAE_cycle_1.jsn` Name of PAE file is <jobname>_<pae_suffix> when coming from Colab server.
- `carry_on = False` Carry on with a job from where it left off. Read in all files that were created instead of making new ones. To use, run the job you want exactly as you did the first time except set `carry_on=True` and specify `carry_on_directory=xxx` where xxx is the `carry_on_directory` from the run you want to continue. If your previous run was /Users/Documents/PredictAndBuild_30, specify: `carry_on_directory= /Users/Documents/PredictAndBuild_30/PredictAndBuild_30_CarryOn`. NOTE: if you specify a CarryOn directory you must also specify `carry_on=True` (False is the default) if you want this to be used.
- `prediction_server = *PhenixServer Colab ManualPrediction LocalServer` Choose where prediction calculations will be done. Normally use the Phenix server (PhenixServer). As a backup you can use Colab (you will need a Google account.) Use ManualPrediction if you are going to do the prediction off-line and copy the result to the location specified by PredictAndBuild. LocalServer is used internally

only.

- `run_directly` = False Run directly. Used internally only
- `max_len_at_server` = 1500 Maximum length of sequence at server. Used internally only
- `dummy_run` = False Write dummy file. Used internally only
- `run_server_test` = False Run test of server and stop (does nothing else and does not require any inputs)
- `copy_of_final_model` = None Copy final model here. Used internally and by `predict_model`. (Also copy map files to same directory). If None, copy to working directory. Use `copy_of_final_model=null` to not write these files at all.
- `copy_of_msa` = None Copy final msa here (only if there is just one). Used internally
- `copy_of_pae` = None Copy final pae here (only if there is just one). Used internally
- `predicted_model_prefix` = None Predicted models will have this prefix. Used by `predict_model`.
- `job_title` = None Job title in PHENIX GUI, not used on command line
- `predict_and_buildmax_rmsd_to_predict_together` = 2 Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
`template_file` = None Template model to be used in prediction.
`get_msa_with_mmseqs2` = True Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply an MSA or to run the mmseqs2 server from the Phenix server or Colab.
`combine_rebuilt_and_autobuild` = True Combine rebuilt and autobuild models and use if better Rfree (X-ray only).
`include_side_in_templates` = False Include side-chains beyond CB in templates as well as just main and CB (two runs).
`only_include_side_in_templates` = None Include side-chains beyond CB in templates only (one run)
`use_templates_in_prediction` = True Use templates if available (i.e., after first cycle)
`required_free_r_to_keep` = 0.49 Skip refinements if R is worse than this value
`autobuild_resolution` = 3.5 Worst resolution where autobuild density modification may be useful.
`autobuild_satisfactory_r_value` = 0.25 Skip autobuild if R is this value or better
`autobuild_for_xray_density_modification` = None Use autobuild rebuilding for density modification. Default is True if resolution is `autobuild_resolution` or better.
`autobuild_n_cycle_rebuild_max` = 3 Maximum autobuild rebuild cycles
`autobuild_n_cycle_build_max` = 1 Maximum autobuild build cycles
`autobuild_n_cycle_refine` = 3 AutoBuild refinement cycles
`autobuild_quick` = False Run AutoBuild in quick mode
`autobuild_min_resolution` = 2 Minimum resolution for autobuilding. Resolution will be cut off at this value if it is better (smaller number).
`density_modify_at_start` = None Density modify cryo-EM maps before analysis. Requires two half maps. Default is True if half maps are supplied and no full map is supplied, otherwise False. Works best if maps contain entire structure with boundary (at least 5 Å) or are original full-sized maps.
`density_modify_with_model` = None Density modify using rebuilt model each cycle. default is True. Requires two half-maps for cryo-EM. Works best for cryo-EM if maps contain entire structure with boundary (at least 5 Å) or are original full-sized maps. Default is True for X-ray and False for cryo-EM
`vrms_value` = None Value of VRMS to use in molecular replacement. This value will be used for all chains if set. Default is to estimate this value for each chain from the average pLDDT value after trimming.
`large_sequence` = 1500 Length of sequence that requires `job_size=large`
`pause_after_mr` = False Pause after MR and refinement (X-ray only). You can go on after checking your results.
`pause_after_docking` = False Pause after docking and refinement (Cryo-EM only). You can go on after checking your results.
`stop_after_predict` = False Stop after prediction. No map file needed if set.
`stop_after_mr` = False Stop after MR.
`parallel_predictions` = 1 Number of predictions to run at once. If more than one, no feedback is provided until the end. If there are waiting jobs in the queue running more than one at a time does not help (your first job may go near the front of the line but the others stay at the end until the first one finishes).
`write_prediction_files_and_wait` = False Write prediction files in working

directory and wait for result to appear. Only used for servers.put_files_for_prediction_here = None
 Program will put prediction files (pdb, msa) in this directory for the rest server to read (may look like
 /app/xxx/xxx)files_for_prediction_will_be_transferred_here = None Program will expect prediction files
 (pdb, msa) from the put_files_for_prediction_here directory will be copied to this directory so the rest
 server can read them. Your system has to automatically do this transfer.precalculated_msas_dir =
 /app/results/precaculated_msas Location of precalculated msas (usually on
 server)precaculated_results_dir = /app/results/precaculated_results Location of precalculated results
 (usually on server)precaculated_pae_dir = /app/results/precac...

- max_rmsd_to_predict_together = 2 Two models are considered similar (and not worth predicting separately) if they have an rmsd after superposition of this value or less (A)
- template_file = None Template model to be used in prediction.
- get_msa_with_mmseqs2 = True Run the mmseqs2 server from this computer to get MSA if not already available. Alternatives are to supply and MSA or to run the mmseqs2 server from the Phenix server or Colab.
- combine_rebuilt_and_autobuild = True Combine rebuilt and autobuild models and use if better Rfree (X-ray only).
- include_side_in_templates = False Include side-chains beyond CB in templates as well as just main and CB (two runs).
- only_include_side_in_templates = None Include side-chains beyond CB in templates only (one run)
- use_templates_in_prediction = True Use templates if available (i.e., after first cycle)
- required_free_r_to_keep = 0.49 Skip refinements if R is worse than this value
- autobuild_resolution = 3.5 Worst resolution where autobuild density modification may be useful.
- autobuild_satisfactory_r_value = 0.25 Skip autobuild if R is this value or better
- autobuild_for_xray_density_modification = None Use autobuild rebuilding for density modification. Default is True if resolution is autobuild_resolution or better.
- autobuild_n_cycle_rebuild_max = 3 Maximum autobuild rebuild cycles
- autobuild_n_cycle_build_max = 1 Maximum autobuild build cycles
- autobuild_n_cycle_refine = 3 AutoBuild refinement cycles
- autobuild_quick = False Run AutoBuild in quick mode
- autobuild_min_resolution = 2 Minimum resolution for autobuilding. Resolution will be cut off at this value if it is better (smaller number).
- density_modify_at_start = None Density modify cryo-EM maps before analysis. Requires two half maps. Default is True if half maps are supplied and no full map is supplied, otherwise False. Works best if maps contain entire structure with boundary (at least 5 A) or are original full-sized maps.
- density_modify_with_model = None Density modify using rebuilt model each cycle. default is True. Requires two half-maps for cryo-EM. Works best for cryo-EM if maps contain entire structure with boundary (at least 5 A) or are original full-sized maps. Default is True for X-ray and False for cryo_EM
- vrms_value = None Value of VRMS to use in molecular replacement. This value will be used for all chains if set. Default is to estimate this value for each chain from the average pLDDT value after trimming.
- large_sequence = 1500 Length of sequence that requires job_size=large

- `pause_after_mr = False` Pause after MR and refinement (X-ray only). You can go on after checking your results.
- `pause_after_docking = False` Pause after docking and refinement (Cryo-EM only). You can go on after checking your results.
- `stop_after_predict = False` Stop after prediction. No map file needed if set.
- `stop_after_mr = False` Stop after MR.
- `parallel_predictions = 1` Number of predictions to run at once. If more than one, no feedback is provided until the end. If there are waiting jobs in the queue running more than one at a time does not help (your first job may go near the front of the line but the others stay at the end until the first one finishes).
- `write_prediction_files_and_wait = False` Write prediction files in working directory and wait for result to appear. Only used for servers.
- `put_files_for_prediction_here = None` Program will put prediction files (pdb, msa) in this directory for the rest server to read (may look like `/app/xxx/xxx`)
- `files_for_prediction_will_be_transferred_here = None` Program will expect prediction files (pdb, msa) from the `put_files_for_prediction_here` directory will be copied to this directory so the rest server can read them. Your system has to automatically do this transfer.
- `precalculated_msas_dir = /app/results/precalculated_msas` Location of precalculated msas (usually on server)
- `precalculated_results_dir = /app/results/precalculated_results` Location of precalculated results (usually on server)
- `precalculated_pae_dir = /app/results/precalculated_pae` Location of precalculated pae files (usually on server)
- `allow_precalculated_result_file = True` Allow using precalculated results if inputs are identical
- `allow_precalculated_msa_file = True` Allow using precalculated msas if inputs are identical
- `files_from_prediction_may_be_found_here = None` Look for output from prediction here (in case full path is not available)
- `rest_server_incoming = None` Location on REST server where files should be placed
- `max_wait_time = 100000` Maximum wait time for `write_prediction_files_and_wait` (sec)
- `stop_if_internet_not_available = True` Stop if attempting to predict models online and internet is not available
- `number_of_models = 5` Number of models to generate for each chain. Normal way to set the value of `random_seed_iterations`. If a good pLDDT is found, no more models are created, where good is `good_enough_plddt` (typically 80)
- `use_supplied_random_seed_in_prediction = False` Use random seed to set up prediction (default is to use a standard random seed).
- `update_unique_sequences = True` Update unique sequences based on MR or docking results
- `minimum_residues = 20` Minimum residues in a chain
- `maximum_residues = 10000` Maximum residues in a sequence that can be used for prediction
- `break_into_chunks_if_length_is = 1500` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.

- `chunk_size = 600` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `overlap_size = 200` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `overlap_match_size = 50` If a sequence is less than `maximum_residues` but at least `break_into_chunks_if_length_is`, break it into chunks of length `chunk_size` with overlap of `overlap_size`, then reassemble using overlap window of `overlap_match_size`.
- `morph_reassembled_model = True` If `break_into_chunks_if_length_is` is applied, after reassembling identify close domains that were in different chunks and repredict them to get their relationships. Then adjust the reassembly to minimize overlaps and match pair relationships. NOTE: This reduces overlaps and can improve domain relationships but can yield distortions of the predicted model.
- `minimum_cycles = 2` Minimum cycles to carry out
- `use_input_scaffold_if_supplied = True` If input scaffold is provided, use it throughout (all cycles)
- `default_maxit_dir = /app/results/rest/` Default place to look for maxit if not supplied. This directory should contain the whole maxit-v11.100-prod-src
- `skip_copying_files = False` Skip copying output files to GUI directory
- `refinementmacro_cycles = 3` Refinement macro_cycles
- `macro_cycles = 3` Refinement macro_cycles
- `rest_serverurl = None` The URL for the Phenix REST server Normally set automatically
`url_type = *prediction ai` Type of server url (prediction or ai). Normally set automatically
`port = None` The port for interacting with the Phenix REST server Normally set automatically
`token = None` Authentication token for accessing the Phenix REST server. Normally set automatically
`timeout = 5` Time to keep trying to connect to a server
`quick_check_interval = 1` Time in seconds between job status checks
`check_interval = 60` Time in seconds between job status checks
`max_tries = 10` Number of tries to get result from server
`max_tries_on_availability = 2` Number of tries to see if server is up
`stop_if_server_not_available = True` Stop if server is not available (wrong url or down)
`job_size = small *medium large` Size of job (small, medium, large)
`requires_gpu = True` Job requires GPU
`running_server_test = False` This is a test
`jobverbose = False` Verbose output
- `url = None` The URL for the Phenix REST server Normally set automatically
- `url_type = *prediction ai` Type of server url (prediction or ai). Normally set automatically
- `port = None` The port for interacting with the Phenix REST server Normally set automatically
- `token = None` Authentication token for accessing the Phenix REST server. Normally set automatically
- `timeout = 5` Time to keep trying to connect to a server
- `quick_check_interval = 1` Time in seconds between job status checks
- `check_interval = 60` Time in seconds between job status checks
- `max_tries = 10` Number of tries to get result from server
- `max_tries_on_availability = 2` Number of tries to see if server is up
- `stop_if_server_not_available = True` Stop if server is not available (wrong url or down)
- `job_size = small *medium large` Size of job (small, medium, large)
- `requires_gpu = True` Job requires GPU

- `running_server_test = False` This is a test job

- `verbose = False` Verbose output

- `input_filesseq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)
`xray_data_file = None` Input X-ray data file (MTZ format).
`xray_data_label = None` Label specifying which data column to use from `xray_data_file`
`xray_test_flag_label = None` Label specifying which test flag column (if any) to use from `xray_data_file`
`truncate_models_to_match_sequences = True` If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
`density_select = None` If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is `True` if map is full-size and `False` if not.
`predicted_model = None` One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
`b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
`msa_list = None` Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
`models_are_already_placed = False` You can specify that all models (predicted_model or model) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a scaffold_model with the same contents as the model.
`model_copies_list = None` List of number of copies to find of each model in predicted_model. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
`processed_model_file = None` Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
`docked_model_file = None` Docked model (e.g., output of `dock_processed_model`). Not available in `predict_and_build` GUI. If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
`morphed_model_file = None` Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
`previous_model_file = None` Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
`scaffold_model = None` Scaffold model. If supplied, used as target for docking of each chain in predicted_model or model. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
`scaffold_minimum_chain_cc = 0.35` Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.
`base_scaffold_...`

- `seq_file = None` Input sequence file. Required if no models supplied. Used to create predicted models. Format should be Fasta or simple sequences separated by blank lines. One sequence per chain to be generated. Must match input models if both are supplied. (Fasta format or sequences separated by blank lines)

- `xray_data_file = None` Input X-ray data file (MTZ format).

- `xray_data_label` = None Label specifying which data column to use from `xray_data_file`
- `xray_test_flag_label` = None Label specifying which test flag column (if any) to use from `xray_data_file`
- `truncate_models_to_match_sequences` = True If input sequences start after or end before supplied predicted models, trim the predicted models to match sequences
- `density_select` = None If set, trim map to region containing molecule (non-zero region) before use. Recommended for cryo-EM maps that have a large map that is mostly unused. Default is True if map is full-size and False if not.
- `predicted_model` = None One or more input models in one or more files (normally predicted models) to be placed. Normally should start at residue 1 and match sequence file exactly.
- `b_value_field_is` = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor. Only applies to protein chains (always B-factor for non-protein).
- `msa_list` = None Multiple sequence alignment (MSA) file. Format is a3m only. First sequence in an MSA file must match a sequence in the sequence file exactly.
- `models_are_already_placed` = False You can specify that all models (`predicted_model` or `model`) are already placed (docked, placed in unit cell). No docking or MR is done if so. Equivalent to specifying a `scaffold_model` with the same contents as the model.
- `model_copies_list` = None List of number of copies to find of each model in `predicted_model`. Default is 1 (must specify all or none of them) . Specify all together like `model_copies_list=1,2,1,1`
- `processed_model_file` = None Processed model file (e.g., from `phenix.process_predicted_model`). If supplied, skip the `process_predicted_model` step and use this file. This model is expected to have one chain for each domain and actual B-values in the B-value field. It is expected that all poorly-predicted residues have been removed.
- `docked_model_file` = None Docked model (e.g., output of `dock_processed_model`). Not available in `predict_and_build` GUI. If supplied, use this file as the docked model. (Skip `process_predicted_model` and docking steps This model is expected to have a single chain with gaps for parts of the model that are not accurately known from the prediction. NOTE: You still need to supply the predicted model as the input model for this procedure in addition to this docked model.
- `morphed_model_file` = None Morphed model (e.g., full model morphed to match output of `dock_processed_model`). If supplied, skip the `process_predicted_model` and docking steps and use this file as the morphed model. This model is expected to have a single chain with no gaps. NOTE: You do not need to supply the predicted model as the input model for this procedure in addition to this morphed model.
- `previous_model_file` = None Previous model (from a previous run or external). Must match sequence of working model exactly. Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `previous_model` is used only as source of model information.
- `scaffold_model` = None Scaffold model. If supplied, used as target for docking of each chain in `predicted_model` or `model`. Must be similar in sequence to models to be docked. If supplied, `model_copies_list` is ignored and docking is done by superposition onto this structure instead of docking into density or MR. If the scaffold model does not have all chains represented, they are added in by docking.
- `scaffold_minimum_chain_cc` = 0.35 Chains in scaffold model that have `scaffold_minimum_cc` are not re-docked, but those with lower CC are.

- `base_scaffold_minimum_chain_cc = -1` Value of `scaffold_minimum_chain_cc` in cases where scaffold is known to be good.
- `fragments_model_file = None` Fragments model (e.g., `map_to_model.pdb`). Used as a source of possible model information and as a hypothesis for docking of working model. If `docked_model_file` is also supplied, `fragments_model` is used only as source of model information.
- `model_to_rebuild_file = None` Model to be rebuilt. Must be already in place (already docked). Must match supplied sequences exactly.
- `symmetry_file = None` Symmetry file (.ncs_spec format or MATRX records) with reconstruction symmetry. Used in identification of unique part of map. NOTE: symmetry file applies to map file in original position. NOTE 2: only proper symmetry (point-group, helical) is allowed NOTE 3: Only applies to cryo-EM maps
- `pae_file = None` Optional input json file with matrix of inter-residue estimated errors (pae file)
- `distance_model_file = None` Distance_model_file. A PDB or mmCIF file containing the model corresponding to the PAE matrix. Only needed if `weight_by_ca_ca_distances` is True.
- `search_model_copies = None` Used internally only
- `search_model = None` used internally only
- `map_modelfull_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
`half_map = None` Input half map files. Usually supply one full map or 2 half maps
`model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `full_map = None` Input map file (for cryo-EM). This can be a boxed or masked map showing just the molecule to dock (best) or a full map with symmetry. If your map has symmetry be sure to set `asymmetric_map = False`. If you have a map with symmetry you can supply a symmetry file if you want. Otherwise symmetry will be automatically determined.
- `half_map = None` Input half map files. Usually supply one full map or 2 half maps
- `model = None` Input predicted model (e.g., AlphaFold model). Assumed to have pLDDT values in B-value field (or RMSD values). May have multiple chains. Normally use `predicted_model` instead.
- `output_filesoutput_model_prefix = None` Output files with superposed models will begin with this prefix
`output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
`pdb_out = None` Used internally only
`temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `output_model_prefix = None` Output files with superposed models will begin with this prefix
- `output_seq_file = None` Sequence file (possibly edited) written to this file by PredictAndBuild
- `pdb_out = None` Used internally only
- `temp_dir = None` Temporary directory. Default is `dock_and_build_xx` where xx creates new directory
- `crystal_inforesolution = None` High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 A lower resolution than your data (i.e, if your data is 2.5 A, try 5 A).
`scattering_table = *None` n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
`wrapping = None` You can specify whether the map is wrapped (can map values

outside bounds to inside with cell translations).asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally onlysequence = None Old-style sequence string (alternative to sequence file).chain_type = *PROTEIN DNA RNA Chain typemsa = None MSA as text. Supply MSA as text or as an msa_file. Used internallytemplates_as_string = None Templates as string (single PDB file with one or more chains)space_group = None Space group (normally read from the data file, applies to X-ray)space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space groupunit_cell = None Unit Cell (normally read from the data file, applies to X-ray)unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""

- resolution = None High-resolution limit for main search. This can be lower resolution than the data. The search is quicker at lower resolution. If your model is poor, try 2-3 Å lower resolution than your data (i.e, if your data is 2.5 Å, try 5 Å).
- scattering_table = *None n_gaussian wk1995 it1992 electron neutron Choice of scattering table for structure factor calculations. Default for X-ray is n_gaussian, for cryoEM is electron.
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- asymmetric_map = None Specifies that this is an asymmetric map and no symmetry is to be supplied or found. Alternative to supplying a symmetry file or symmetry. Applies to cryo-EM reconstructions only.
- solvent_content = None Solvent fraction (content) of the cell. You can specify the fraction of the volume of this cell that is taken up by the macromolecule. Normally set automatically. Values go from 0 to 1.
- pdb70_text = None Database of PDB entries to be used as templates (pdb70 file). Normally used internally only
- sequence = None Old-style sequence string (alternative to sequence file).
- chain_type = *PROTEIN DNA RNA Chain type
- msa = None MSA as text. Supply MSA as text or as an msa_file. Used internally
- templates_as_string = None Templates as string (single PDB file with one or more chains)
- space_group = None Space group (normally read from the data file, applies to X-ray)
- space_group_alternatives = *HAND ALL NONE Space group alternatives for MR: ALL: all space groups in point group HAND: enantiomer and listed space group LIST: list supplied (not available) NONE: only supplied space group
- unit_cell = None Unit Cell (normally read from the data file, applies to X-ray)
- unique_sequencesFile names and data labels.sequence = "Enter or edit sequence" copies = 1 Copies of this sequencelabel = ""
- sequence = "Enter or edit sequence"
- copies = 1 Copies of this sequence
- label = ""
- iterationcycles = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quickcycle = 1 Iteration cycle

- cycles = None Cycles of prediction and rebuilding (default is 10 for Thorough, 3 for Standard and 1 for Quick
- cycle = 1 Iteration cycle
- process_predicted_modelremove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)continuous_chain = False When removing low-confidence residues, only trim from endsplitsplit_model_by_compact_regions = True Split model into compact regions after removing low-confidence residues.maximum_domains = 3 Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make domain_size smaller) to get more domains. If model is processed in chunks, maximum_domains will apply to each chunk.domain_size = 15 Approximate size of domains to be found (A units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making domain_size bigger (maximum is 70 A).adjust_domain_size = True If more than maximum_domains are initially found, increase domain_size in increments of 5 A and take the value that gives the smallest number of domains, but at least maximum_domains.minimum_domain_length = None Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or alt_pae_params=False and 20 for pae_matrix with alt_pae_params=Truemaximum_fraction_close = 0.3 Maximum fraction of CA in one domain close to one in another before mergingthemminimum_sequential_residues = None Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or alt_pae_params=False and 4 for pae_matrix with alt_pae_params=Trueminimum_remainder_sequence_length = 15 used to choose whether the sequence of a removed segment is written to the remainder sequence file.b_value_field_is = *plddt rmsd b_value The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (A) or a B-factorinput_plddt_is_fractional = None You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.minimum_plddt = None If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.maximum_rmsd = 1.5 If low-confidence residues are removed, the cutoff is defined by minimum_plddt or maximum_rmsd, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of input_plddt_is_fractional.default_maximum_rmsd = 1.5 Default value of maximum_rmsd, used if maximum_rmsd is not setsubtract_minimum_b = False If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low-confidence residues.pae_power = None If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 alt_pae_params=False and 2 for alt_pae_params=Truepae_cutoff = None If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with $\text{pae} < \text{pae_cutoff}$. Default is 5 alt_pae_params=False and 4 for alt_pae_params=Truepae_graph_resolution = None If PAE matrix (predicted alignment error matrix) is supplied, pae_graph_resolution regulates how aggressively the clustering algorithm is. Smaller values lead to larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=Truealt_pae_params = False If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolutio...
- remove_low_confidence_residues = True Remove low-confidence residues (based on minimum plddt or maximum_rmsd, whichever is specified)

- `continuous_chain = False` When removing low-confidence residues, only trim from ends
- `split_model_by_compact_regions = True` Split model into compact regions after removing low-confidence residues.
- `maximum_domains = 3` Maximum domains to obtain. You can use this to merge the closest domains at the end of splitting the model. Make it bigger (and optionally make `domain_size` smaller) to get more domains. If model is processed in chunks, `maximum_domains` will apply to each chunk.
- `domain_size = 15` Approximate size of domains to be found (Å units). This is the resolution that will be used to make a domain map. If you are getting too many domains, try making `domain_size` bigger (maximum is 70 Å).
- `adjust_domain_size = True` If more than `maximum_domains` are initially found, increase `domain_size` in increments of 5 Å and take the value that gives the smallest number of domains, but at least `maximum_domains`.
- `minimum_domain_length = None` Minimum length of a domain to keep (reject at end if smaller). Default is 10 if no pae matrix or `alt_pae_params=False` and 20 for pae_matrix with `alt_pae_params=True`
- `maximum_fraction_close = 0.3` Maximum fraction of CA in one domain close to one in another before merging them
- `minimum_sequential_residues = None` Minimum length of a short segment to keep (reject at end). Default is 5 if no pae matrix or `alt_pae_params=False` and 4 for pae_matrix with `alt_pae_params=True`
- `minimum_remainder_sequence_length = 15` used to choose whether the sequence of a removed segment is written to the remainder sequence file.
- `b_value_field_is = *plddt rmsd b_value` The B-factor field in predicted models can be pLDDT (confidence, 0-1 or 0-100) or rmsd (Å) or a B-factor
- `input_plddt_is_fractional = None` You can specify if the input plddt values (in B-factor field) are fractional (0-1) or not (0-100). By default if all values are between 0 and 1 it is fractional.
- `minimum_plddt = None` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `maximum_rmsd = 1.5` If low-confidence residues are removed, the cutoff is defined by `minimum_plddt` or `maximum_rmsd`, whichever is defined (you cannot define both). A minimum plddt of 0.70 corresponds to a maximum rmsd of 1.5. Minimum plddt values are fractional or not depending on the value of `input_plddt_is_fractional`.
- `default_maximum_rmsd = 1.5` Default value of `maximum_rmsd`, used if `maximum_rmsd` is not set
- `subtract_minimum_b = False` If set, subtract the lowest B-value from all B-values just before writing out the final files. Does not affect the cutoff for removing low- confidence residues.
- `pae_power = None` If PAE matrix (predicted alignment error matrix) is supplied, each edge in the graph will be weighted proportional to $(1/\text{pae}^{\text{pae_power}})$. Use this to try and get the number of domains that you want (try 1, 0.5, 1.5, 2). Default is 1 `alt_pae_params=False` and 2 for `alt_pae_params=True`
- `pae_cutoff = None` If PAE matrix (predicted alignment error matrix) is supplied, graph edges will only be created for residue pairs with `pae < pae_cutoff`. Default is 5 `alt_pae_params=False` and 4 for `alt_pae_params=True`
- `pae_graph_resolution = None` If PAE matrix (predicted alignment error matrix) is supplied, `pae_graph_resolution` regulates how aggressively the clustering algorithm is. Smaller values lead to

larger clusters. Value should be larger than zero, and values larger than 5 are unlikely to be useful, Default is 0.5 alt_pae_params=False and 4 for alt_pae_params=True

- alt_pae_params = False If PAE matrix is supplied, use alternative set of defaults (minimum_domain_length=20 minimum_sequential_residues=10 pae_power=2 pae_cutoff=4 pae_graph_resolution=4). Standard parameters (alt_pae_params=False) are: (minimum_domain_length=10 minimum_sequential_residues=5 pae_power=1 pae_cutoff=5 pae_graph_resolution=0.5).
- weight_by_ca_ca_distance = False Adjust the edge weighting for each residue pair according to the distance between CA residues. If this is True, then distance_model can be provided, otherwise supplied model will be used. See also distance_power
- distance_power = 1 If weight_by_ca_ca_distance is True, then edge weights will be multiplied by $1/\text{distance}^{**\text{distance_power}}$.
- stop_if_no_residues_obtained = True Raise Sorry and stop if processing yields no residues
- keep_all_if_no_residues_obtained = False Keep everything if processing yields no residues
- vrms_from_rmsd_intercept = 0.25 Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.
- vrms_from_rmsd_slope = 1.0 Estimate of vrms (error in model) from pLDDT will be based on $\text{vrms_from_rmsd_intercept} + \text{vrms_from_rmsd_slope} * \text{pLDDT}$ where mean pLDDT of non-low_confidence_residues is used.
- break_into_chunks_if_length_is = 1500 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae matrix
- chunk_size = 600 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae_matrix
- overlap_size = 200 If a sequence is at least break_into_chunks_if_length_is, break it into chunks of length chunk_size with overlap of overlap_size for domain identification using split_model_by_compact_regions without a pae_matrix
- searchdock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together. create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise. minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain. weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain. choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain low_res_search = True Try to fit by searching at low resolution first dock_with_mr = True Try

using MR to dock firstssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.refine_cycles = 3 rigid-body refinement cyclesssm_search_min_cc = 0.30 Stop ssm search if this cc achievedbackup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each timebackup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each timeresolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similarmax_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each otherradius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of modeluse_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.rigid_body_refinement = True Run rigid-body refinement on final modelrigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains)rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up moleculeigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specifiedappend_to_fixed_model = True Append placed search model to fixed model (if any) after searchmin_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.run_in_boxes = True Run on sub-boxes and combine at the endtarget_box_size = 60 Try to get boxes about this big on a side (grid units)target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_sizebox_to_run = None Run only this boxbox_overlap_scale = 1 box overlap (ov...

- dock_chains_individually = False Dock each chain (identified by chain_id field) individually. This is useful if the chains might have different orientations in the map compared to the search model, such as in a search model that is a predicted model. Normally used along with create_unique_chain_at_end=True for predicted models to put the model back together.

- create_unique_chain_at_end = None Take all docked chains and try to create one chain that has no duplicate residue numbers and that has distances between ends of fragments consistent with the number of residues between them. If symmetry has been applied, create N copies representing that symmetry. Default is True if dock_chains_individually = True and False otherwise.

- minimum_cc_to_keep_domain = 0.2 Do not keep domains in create_unique_chain_at_end if cc is less than minimum_cc_to_keep_domain.

- weight_sequential_fragments_by_distance = None Try to dock a chain in a way that minimizes the distance between its first residue and the previously-docked residue with the highest residue number less than this, and similarly for the last residue and the next available placed residue. Default is True if create_unique_chain_at_end is set. Normally use only if you have a model with a single chain and you are docking pieces of that chain.

- choose_better_of_individual_and_group_docking = None Dock entire search model (normally one only) and also by chain...pick better-fitting of the two for each chain

- low_res_search = True Try to fit by searching at low resolution first

- dock_with_mr = True Try using MR to dock first

- ssm_search = None Try to fit by searching for secondary structure. Default is False for dock_in_map and True for dock_and_rebuild if dock_with_mr is not used.

- refine_cycles = 3 rigid-body refinement cycles
- ssm_search_min_cc = 0.30 Stop ssm search if this cc achieved
- backup_resolution_cutoff_searches = 2 If initial fitting does not work, try up to this many times increasing resolution by backup_resolution_ratio each time
- backup_resolution_ratio = 1.67 If initial fitting does not work, try up to backup_resolution_cutoff_searches times increasing the resolution by backup_resolution_ratio each time
- resolution_radius_scale = 0.5 Resolution for low-res search will be resolution_radius_scale times the radius of gyration of the search model.
- align_moments = False Try to fit by aligning moments of inertia if size of density region and molecule are similar
- max_radius_ratio = 2. Radius of gyration of molecule and density must be within max_radius_ratio of each other
- radius_scale = 1.5 Mask for density and atoms will be radius_scale times the radius of gyration of model
- use_symmetry = True If search_model_copies or search_map_copies are the same for each model and greater than one and allow_symmetry is set, use symmetry in the map to place all copies after the first one.
- skip_if_low_cc = True Skip solution before rigid-body refinement if map-model CC is less than half of min_cc.
- rigid_body_refinement = True Run rigid-body refinement on final model
- rigid_body_refinement_single_unit = True Run rigid-body refinement with just one unit (do not break up into chains)
- rigid_body_refinement_split_method = *chain_id segid When splitting up molecule for rigid-body refinement (if rigid_body_refinement_single_unit=False), use either chain_id or segid to split up molecule
- rigid_body_refinement_resolution = None Run rigid-body refinement at this resolution if specified
- append_to_fixed_model = True Append placed search model to fixed model (if any) after search
- min_cc = 0.4 If quick run, stop if minimum CC is achieved in local search. Also always skip if starting CC_mask is less than 1/4 min_cc.
- run_in_boxes = True Run on sub-boxes and combine at the end
- target_box_size = 60 Try to get boxes about this big on a side (grid units)
- target_boxes = None Try to get this many boxes. Default is nproc unless this makes box size much smaller than target_box_size
- box_to_run = None Run only this box
- box_overlap_scale = 1 box overlap (overlap of boxes) will be box_overlap_scale times the density radius
- edge_ratio = 10 box edge box_overlap times edge_ratio
- density_radius = None Radius for density to be cut out and compared. Default is 6 times the resolution.
- model_radius = 3 Radius for removing density near fixed_model
- zero_value = 0 Value to set map in regions overlapping fixed model
- density_peaks = 20 Number of NCS-related peaks of density to check
- delta_phi = 20 Angular spacing of search

- `max_rot` = None Maximum rotations to try
- `rotz_only` = None Rotate only around Z
- `single_positions_to_try` = 10 Number of offset positions to try in optimizing orientation. Positions along the chain are selected as centers and a local fit near each position is carried out. The resulting offsets relative to the original placement are used to optimize the overall orientation and position.
- `max_position_shift_frac` = 0.05 Maximum fractional positional shift in `single_positions` run
- `min_relative_cc` = 0.67 Minimum local CC relative to original CC to keep a local search. This is a way to reject local searches that are completely wrong.
- `sieve_fit` = None Use `sieve_fit` fraction of single positions in fitting. If None, use all
- `ncs_copies_max` = None Maximum number of matching models to write. If more than one they will be written as MODEL records in the output PDB file. You can get them individually afterwards with `phenix.pdbtools placed_model.pdb keep="model 1"` etc.
- `start_rot` = None Three numbers `rotx`, `rotz`, `roty` defining the starting rotation of the search model. Normally used along with `delta_phi=1000` or `max_rot=1` to generate exactly one defined rotation.
- `search_center` = None Optional coordinates in search model for centering search. Note this is different from `target_search_center` which is the location xyz in the map to look.
- `search_center_selection` = None Optional selection defining coordinates in search model for centering search.
- `target_search_center` = None Optional coordinates in reference map where `search_center` should be approximately located after superimposing maps. Used to eliminate possible superpositions that place the search center elsewhere. Overlap scores decreased based on distance to `target_search_center/density_radius`.
- `model_search_position` = None Optional coordinates (usually part of search model) for matching to `target_search_position` after transformation. These can be specified in addition to `target_search_center`. Can have multiple `model_search_position` and `target_search_position` pairs by specifying each multiple times in order. NOTE: Not compatible with map search
- `target_search_position` = None Target positions for `model_search_position` after transformation.
- `search_position_radius` = None Radius for comparison of `target_search_position` and transformed `search_position` values. Default is `density_radius`. If specified, must be a single value or the same number of values as entries in `model_search_position` and `target_search_position`
- `rot_id_n` = None Number of rotation groups. Along with `rot_id_group`, allows defining groups of rotations to be carried out in one run. `short_caption` = Number of rotation groups
- `rot_id_group` = None rotation group to include. See `rot_id_n`.
- `map_box` = True Run `map_box` to extract useful part of map before search
- `fix_search_position` = False You can choose to not move your model center of mass to the origin by fixing the search position
- `search_box_size` = None You can choose the size of the search box for local searches. Normally set by default corresponding to 3 times `density_radius`
- `lower_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
- `upper_bounds` = None You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.

- `keep_search_order = None` Keep search order as input
- `remove_water = False` Remove waters and other hetero atoms from input files
- `prediction_method = *alphafold` Prediction method to use
- `template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword `upload_msa_file=True` (structure_search does not generate MSAs).
- `starting_alphafold_model = None` Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an `output_directory` with just an AlphaFold model named as the expected first AlphaFold model and specifying `carry_on=True`.
- `input_directory = ColabInputs` Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- `output_directory = ColabOutputs` Output directory. Copy outputs to `output_directory`. Used to restart with `carry on`.
- `save_outputs_in_google_drive = False` If run on Colab, copy outputs to `output_directory` in Google drive
- `content_dir = None` Content directory. Default is working directory
- `maxit_path = None` Path to maxit (pdb to cif) converter. Optional.
- `data_dir = /mnt` Data directory (location of AlphaFold parameters)
- `upload_file_with_jobname_resolution_sequence_lines = None` Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- `maximum_cycles = 10` Maximum cycles to carry out
- `cycle_rmsd_to_resolution_ratio = 0.25` Stop iteration if rmsd between subsequent AlphaFold models is less than `cycle_rmsd_to_resolution_ratio` times the resolution for two cycles in a row
- `significant_increase_in_residues = 5` Significant increase in residues in model
- `password = None` Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- `version = dev-4502` Version of Phenix to run (Colab only)
- `query_sequence = None` Sequence
- `resolution = None` Resolution of map (Å). Internal use only. Normally set instead `crystal_info.resolution`.
- `jobname = None` Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- `use_msa = True` Use multiple sequence alignments at some points
- `skip_all_msa_after_first_cycle = False` Skip multiple sequence alignments after first cycle
- `include_templates_from_pdb = True` Include templates from PDB. Note special behavior when running `predict_and_build`: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- `maximum_templates_from_pdb = 20` Maximum templates from PDB to include
- `release_date = None` release date for templates from pdb (only use up to this date. Format is 2020-05-14).
- `upload_msa_file = False` Supply MSA directly (.a3m format). You can supply the MSA as a file in your `input_directory` and it will be used instead of using `mmseqs2` to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.
- `upload_manual_templates = False` Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_tem...`
- `prediction_method = *alphafold` Prediction method to use
- `template_search_method = *mmseqs2` structure_search Method to identify templates from PDB and generate MSAs. If `structure_search` is set, you can specify what PDB databases to use (same as in `phenix.structure_search`). If `structure_search` is set, you must supply your own MSA file with the keyword

upload_msa_file=True (structure_search does not generate MSAs).

- starting_alphafold_model = None Starting AlphaFold model. You can supply an AlphaFold model and skip the initial AlphaFold step. This is equivalent to setting up an output_directory with just an AlphaFold model named as the expected first AlphaFold model and specifying carry_on=True.
- input_directory = ColabInputs Input directory containing density map. The map filename must start with the same characters as the jobname (only including characters before the first underscore). If you are supplying an MSA file it goes here as well.
- output_directory = ColabOutputs Output directory. Copy outputs to output_directory. Used to restart with carry on.
- save_outputs_in_google_drive = False If run on Colab, copy outputs to output_directory in Google drive
- content_dir = None Content directory. Default is working directory
- maxit_path = None Path to maxit (pdb to cif) converter. Optional.
- data_dir = /mnt Data directory (location of AlphaFold parameters)
- upload_file_with_jobname_resolution_sequence_lines = None Upload a file with a set of jobs (Colab only). Each line in the file is a jobname, resolution, and sequence
- maximum_cycles = 10 Maximum cycles to carry out
- cycle_rmsd_to_resolution_ratio = 0.25 Stop iteration if rmsd between subsequent AlphaFold models is less than cycle_rmsd_to_resolution_ratio times the resolution for two cycles in a row
- significant_increase_in_residues = 5 Significant increase in residues in model
- password = None Phenix download password (Colab only). The password used to download Phenix at your institution. Updated weekly, so you may need to request a new one frequently.
- version = dev-4502 Version of Phenix to run (Colab only)
- query_sequence = None Sequence
- resolution = None Resolution of map (Å). Internal use only. Normally set instead crystal_info.resolution.
- jobname = None Name of this job. The first characters before any underscore must be unique and will define the first characters of the corresponding map file in the input directory. The job name will normally also be the name of the working directory. Used internally only
- use_msa = True Use multiple sequence alignments at some point
- skip_all_msa_after_first_cycle = False Skip multiple sequence alignments after first cycle
- include_templates_from_pdb = True Include templates from PDB. Note special behavior when running predict_and_build: applies only on first cycle, and if PhenixServer or Colab used prediction will be run with and without templates and PDB and highest plddt model will be kept
- maximum_templates_from_pdb = 20 Maximum templates from PDB to include
- release_date = None release date for templates from pdb (only use up to this date. Format is 2020-05-14.
- upload_msa_file = False Supply MSA directly (.a3m format). You can supply the MSA as a file in your input_directory and it will be used instead of using mmseqs2 to generate an MSA. Your file name must end in .a3m. The format is two lines per sequence, the first starts with a greater-than sign and is ignored, the second is a sequence of letters or minus signs. All sequence lines must have the same length. The first sequence must be the target.

- `upload_manual_templates = False` Supply templates for AlphaFold prediction. Used in the same way as templates from the PDB unless `uploaded_templates_are_map_to_model` is set. May be .cif or .pdb files. Templates must start with characters matching the first characters in the jobname (before the first underscore), the remainder of the file name can be anything but just end in .cif or .pdb. The files must be in the `input_directory` or be uploaded (in Colab only).
- `uploaded_templates_are_map_to_model = False` The manual templates are models that may or may not have the sequence of the alphafold models. These are used only as suggestions for placement of the main chain.
- `upload_maps = True` Use maps (required to be True)
- `random_seed = 7231771` Random seed. Used internally
- `random_seed_iterations = 1` Random seed iterations of AlphaFold in first cycle. The model with the highest pLDDT will be used. If running `predict_and_build` and `include_templates_from_pdb` is True this many iterations will be carried out with and without templates . In `predict_and_build` and `predict_chain` set `number_of_models` instead.
- `minimum_random_seed_iterations = 1` Random seed iterations of AlphaFold after first cycle. The model with the highest pLDDT will be used
- `big_improvement = 10` How much improvement in pLDDT is worth going through all randomization cycles
- `good_enough_plddt = 80` Value of pLDDT that is good enough to not make any more models
- `nproc = 4` Number of processors to use. Internal use only. Normally set instead `control.nproc`
- `debug = False` Debugging run (print traceback error messages)
- `get_msa_only = False` Just get MSA and save it on server, no prediction. Does not return the MSA
- `carry_on = False` Carry on from where previous run ended. Used (usually in Colab) to go on after a crash or timeout. Requires that files are saved in the output directory
- `cif_dir = None` Location of templates (normally set automatically)
- `template_hit_list = None` List of templates (normally set automatically)
- `jobnames = None` List of jobnames (normally set automatically)
- `resolutions = None` List of resolutions (normally set automatically)
- `include_templates_from_pdb_list = None` List of `include_templates_from_pdb` ((normally set automatically)
- `include_side_in_templates_list = None` List of `include_side_in_templates_list` ((normally set automatically)
- `map_filename_dict = None` Map filename dict (used internally only)
- `msa_filename_dict = None` MSA filename dict (used internally only)
- `cif_filename_dict = None` CIF filename dict (used internally only)
- `query_sequences = None` List of query sequences (one per jobname) (used internally only)
- `manual_templates_uploaded = None` List of manual templates (used internally only)
- `upload_dir = None` Upload directory (used internally only)
- `working_directory = None` Working directory (used internally only)
- `maps_uploaded = None` List of maps (used internally only)
- `msas_uploaded = None` List of msas (used internally only)

- num_models = 1 Number of models (used internally only)
- homooligomer = 1 Number of copies (used internally only)
- cycle = None Cycle number (internal use only)
- host_url = https://api.colabfold.com Host url for mmseqs
- use_env = None Use env (internal use only)
- use_custom_msa = None Use custom msa (internal use only)
- use_templates = None Use templates (internal use only)
- template_paths = None Template paths (internal use only)
- mtm_file_name = None map_to_model file name (internal use only)
- cycle_model_file_name = None cycle_model_file_name (internal use only)
- previous_final_model_name = None previous_final_model_name (internal use only)
- msa = None MSA (internal use only)
- msa_is_msa_object = None MSA info (internal use only)
- deletion_matrix = None Deletion matrix (internal use only)
- structure_search_paramsstructure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure searchstructure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results

from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optiosPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCIF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be \$PDB_MIRRORDIR/data/structures/divided/mmcif directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and...

- structure_searchpdb_file = None Enter a PDB file name sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB. output_prefix = 'output' Provide an output prefix if needed blastpath = None Enter path to blastall executable sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure searchstructure_only = False Do only a Ramachandran-based structure search. db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2db = 'rcsb' Database used in search. rcsb, scop95, or AF2.get_ligand = False Use get_ligand=True to retrieve ligands.get_rama_code_only = False Generate Rama code for input pdb/cif only. This is for developers only.get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.use_pdb100aa = False Use PDB100 sequence database for sequence search.use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directorycustom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collectedatom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc. get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. PDBs deposited after this year will be discarded.deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. PDBs deposited before this year will be discarded.batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processedmin_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.write_results = True Set to False if no output results/log files is needed. Useful when calling Structure_Search within another program and only want to pass pdb objects.trim_hit_pdb = False Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.pickle_hits = False Pickle blast hit results from xml output.coot_display = False (default) Display output pdb files in coot.ask_coot = True prompt for coot display optiosPDB_MIRRORDIR = None Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in PDB_MIRRORDIR/data/structures/divided/pdb directory. If you use PDB's rsync script, this variable would be the same as the \$MIRRORDIR set in the scriptPDB_MIRROR_MMCIF = None Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory ## where ## are the second and third letters in the PDB id. This keyword should be

\$PDB_MIRRORDIR/data/structures/divided/mmcif directory. PDB_MIRROR_PDB = None Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory ## where ## are the second and third letters in the ...

- pdb_file = None Enter a PDB file name
- sequence = None Optional Fasta sequence file. Only needed for a quick sequence search against RCSB without a PDB.
- output_prefix = 'output' Provide an output prefix if needed
- blastpath = None Enter path to blastall executable
- sequence_only = False Do a Blast search against PDBaa sequence instead of doing a Ramachandran-based structure search
- structure_only = False Do only a Ramachandran-based structure search.
- db_used = 'rcsb' structure database used in search. rcsb, scop95, or AF2
- db = 'rcsb' Database used in search. rcsb, scop95, or AF2.
- get_ligand = False Use get_ligand=True to retrieve ligands.
- get_ramcode_only = False Generate Rama code for input pdb/cif only. This is for developers only.
- get_xml_only = False Get BLAST XML output returned as a string object. No coordinate superposition will be performed. Developers only.
- use_pdb100aa = False Use PDB100 sequence database for sequence search.
- use_custom_db = False Use custom database specified by custom_db_files/custom_db_dir.
- custom_db_dir = None The directory of pdb/cif files to make custom database. Default is current directory
- custom_db_files = None Filenames of the pdb/cif files separated by spaces for database. If none specified, all pdb/cif in the custom_db_dir will be collected
- atom_selection = 'all' Choose part of the pdb used in the search (default=all). for example: chain B, resseq 113:219, ... etc.
- get_pdb = 10 get_pdb=N will collect and superpose the top N homologous pdbs. Use get_pdb=0 to disable this option.
- deposited_before = 0 Specify the latest year of matching structures to be considered for scoring. Pdb's deposited after this year will be discarded.
- deposited_after = 0 Specify the earliest year of matching structures to be considered for scoring. Pdb's deposited before this year will be discarded.
- batch_size = 0 Process the pdbs in batch of <batch_size> until <min_match> hits are identified or until all <get_pdb> pdbs are processed
- min_match = 0 Finish structure_search when <min_match> matches are found. Usually uses with <trim_ends> to exit the search once find suitable pdbs.
- keep_all_pdb = False Keep all the PDB files, including full PDB, PDB_Chain and superposed PDB_Chain. Default is False which will keep only superposed PDB_Chain files in the directory specified in the output message.
- trim_ends = False Remove terminal residues of hit pdbs extending beyond those of the target pdb.
- write_pdb = True Set to False if no output pdb file is needed. Sometimes useful if use Structure_Search within another program and only want to pass pdb objects.

- `write_results = True` Set to False if no output results/log files is needed. Useful when calling `Structure_Search` within another program and only want to pass pdb objects.
- `trim_hit_pdb = False` Remove extra domains, extended loops, and unfit portions of hit pdbs after superposed to the target pdb.
- `pickle_hits = False` Pickle blast hit results from xml output.
- `coot_display = False` (default) Display output pdb files in coot.
- `ask_coot = True` prompt for coot display options
- `PDB_MIRRORDIR = None` Enter the top directory of local RCSB PDB mirror. The program will try to retrieve PDBs and/or structure factors from this mirror first. Note this assumes the directory trees under it follows those in RCSB -- pdb files as 'pdb####.ent.gz' in `PDB_MIRRORDIR/data/structures/divided/pdb` directory. If you use PDB's rsync script, this variable would be the same as the `$MIRRORDIR` set in the script
- `PDB_MIRROR_MMCIF = None` Enter the parent directory of the mmCIF files in the local PDB mirror. MMCIFs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/mmcif` directory.
- `PDB_MIRROR_PDB = None` Enter the parent directory of the PDB files in the local PDB mirror. PDBs will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be `$PDB_MIRRORDIR/data/structures/divided/pdb` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and others together. However, users may choose to specify `PDB_MIRROR_PDB` directly
- `PDB_MIRROR_STRUCTURE_FACTORS = None` Enter the parent directory of the PDB files in the local PDB mirror. structure factors s will be retrieved from subdirectory `##` where `##` are the second and third letters in the PDB id. This keyword should be the same as the `$PDB_MIRRORDIR/data/structures/divided/structure_factors` directory. We recommend setting `PDB_MIRRORDIR` and it will take care of both `PDB_MIRROR_PDB` and `PDB_MIRROR_STRUCTURE_FACTORS` together. However, users may choose to specify `PDB_MIRROR_STRUCTURE_FACTORS` directly
- `local_pdb_dir = None` Enter the path directly to your local PDB repository.
- `verbose = False` verbose output
- `debug = False` debugging output
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `guiGUI-specific parameter required for output directory` `output_dir = None`
- `output_dir = None`
- `buildrebuild_strategy = Thorough` Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding. `refine_only = None` Refine only, no rebuilding steps (set automatically with Quick) `refine_only_resolution = 3.5` Default cutoff for using Refine only `run_fit_loops = True` Run standard fit loops `run_iterative_morph = True` Run iterative morphing `run_trace_loops_through_density = True` Run loop fitting with trace_through_density algorithm `run_refine = True` Refine morphed model. Required for `run_fit_loops` or `run_trace_loops_through_density` `run_iterative_resolution_refine = True` Refine morphed model with iterative resolution method `run_extend = True` Extend ends of morphed model `extract_unique = True` Extract unique part of map. Applies after `density_select` (if set). `use_symmetry_in_extract_unique = True` Use symmetry from symmetry file (if available) when extracting unique part of

map.acceptable_docking_cc = 0.5 Acceptable docking CC for a chain
 minimum_docking_cc = 0.15 Minimum docking CC for a chain
 minimum_cutoff = 0.1 Minimum cutoff for estimating density in better
 parts of model
 reasonable_cc_ratio = 0.80 Acceptable CC (ratio) for a chain relative to average. Used to
 make sure a chain entirely in very bad density is not kept
 reasonable_cc_diff = 0.15 Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not
 kept
 shift_field_distance = None Shift field characteristic distance (default = 10 Å) for morphing
 cc_sd_ratio = 3. Keep residues with CC at least within cc_sd_ratio of the mean CC for good residues
 cc_sd_ratio_end = 2. Keep residues with CC at least within cc_sd_ratio_end of the mean CC for good residues (applies to
 end of fragments)
 cc_sd_ratio_ok = 2. Residues with CC at least within cc_sd_ratio_ok of the mean
 CC for good residues are ok (do not delete on the basis of plddt)
 max_gap_ratio = 3. Allow CA-CA distance to be up to max_gap_ratio times expected
 maximum_connectivity_deviation = 15 Maximum connectivity deviation. Reject a solution with bigger deviation than this plus 2 *
 resolution.
 keep_fraction_of_best = 0.5 Acceptable CC to keep as ratio to best found. Applies if best found
 is at least acceptable_docking_cc.
 keep_maximum_entries = 10 Maximum dock positions to keep
 rmsd_for_similar_placement = None RMSD value indicating that two placements are similar. default
 is resolution of map
 rigid_body_refine_cycles = 1 Refinement cycles for rigid-body
 refinement
 overlap_ca_ca_distance = 3 Overlap distance for CA-CA atoms (or P-P)
 ca_ca_distance = 3.8 CA-CA distance (or P-P)
 allowed_fraction_overlapping = 0.10 Allowed fraction
 overlapping
 maximum_combinations = 100000 Maximum combinations of placements to
 consider
 proceed_with_any_symmetry = False Run even with symmetry that is not point-group or helical.
 Not recommended as symmetry may not work properly
 box_cushion = 20 Size of buffer around model when boxing
 refine_cycles = 3 Refinement cycles
 loop_refine_cycles = 5 Refinement cycles for loops
 loop_backup_residues = 3 Number of tries removing one residue at a time from each end of existing
 ends of loop if no loop is found with initial gap
 residues_to_trim = 5 Residues to trim on each end of all trimmed fragments
 find_ncs_from_model = True Find NCS (symmetry) from working models and apply in
 density modification. Also turns on finding NCS in any autobuild density modification (including
 density-based search)
 allow_split_more_than_one_chain_for_mr = False Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as
 reassembly may not work with split multiple chains.
 acceptable_cc_ratio = 0.8 Keep segments with CC at least equal to average of base segments times minimum_cc_ratio minus difference between segment
 confidence...

- rebuild_strategy = Thorough Standard Quick Rebuilding strategy. Standard is up to 3 cycles, refining and replacing poorly-fitting loops in predicted models each cycle, using autobuild for density modification (if Xray). Quick is refinement only, one cycle. Thorough is up to 10 cycles, extensive rebuilding.
- refine_only = None Refine only, no rebuilding steps (set automatically with Quick)
- refine_only_resolution = 3.5 Default cutoff for using Refine only
- run_fit_loops = True Run standard fit_loops
- run_iterative_morph = True Run iterative morphing
- run_trace_loops_through_density = True Run loop fitting with trace_through_density algorithm
- run_refine = True Refine morphed model. Required for run_fit_loops or run_trace_loops_through_density
- run_iterative_resolution_refine = True Refine morphed model with iterative resolution method
- run_extend = True Extend ends of morphed model
- extract_unique = True Extract unique part of map. Applies after density_select (if set).

- `use_symmetry_in_extract_unique = True` Use symmetry from symmetry file (if available) when extracting unique part of map.
- `acceptable_docking_cc = 0.5` Acceptable docking CC for a chain
- `minimum_docking_cc = 0.15` Minimum docking CC for a chain
- `minimum_cutoff = 0.1` Minimum cutoff for estimating density in better parts of model
- `reasonable_cc_ratio = 0.80` Acceptable CC (ratio) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `reasonable_cc_diff = 0.15` Acceptable CC (difference) for a chain relative to average. Used to make sure a chain entirely in very bad density is not kept
- `shift_field_distance = None` Shift field characteristic distance (default = 10 Å) for morphing
- `cc_sd_ratio = 3`. Keep residues with CC at least within `cc_sd_ratio` of the mean CC for good residues
- `cc_sd_ratio_end = 2`. Keep residues with CC at least within `cc_sd_ratio_end` of the mean CC for good residues (applies to end of fragments)
- `cc_sd_ratio_ok = 2`. Residues with CC at least within `cc_sd_ratio_ok` of the mean CC for good residues are ok (do not delete on the basis of plddt)
- `max_gap_ratio = 3`. Allow CA-CA distance to be up to `max_gap_ratio` times expected
- `maximum_connectivity_deviation = 15` Maximum connectivity deviation. Reject a solution with bigger deviation than this plus $2 * \text{resolution}$.
- `keep_fraction_of_best = 0.5` Acceptable CC to keep as ratio to best found. Applies if best found is at least `acceptable_docking_cc`.
- `keep_maximum_entries = 10` Maximum dock positions to keep
- `rmsd_for_similar_placement = None` RMSD value indicating that two placements are similar. default is resolution of map
- `rigid_body_refine_cycles = 1` Refinement cycles for rigid-body refinement
- `overlap_ca_ca_distance = 3` Overlap distance for CA-CA atoms (or P-P)
- `ca_ca_distance = 3.8` CA-CA distance (or P-P)
- `allowed_fraction_overlapping = 0.10` Allowed fraction overlapping
- `maximum_combinations = 100000` Maximum combinations of placements to consider
- `proceed_with_any_symmetry = False` Run even with symmetry that is not point-group or helical. Not recommended as symmetry may not work properly
- `box_cushion = 20` Size of buffer around model when boxing
- `refine_cycles = 3` Refinement cycles
- `loop_refine_cycles = 5` Refinement cycles for loops
- `loop_backup_residues = 3` Number of tries removing one residue at a time from each end of existing ends of loop if no loop is found with initial gap
- `residues_to_trim = 5` Residues to trim on each end of all trimmed fragments
- `find_ncs_from_model = True` Find NCS (symmetry) from working models and apply in density modification. Also turns on finding NCS in any autobuild density modification (including density-based search)

- `allow_split_more_than_one_chain_for_mr = False` Allow splitting chains into domains for mr (X-ray only) even if there are multiple chains. Normally only split if a single chain as reassembly may not work with split multiple chains.
- `acceptable_cc_ratio = 0.8` Keep segments with CC at least equal to average of base segments times `minimum_cc_ratio` minus difference between segment confidence (`plDDT`) and confidence cutoff (typically 0.7)
- `low_res_if_multiple_solutions = 3.5` Try phaser MR at increasing lower resolutions up to this value if multiple solutions are found
- `delta_low_res = 0.5` Resolution increment for `low_res_if_multiple_solutions`
- `controlstop_after_dock = None` Stop after docking step
`stop_after_morph = False` Stop after morphing step
`read_files = False` Read existing output files and use them if present
`write_files = True` Write output files
`nproc = 1` Number of processors to use on your local machine
`ignore_symmetry_conflicts = True` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
`random_seed = 171731` Random seed
`max_dirs = 1000` Maximum number of directories (`dock_and_build_xxx`)
`verbose = False` Verbose output
`quick = True` Run quickly
- `gui` GUI-specific parameter required for output directory
`output_dir = None`

mmtbx.prepare_pdb_deposition

[prepare_pdb_deposition.html](#)

Description

This tool is primarily for adding sequence information to the mmCIF output from phenix.refine to make the file suitable for deposition into the Protein Data Bank. The minimum inputs for this use case are the model from phenix.refine and a sequence file.

If you are starting with a model in PDB format, we recommended that you run phenix.refine on your model and data to generate the necessary fields related to the data, then run this tool to add the sequence information. If your model contains ligands, additional restraints files for the ligands may be necessary.

The input sequence should just be the one letter canonical sequence. For non-standard amino acids or residues, this tool will replace the one letter code with the three letter code surrounded by the parentheses. For example, if there is a selenomethionine, your input sequence should represent that residue as "M". This tool will replace the "M" with "(MSE)" in the correct field in the output mmCIF file.

However, this feature is dependent on curated monomer libraries, so if there is an ambiguity in the parent residue (e.g. methionine is the parent residue for selenomethionine), the conversion will not be automatic. For this situation, you can manually specify which residues to convert from the one letter code to the three letter code with the "custom_residues" parameter. You just need to provide a space-separated sequence of residues. Two examples where this is necessary are SUI in PDB code 2jjj and OTY in PDB code 4jye.

Command line usage examples

```
% mmtbx.prepare_pdb_deposition model.cif sequence.fa % mmtbx.prepare_pdb_deposition 2jjj.cif  
sequence.fa custom_residues="SUI OTY" % mmtbx.prepare_pdb_deposition 4jye.cif sequence.fa  
custom_residues=OTY
```

```
% mmtbx.prepare_pdb_deposition model.cif sequence.fa
```

```
% mmtbx.prepare_pdb_deposition 2jjj.cif sequence.fa custom_residues="SUI OTY"
```

```
% mmtbx.prepare_pdb_deposition 4jye.cif sequence.fa custom_residues=OTY
```

GUI

This tool is available in the Phenix GUI in the "PDB deposition" section.

List of all available keywords

custom_residues = None Space-separated list of three letter codes to be included in the
entity_poly.pdbx_one_letter_code mmCIF loop (e.g. custom_residues='SUI PYL')keep_original_loops = True
Preserves mmCIF data from the input model file (if available) that is not overwritten by other
inputalign_columns = False When set to True, the columns are aligned. This will take longer because the
column widths need to be determined before outputting.job_title = None Job title in PHENIX GUI, not used
on command linevalidationsequencealignmentssimilarity_matrix = blosum50 dayhoff *identity
min_allowable_identity = 0.5 pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True
show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all
SF4 F3S DVT disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position =

False correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones) disulfide_bond_exclusions_selection_string = None exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond. link_distance_cutoff = 3 Length of link between the linked residues disulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides = True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers only** nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing. const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only** vdw_1_4_factor = 0.8 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False restraints_library_cdl = True Use Conformation Dependent Library (CDL) for geometry restraints include_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF₄, FES, F3S, ... cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints cdl_svl = False rdl = False rdl_selection = *all TRP hpd1 = False cdl_nucleotides enable = False Use RestraintsLib for DNA and RNA for geometry restraints esd = *phenix csd factor = 2.0 user_supplied_path = None Root directory of user supplied restraints action = *pre post Choice to look here before GeoStd, the default, or after secondary structure ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False. from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp. max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True. max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains enabled = False Turn on secondary structure restraints (main switch) protein enabled = True Turn on secondary structure restraints fo...

- custom_residues = None Space-separated list of three letter codes to be included in the entity_poly.pdbx_one_letter_code mmCIF loop (e.g. custom_residues='SUI PYL')
- keep_original_loops = True Preserves mmCIF data from the input model file (if available) that is not overwritten by other input
- align_columns = False When set to True, the columns are aligned. This will take longer because the column widths need to be determined before outputting.
- job_title = None Job title in PHENIX GUI, not used on command line
- mmtbx_validation_sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- validation_sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- sequences sequence_alignment_similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5
- sequence_alignments similarity_matrix = blosum50 dayhoff *identity min_allowable_identity = 0.5

- similarity_matrix = blosum50 dayhoff *identity
- min_allowable_identity = 0.5
- pdb_interpretationsort_atoms = True use_ncs_to_build_restraints = True show_restraints_histograms = True flip_symmetric_amino_acids = True superpose_ideal_ligand = *None all SF4 F3S DVT
 disable_uc_volume_vs_n_atoms_check = False allow_polymer_cross_special_position = False
 correct_hydrogens = True c_beta_restraints = True use_neutron_distances = False Use neutron X-H
 distances (which are longer than X-ray ones)disulfide_bond_exclusions_selection_string = None
 exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it
 may coordinate then this SG is excluded from SS bond.link_distance_cutoff = 3 Length of link between
 the linked residuesdisulfide_distance_cutoff = 3 add_angle_and_dihedral_restraints_for_disulfides =
 True dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
 chir_volume_esd = 0.2 max_reasonable_bond_distance = 50.0 nonbonded_distance_cutoff = None
 default_vdw_distance = 1 min_vdw_distance = 1 nonbonded_buffer = 1 **EXPERIMENTAL, developers
 only**nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16
 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are
 missing.const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**vdw_1_4_factor = 0.8
 min_distance_sym_equiv = 0.5 custom_nonbonded_symmetry_exclusions = None
 translate_cns_dna_rna_residue_names = None proceed_with_excessive_length_bonds = False
 restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
 restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL)
 for tetrahedral Zn++ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False
 Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl =
 False rdl_selection = *all TRP hpdI = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and
 RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of
 user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or
 aftersecondary_structuress_by_chain = True Only applies if search_method = from_ca. Find secondary
 structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much
 slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many
 fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make
 from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if
 search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same
 for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a
 representative of all chains with the same sequence. Alternative is to examine each chain individually.
 Can be much slower with use_representative_of_chain=False if there are many symmetry copies.
 Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method =
 from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure
 restraints (main switch)proteinenabled = True Turn on secondary structure restraints for
 proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary
 structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5
 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true,
 h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True
 helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection =
 None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10
 helices are used for hydrogen-bond res...
- sort_atoms = True
- use_ncs_to_build_restraints = True
- show_restraints_histograms = True

- flip_symmetric_amino_acids = True
- superpose_ideal_ligand = *None all SF4 F3S DVT
- disable_uc_volume_vs_n_atoms_check = False
- allow_polymer_cross_special_position = False
- correct_hydrogens = True
- c_beta_restraints = True
- use_neutron_distances = False Use neutron X-H distances (which are longer than X-ray ones)
- disulfide_bond_exclusions_selection_string = None
- exclusion_distance_cutoff = 3 If SG of CYS forming SS bond is closer than this distance to an atom that it may coordinate then this SG is excluded from SS bond.
- link_distance_cutoff = 3 Length of link between the linked residues
- disulfide_distance_cutoff = 3
- add_angle_and_dihedral_restraints_for_disulfides = True
- dihedral_function_type = *determined_by_sign_of_periodicity all_sinusoidal all_harmonic
- chir_volume_esd = 0.2
- max_reasonable_bond_distance = 50.0
- nonbonded_distance_cutoff = None
- default_vdw_distance = 1
- min_vdw_distance = 1
- nonbonded_buffer = 1 **EXPERIMENTAL, developers only**
- nonbonded_weight = None Weighting of nonbonded restraints term. By default, this will be set to 16 if explicit hydrogens are used (this was the default in earlier versions of Phenix), or 100 if hydrogens are missing.
- const_shrink_donor_acceptor = 0 **EXPERIMENTAL, developers only**
- vdw_1_4_factor = 0.8
- min_distance_sym_equiv = 0.5
- custom_nonbonded_symmetry_exclusions = None
- translate_cns_dna_rna_residue_names = None
- proceed_with_excessive_length_bonds = False
- restraints_librarycdl = True Use Conformation Dependent Library (CDL) for geometry
- restraintsinclude_modified_amino_acid_in_cdl = False mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF4, FES, F3S, ...cis_pro_eh99 = True omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraintscdl_svl = False rdl = False rdl_selection = *all TRP hpdI = False cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0 user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- cdl = True Use Conformation Dependent Library (CDL) for geometry restraints
- include_modified_amino_acid_in_cdl = False
- mcl = True Use Metal Coordination Library (MCL) for tetrahedral Zn⁺⁺ and iron-sulfur clusters SF4, FES, F3S, ...

- cis_pro_eh99 = True
- omega_cdl = False Use Omega Conformation Dependent Library (omega-CDL) for geometry restraints
- cdl_svl = False
- rdl = False
- rdl_selection = *all TRP
- hpdl = False
- cdl_nucleotidesenable = False Use RestraintsLib for DNA and RNA for geometry restraintsesd = *phenix csd factor = 2.0
- enable = False Use RestraintsLib for DNA and RNA for geometry restraints
- esd = *phenix csd
- factor = 2.0
- user_suppliedpath = None Root directory of user supplied restraintsaction = *pre post Choice to look here before GeoStd, the default, or after
- path = None Root directory of user supplied restraints
- action = *pre post Choice to look here before GeoStd, the default, or after
- secondary_structuresss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identificationuse_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chainsenabled = False Turn on secondary structure restraints (main switch)proteinenabled = True Turn on secondary structure restraints for proteinsearch_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.distance_ideal_n_o = 2.9 Target length for N-O hydrogen bonddistance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be establishedremove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be establishedrestrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restraining this particular helixselection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuehbonddonor = None acceptor = None sheetenabled = True Restraining this particular sheetfirst_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular valuestrandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None hbonddonor = None acceptor = None nucleic_acidenabled = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The

smaller number is tighter restraints.
base_pairenabled = True Restraint this particular base-pair
base1 = None Selection string selecting at least one atom in the desired residue
base2 = None Selection string selecting at least one atom in the desired residue
saenger_class = 0 Type of base-pair
restrain_planarity = False Apply planarity restraint to this base-pair
planarity_sigma = 0.176
restrain_hbonds = True Restrain hydrogen bonds length for this base-pair
restrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pair
restrain_parallelity = True Apply parallelity restraint to this base-pair
parallelity_target = 0 parallelity_sigma = 0.0335
stacking_pairenabled = True Restraint this particular base-pair
base1 = None Selection string selecting at least one atom in the desired residue
base2 = None Selection string selecting at least one atom in the desired residue
angle = 0 sigma = 0.027

- ss_by_chain = True Only applies if search_method = from_ca. Find secondary structure only within individual chains. Alternative is to allow H-bonds between chains. Can be much slower with ss_by_chain=False. If your model is complete use ss_by_chain=True. If your model is many fragments, use ss_by_chain=False.
- from_ca_conservative = False various parameters changed to make from_ca method more conservative, hopefully to closer resemble ksdssp.
- max_rmsd = 1 Only applies if search_method = from_ca. Maximum rmsd to consider two chains with identical sequences as the same for ss identification
- use_representative_chains = True Only applies if search_method = from_ca. Use a representative of all chains with the same sequence. Alternative is to examine each chain individually. Can be much slower with use_representative_of_chain=False if there are many symmetry copies. Ignored unless ss_by_chain is True.
- max_representative_chains = 100 Only applies if search_method = from_ca. Maximum number of representative chains
- enabled = False Turn on secondary structure restraints (main switch)
- proteinenabled = True Turn on secondary structure restraints for protein
search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established
restrain_hbond_angles = True helixserial_number = None helix_identifier = None enabled = True Restrain this particular helix
selection = None helix_type = *alpha pi 3_10 unknown Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices are used for hydrogen-bond restraints.
sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
hbonddonor = None acceptor = None sheetenabled = True Restrain this particular sheet
first_strand = None sheet_id = None sigma = 0.05 slack = 0 top_out = False angle_sigma_scale = 1 Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10.
angle_sigma_set = None Use this parameter to set sigmas for h-bond angles to a particular value
strandselection = None sense = parallel antiparallel *unknown bond_start_current = None bond_start_previous = None
hbonddonor = None acceptor = None
- enabled = True Turn on secondary structure restraints for protein
- search_method = *ksdssp from_ca cablam Particular method to search protein secondary structure.
- distance_ideal_n_o = 2.9 Target length for N-O hydrogen bond
- distance_cut_n_o = 3.5 Hydrogen bond with length exceeding this value will not be established
- remove_outliers = True If true, h-bonds exceeding distance_cut_n_o length will not be established

- `restrain_hbond_angles = True`
- `helixserial_number = None` `helix_identifier = None` `enabled = True` Restrain this particular helix
`selection = None` `helix_type = *alpha pi 3_10 unknown` Type of helix, defaults to alpha. Only alpha, pi, and 3_10
helices are used for hydrogen-bond restraints. `sigma = 0.05` `slack = 0` `top_out = False` `angle_sigma_scale`
`= 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to 10. `angle_sigma_set`
`= None` Use this parameter to set sigmas for h-bond angles to a particular value `hbonddonor = None`
`acceptor = None`
- `serial_number = None`
- `helix_identifier = None`
- `enabled = True` Restrain this particular helix
- `selection = None`
- `helix_type = *alpha pi 3_10 unknown` Type of helix, defaults to alpha. Only alpha, pi, and 3_10 helices
are used for hydrogen-bond restraints.
- `sigma = 0.05`
- `slack = 0`
- `top_out = False`
- `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to
10.
- `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
- `hbonddonor = None` `acceptor = None`
- `donor = None`
- `acceptor = None`
- `sheetenabled = True` Restrain this particular sheet `first_strand = None` `sheet_id = None` `sigma = 0.05`
`slack = 0` `top_out = False` `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original
sigmas range from 5 to 10. `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles
to a particular value `strandselection = None` `sense = parallel antiparallel *unknown` `bond_start_current =`
`None` `bond_start_previous = None` `hbonddonor = None` `acceptor = None`
- `enabled = True` Restrain this particular sheet
- `first_strand = None`
- `sheet_id = None`
- `sigma = 0.05`
- `slack = 0`
- `top_out = False`
- `angle_sigma_scale = 1` Multiply sigmas for h-bond angles by this value. Original sigmas range from 5 to
10.
- `angle_sigma_set = None` Use this parameter to set sigmas for h-bond angles to a particular value
- `strandselection = None` `sense = parallel antiparallel *unknown` `bond_start_current = None`
`bond_start_previous = None`
- `selection = None`
- `sense = parallel antiparallel *unknown`

- `bond_start_current` = None
- `bond_start_previous` = None
- `hbonddonor` = None `acceptor` = None
- `donor` = None
- `acceptor` = None
- `nucleic_acidenabled` = True Turn on secondary structure restraints for nucleic acidshbond_distance_cutoff = 3.4 Hydrogen bonds with length exceeding this limit will not be establishedscale_bonds_sigma = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.base_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335 stacking_pairenabled = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residueangle = 0 sigma = 0.027
- `enabled` = True Turn on secondary structure restraints for nucleic acids
- `hbond_distance_cutoff` = 3.4 Hydrogen bonds with length exceeding this limit will not be established
- `scale_bonds_sigma` = 1. All sigmas for h-bond length will be multiplied by this number. The smaller number is tighter restraints.
- `base_pairenabled` = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the desired residuesaenger_class = 0 Type of base-pairingrestrain_planarity = False Apply planarity restraint to this base-pairplanarity_sigma = 0.176 restrain_hbonds = True Restrain hydrogen bonds length for this base-pairrestrain_hb_angles = True Restrain angles around hydrogen bonds for this base-pairrestrain_parallelity = True Apply parallelity restraint to this base-pairparallelity_target = 0 parallelity_sigma = 0.0335
- `enabled` = True Restraint this particular base-pair
- `base1` = None Selection string selecting at least one atom in the desired residue
- `base2` = None Selection string selecting at least one atom in the desired residue
- `saenger_class` = 0 Type of base-pairing
- `restrain_planarity` = False Apply planarity restraint to this base-pair
- `planarity_sigma` = 0.176
- `restrain_hbonds` = True Restrain hydrogen bonds length for this base-pair
- `restrain_hb_angles` = True Restrain angles around hydrogen bonds for this base-pair
- `restrain_parallelity` = True Apply parallelity restraint to this base-pair
- `parallelity_target` = 0
- `parallelity_sigma` = 0.0335
- `stacking_pairenabled` = True Restraint this particular base-pairbase1 = None Selection string selecting at least one atom in the desired residuebase2 = None Selection string selecting at least one atom in the

desired residueangle = 0 sigma = 0.027

- enabled = True Restraint this particular base-pair
- base1 = None Selection string selecting at least one atom in the desired residue
- base2 = None Selection string selecting at least one atom in the desired residue
- angle = 0
- sigma = 0.027
- reference_coordinate_restraintsRestrains coordinates in Cartesian space to stay near their starting positions. This is intended for use in generating simulated annealing omit maps, to prevent refined atoms from collapsing in on the region missing atoms. For conserving geometry quality at low resolution, the more flexible reference model restraints should be used instead.enabled = False exclude_outliers = True selection = all sigma = 0.2 limit = 1.0 top_out = False reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- enabled = False
- exclude_outliers = True
- selection = all
- sigma = 0.2
- limit = 1.0
- top_out = False
- reference_is_average_alt_confs = False Sets the reference coordinate to the average of two alt. confs. Ignores selection option.
- automatic_restraintsside_chain_parallelity = False
- side_chain_parallelity = False
- automatic_linkinglink_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.link_none = False link_metals = Auto link_residues = True link_amino_acid_rna_dna = False link_carbohydrates = True link_ligands = True link_small_molecules = False metal_coordination_cutoff = 3.0 amino_acid_bond_cutoff = 1.9 inter_residue_bond_cutoff = 2.2 buffer_for_second_row_elements = 0.5 carbohydrate_bond_cutoff = 1.99 ligand_bond_cutoff = 1.99 small_molecule_bond_cutoff = 1.98 exclude_hydrogens_from_bonding_decisions = False
- link_all = False If True, bond restraints will be generated for any appropriate ligand-protein or ligand-nucleic acid covalent bonds. This includes sugars, amino acid modifications, and other prosthetic groups.
- link_none = False
- link_metals = Auto
- link_residues = True
- link_amino_acid_rna_dna = False
- link_carbohydrates = True
- link_ligands = True
- link_small_molecules = False
- metal_coordination_cutoff = 3.0
- amino_acid_bond_cutoff = 1.9

- inter_residue_bond_cutoff = 2.2
- buffer_for_second_row_elements = 0.5
- carbohydrate_bond_cutoff = 1.99
- ligand_bond_cutoff = 1.99
- small_molecule_bond_cutoff = 1.98
- exclude_hydrogens_from_bonding_decisions = False
- include_in_automatic_linkingselection_1 = None selection_2 = None bond_cutoff = 4.5
- selection_1 = None
- selection_2 = None
- bond_cutoff = 4.5
- exclude_from_automatic_linkingselection_1 = None selection_2 = None
- selection_1 = None
- selection_2 = None
- apply_cis_trans_specificationcis_trans_mod = cis *trans residue_selection = None Residues containing C-alpha atom of omega dihedral
- cis_trans_mod = cis *trans
- residue_selection = None Residues containing C-alpha atom of omega dihedral
- apply_cif_restraintsrestraints_file_name = None residue_selection = None
- restraints_file_name = None
- residue_selection = None
- apply_cif_modificationdata_mod = None residue_selection = None
- data_mod = None
- residue_selection = None
- apply_cif_linkdata_link = None residue_selection_1 = None residue_selection_2 = None
- data_link = None
- residue_selection_1 = None
- residue_selection_2 = None
- peptide_linkramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!!
 Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.cis_threshold = 45
 apply_all_trans = False discard_omega = False discard_psi_phi = True apply_peptide_plane = False
 omega_esd_override_value = None rama_weight = 1.0 scale_allowed = 1.0 rama_potential = *oldfield
 emsley rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.restrain_rama_outliers = True Apply restraints to Ramachandran outliersrestrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran
 plotrestrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027

- ramachandran_restraints = False !!! OBSOLETE. Kept for backward compatibility only !!! Restrains peptide backbone to fall within allowed regions of Ramachandran plot. Although it does not eliminate outliers, it can significantly improve the percent favored and percent outliers at low resolution. Probably not useful (and maybe even harmful) at resolutions much higher than 3.5Å.
- cis_threshold = 45
- apply_all_trans = False
- discard_omega = False
- discard_psi_phi = True
- apply_peptide_plane = False
- omega_esd_override_value = None
- rama_weight = 1.0
- scale_allowed = 1.0
- rama_potential = *oldfield emsley
- rama_selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- restrain_rama_outliers = True Apply restraints to Ramachandran outliers
- restrain_rama_allowed = True Apply restraints to residues in allowed region on Ramachandran plot
- restrain_allowed_outliers_with_emsley = False In case of restrain_rama_outliers=True and/or restrain_rama_allowed=True still restrain these residues with emsley. Make sense only in case of using oldfield potential.
- oldfieldesd = 10.0 weight_scale = 1.0 dist_weight_max = 10.0 weight = None plot_cutoff = 0.027
- esd = 10.0
- weight_scale = 1.0
- dist_weight_max = 10.0
- weight = None
- plot_cutoff = 0.027
- ramachandran_plot_restraintsenabled = False favored = *oldfield emsley emsley8k phi_psi_2 allowed = *oldfield emsley emsley8k phi_psi_2 outlier = *oldfield emsley emsley8k phi_psi_2 selection = None Selection of part of the model for which Ramachandran restraints will be set up.inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use botholdfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: $(w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) \frac{1}{esd^2} * \max(2.0, \min(current_distance_to_allowed, dist_weight_max)) * weight_scale \max(2.0, current_distance_to_allowed) \frac{1}{esd^2} * weight_scale * \max(distance_to_allowed_cutoff, current_distance_to_allowed) weight_scale(=0.01) * \max(distance_weight_min(=2.), \min(distance_weight_max(=10.), current_distance_to_allowed))weight_scale = 0.01$ distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027 emsleyweight = 1.0 scale_allowed = 1.0 emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0 phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- enabled = False

- favored = *oldfield emsley emsley8k phi_psi_2
- allowed = *oldfield emsley emsley8k phi_psi_2
- outlier = *oldfield emsley emsley8k phi_psi_2
- selection = None Selection of part of the model for which Ramachandran restraints will be set up.
- inject_emsley8k_into_oldfield_favored = True Backdoor to disable temporary dirty hack to use both
- oldfieldweight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) $1 / \text{esd}^2 * \max(2.0, \min(\text{current_distance_to_allowed}, \text{dist_weight_max})) * \text{weight_scale}$ $\max(2.0, \text{current_distance_to_allowed}) 1 / \text{esd}^2 * \text{weight_scale} * \max(\text{distance_to_allowed_cutoff}, \text{current_distance_to_allowed}) \text{weight_scale}(=0.01) *$ $\max(\text{distance_weight_min}(=2.), \min(\text{distance_weight_max}(=10.), \text{current_distance_to_allowed})) \text{weight_scale} = 0.01$ distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.plot_cutoff = 0.027
- weight = 0. Direct weight value. If 0 the weight will be calculated as following: (w, op.esd, op.dist_weight_max, 2.0, op.weight_scale) $1 / \text{esd}^2 * \max(2.0, \min(\text{current_distance_to_allowed}, \text{dist_weight_max})) * \text{weight_scale}$ $\max(2.0, \text{current_distance_to_allowed}) 1 / \text{esd}^2 * \text{weight_scale} * \max(\text{distance_to_allowed_cutoff}, \text{current_distance_to_allowed}) \text{weight_scale}(=0.01) *$ $\max(\text{distance_weight_min}(=2.), \min(\text{distance_weight_max}(=10.), \text{current_distance_to_allowed}))$
- weight_scale = 0.01
- distance_weight_min = 2.0 minimum coefficient when scaling depending on how far the residue is from allowed region.
- distance_weight_max = 10.0 maximum coefficient when scaling depending on how far the residue is from allowed region.
- plot_cutoff = 0.027
- emsleyweight = 1.0 scale_allowed = 1.0
- weight = 1.0
- scale_allowed = 1.0
- emsley8kweight_favored = 5.0 weight_allowed = 10.0 weight_outlier = 10.0
- weight_favored = 5.0
- weight_allowed = 10.0
- weight_outlier = 10.0
- phi_psi_2favored_strategy = *closest highest_probability random weighted_random allowed_strategy = *closest highest_probability random weighted_random outlier_strategy = *closest highest_probability random weighted_random
- favored_strategy = *closest highest_probability random weighted_random
- allowed_strategy = *closest highest_probability random weighted_random
- outlier_strategy = *closest highest_probability random weighted_random
- rna_sugar_pucker_analysisbond_min_distance = 1.2 bond_max_distance = 1.8 epsilon_range_min = 155.0 epsilon_range_max = 310.0 delta_range_2p_min = 129.0 delta_range_2p_max = 162.0 delta_range_3p_min = 65.0 delta_range_3p_max = 104.0 p_distance_c1p_outbound_line_2p_max = 2.9 o3p_distance_c1p_outbound_line_2p_max = 2.4 bond_detection_distance_tolerance = 0.5 enable = True

- bond_min_distance = 1.2
- bond_max_distance = 1.8
- epsilon_range_min = 155.0
- epsilon_range_max = 310.0
- delta_range_2p_min = 129.0
- delta_range_2p_max = 162.0
- delta_range_3p_min = 65.0
- delta_range_3p_max = 104.0
- p_distance_c1p_outbound_line_2p_max = 2.9
- o3p_distance_c1p_outbound_line_2p_max = 2.4
- bond_detection_distance_tolerance = 0.5
- enable = True
- show_histogram_slotsbond_lengths = 5 nonbonded_interaction_distances = 5
bond_angle_deviations_from_ideal = 5 dihedral_angle_deviations_from_ideal = 5
chiral_volume_deviations_from_ideal = 5
- bond_lengths = 5
- nonbonded_interaction_distances = 5
- bond_angle_deviations_from_ideal = 5
- dihedral_angle_deviations_from_ideal = 5
- chiral_volume_deviations_from_ideal = 5
- show_max_itemsnot_linked = 5 bond_restraints_sorted_by_residual = 5
nonbonded_interactions_sorted_by_model_distance = 5 bond_angle_restraints_sorted_by_residual = 5
dihedral_angle_restraints_sorted_by_residual = 3 chirality_restraints_sorted_by_residual = 3
planarity_restraints_sorted_by_residual = 3 residues_with_excluded_nonbonded_symmetry_interactions
= 12 fatal_problem_max_lines = 10
- not_linked = 5
- bond_restraints_sorted_by_residual = 5
- nonbonded_interactions_sorted_by_model_distance = 5
- bond_angle_restraints_sorted_by_residual = 5
- dihedral_angle_restraints_sorted_by_residual = 3
- chirality_restraints_sorted_by_residual = 3
- planarity_restraints_sorted_by_residual = 3
- residues_with_excluded_nonbonded_symmetry_interactions = 12
- fatal_problem_max_lines = 10
- ncs_groupThe definition of one NCS group. Note, that almost always in refinement programs they will
be checked and filtered if needed.reference = None Residue selection string for the complete master
NCS copyselection = None Residue selection string for each NCS copy location in ASU
- reference = None Residue selection string for the complete master NCS copy
- selection = None Residue selection string for each NCS copy location in ASU

- `ncs_searchSet` of parameters for NCS search procedure. Some of them also used for filtering user-supplied `ncs_group.enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program). `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS. `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues. `chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies. `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues. `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically. `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms. `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `enabled = False` Enable NCS restraints or constraints in refinement (in some cases may be switched on inside refinement program).
- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.
- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 2.` limit of rms difference between chains to be considered as copies
- `residue_match_radius = 4.0` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `clash_guard.nonbonded_distance_threshold = 0.5` `max_number_of_distances_below_threshold = 100` `max_fraction_of_distances_below_threshold = 0.1`
- `nonbonded_distance_threshold = 0.5`
- `max_number_of_distances_below_threshold = 100`
- `max_fraction_of_distances_below_threshold = 0.1`
- `geometry_restraintseditsexcessive_bond_distance_limit = 10` `bondaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `symmetry_operation = None` The bond is between `atom_1` and `symmetry_operation * atom_2`, with `atom_1` and `atom_2` given in fractional coordinates. Example: `symmetry_operation = -x-1,-y,z` `distance_ideal = None` `sigma = None` `slack = None` `limit = -1.0` `top_out = False` `angleaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `atom_selection_3 = None` `angle_ideal = None` `sigma = None` `dihedralaction = *add delete change` `atom_selection_1 = None` `atom_selection_2 = None` `atom_selection_3 = None` `atom_selection_4 = None` `angle_ideal = None` `alt_angle_ideals = None` `sigma = None` `periodicity = 1` `planarityaction = *add delete change` `atom_selection = None` `sigma = None` `parallelityaction = *add delete change` `atom_selection_1 =`

None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0 removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None

- editsexcessive_bond_distance_limit = 10 bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1 planarityaction = *add delete change atom_selection = None sigma = None parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0

- excessive_bond_distance_limit = 10

- bondaction = *add delete change atom_selection_1 = None atom_selection_2 = None symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z distance_ideal = None sigma = None slack = None limit = -1.0 top_out = False

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- symmetry_operation = None The bond is between atom_1 and symmetry_operation * atom_2, with atom_1 and atom_2 given in fractional coordinates. Example: symmetry_operation = -x-1,-y,z

- distance_ideal = None

- sigma = None

- slack = None

- limit = -1.0

- top_out = False

- angleaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None angle_ideal = None sigma = None

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None

- angle_ideal = None

- sigma = None

- dihedralaction = *add delete change atom_selection_1 = None atom_selection_2 = None atom_selection_3 = None atom_selection_4 = None angle_ideal = None alt_angle_ideals = None sigma = None periodicity = 1

- action = *add delete change

- atom_selection_1 = None

- atom_selection_2 = None

- atom_selection_3 = None
- atom_selection_4 = None
- angle_ideal = None
- alt_angle_ideals = None
- sigma = None
- periodicity = 1
- planarityaction = *add delete change atom_selection = None sigma = None
- action = *add delete change
- atom_selection = None
- sigma = None
- parallelityaction = *add delete change atom_selection_1 = None atom_selection_2 = None sigma = 0.027 target_angle_deg = 0
- action = *add delete change
- atom_selection_1 = None
- atom_selection_2 = None
- sigma = 0.027
- target_angle_deg = 0
- removeangles = None dihedrals = None chiralities = None planarities = None parallelities = None
- angles = None
- dihedrals = None
- chiralities = None
- planarities = None
- parallelities = None
- reference_modelenabled = False Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.file = None use_starting_model_as_reference = False sigma = 1.0 limit = 15.0 hydrogens = False Include dihedrals with hydrogen atomsmain_chain = True Include dihedrals formed by main chain atomsside_chain = True Include dihedrals formed by side chain atomsfix_outliers = True Try to fix rotamer outliers in refined modelstrict_rotamer_matching = False Make sure that rotamers in refinement model matches those in reference model even when they are not outliersauto_shutoff_for_ncs = False Do not apply to parts of structure covered by NCS restraintssecondary_structure_only = False Only apply reference model restraints to secondary structure elements (helices and sheets)reference_groupreference = None selection = None file_name = None this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the usersearch_optionsSet of parameters for NCS search procedure. Some of them also used for filtering user-supplied ncs_group.exclude_selection = "element H or element D or water" Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS.chain_similarity_threshold = 0.85 Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residueschain_max_rmsd = 100. limit of rms difference between chains to be considered as copiesresidue_match_radius = 1000 Maximum allowed distance difference between pairs of matching

atoms of two residues `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically. `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms `validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `enabled = False` Restrains the dihedral angles to a high-resolution reference structure to reduce overfitting at low resolution. You will need to specify a reference PDB file (in the input list in the main window) to use this option.

- `file = None`

- `use_starting_model_as_reference = False`

- `sigma = 1.0`

- `limit = 15.0`

- `hydrogens = False` Include dihedrals with hydrogen atoms

- `main_chain = True` Include dihedrals formed by main chain atoms

- `side_chain = True` Include dihedrals formed by side chain atoms

- `fix_outliers = True` Try to fix rotamer outliers in refined model

- `strict_rotamer_matching = False` Make sure that rotamers in refinement model matches those in reference model even when they are not outliers

- `auto_shutoff_for_ncs = False` Do not apply to parts of structure covered by NCS restraints

- `secondary_structure_only = False` Only apply reference model restraints to secondary structure elements (helices and sheets)

- `reference_group` preference = None selection = None `file_name = None` this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- `reference = None`

- `selection = None`

- `file_name = None` this is to used internally to disambiguate cases where multiple reference models contain the same chain ID. This normally does not need to be set by the user

- `search_options` Set of parameters for NCS search procedure. Some of them also used for filtering

- `user-supplied ncs_group.exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms defined by this selection will be in NCS. `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues `chain_max_rmsd = 100`. limit of rms difference between chains to be considered as copies `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically. `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms `validate_user_supplied_groups = True` Enable validation of user-supplied ncs_group. Need to exercise a lot of caution turning this off. This option is for developers only.

- `exclude_selection = "element H or element D or water"` Atoms selected by this selection will be excluded from the model before any NCS search and/or filtering procedures. There is no way atoms

defined by this selection will be in NCS.

- `chain_similarity_threshold = 0.85` Threshold for sequence similarity between matching chains. A smaller value may cause more chains to be grouped together and can lower the number of common residues
- `chain_max_rmsd = 100`. limit of rms difference between chains to be considered as copies
- `residue_match_radius = 1000` Maximum allowed distance difference between pairs of matching atoms of two residues
- `try_shortcuts = False` Try very quick check to speed up the search when chains are identical. If failed, regular search will be performed automatically.
- `minimum_number_of_atoms_in_copy = 3` Do not create ncs groups where master and copies would contain less than specified amount of atoms
- `validate_user_supplied_groups = True` Enable validation of user-supplied `ncs_group`. Need to exercise a lot of caution turning this off. This option is for developers only.
- `amberParameters` for using Amber in refinement.
 - `use_amber = False` Use Amber for all the gradients in refinement
 - `topology_file_name = None` A topology file needed by Amber. Can be generated using `phenix.AmberPrep`.
 - `coordinate_file_name = None` A coordinate file needed by Amber. Can be generated using `phenix.AmberPrep`.
 - `order_file_name = None` A file that maps amber atom numbers to phenix atom numbers.
 - `wxc_factor = 1`.
 - `restraint_wt = 0`.
 - `restraintmask = "`
 - `reference_file_name = "`
 - `bellymask = "` If given, turn on belly in sander
 - `qmmask = "` If given, turn on QM/MM with the given mask
 - `qmcharge = 0` Charge of the QM/MM region
 - `netcdf_trajectory_file_name = "` If given, turn on writing netcdf trajectory
 - `print_amber_energies = False` Print details of Amber energies during refinement
- `qiQMworking_directory = None` `qm_restraintsselection = None` selection for core of atoms to calculate new restraints via a QM geometry minimisation
 - `run_in_macro_cycles = *first_only first_and_last all last_only` test the steps of the refinement that the restraints generation is run
 - `buffer = 3.5` distance to include entire residues into the environment of the core
 - `calculate = *in_situ_opt` starting_energy final_energy starting_strain final_strain starting_bound final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose QM calculations to run
 - `write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer` which ligand or cluster files to

writeprotein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta
 torsions the parts of protein residues that are frozen when an amino acid is the main
 selectionremove_water = False restraints_filename = Auto restraints filename is based on model name if
 not specifiedcleanup = all *most None ignore_x_h_distance_protein = False skip check on transfer of
 proton during QM optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for
 entities such as MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False
 include the side chains of protein in the QM optimisationinclude_inter_residue_restraints = False
 do_not_update_restraints = False For testing and maybe getting strain energy of standard
 restraintsbuffer_selection = None use this instead of distance from
 selectionsselectionspecific_atom_chargesatom_selection = None charge = None
 specific_atom_multiplicitiesatom_selection = None multiplicity = None
 freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity
 = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1
 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- working_directory = None

- qm_restraintsselection = None selection for core of atoms to calculate new restraints via a QM
 geometry minimisationrun_in_macro_cycles = *first_only first_and_last all last_only test the steps of the
 refinement that the restraints generation is runbuffer = 3.5 distance to include entire residues into the
 environment of the corecalculate = *in_situ_opt starting_energy final_energy starting_strain final_strain
 starting_bound final_bound starting_binding final_binding starting_higher_single_point
 final_higher_single_point Choose QM calculations to runwrite_files = *restraints pdb_core pdb_buffer
 pdb_final_core pdb_final_buffer which ligand or cluster files to writeprotein_optimisation_freeze = *all
 None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that
 are frozen when an amino acid is the main selectionremove_water = False restraints_filename = Auto
 restraints filename is based on model name if not specifiedcleanup = all *most None
 ignore_x_h_distance_protein = False skip check on transfer of proton during QM
 optimisationignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as
 MgF3capping_groups = True include_nearest_neighbours_in_optimisation = False include the side
 chains of protein in the QM optimisationinclude_inter_residue_restraints = False
 do_not_update_restraints = False For testing and maybe getting strain energy of standard
 restraintsbuffer_selection = None use this instead of distance from
 selectionsselectionspecific_atom_chargesatom_selection = None charge = None
 specific_atom_multiplicitiesatom_selection = None multiplicity = None
 freeze_specific_atomsatom_selection = None packageprogram = *mopac test charge = Auto multiplicity
 = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1
 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None

- selection = None selection for core of atoms to calculate new restraints via a QM geometry minimisation

- run_in_macro_cycles = *first_only first_and_last all last_only test the steps of the refinement that the
 restraints generation is run

- buffer = 3.5 distance to include entire residues into the environment of the core

- calculate = *in_situ_opt starting_energy final_energy starting_strain final_strain starting_bound
 final_bound starting_binding final_binding starting_higher_single_point final_higher_single_point Choose
 QM calculations to run

- write_files = *restraints pdb_core pdb_buffer pdb_final_core pdb_final_buffer which ligand or cluster
 files to write

- protein_optimisation_freeze = *all None main_chain main_chain_to_beta main_chain_to_delta torsions the parts of protein residues that are frozen when an amino acid is the main selection
- remove_water = False
- restraints_filename = Auto restraints filename is based on model name if not specified
- cleanup = all *most None
- ignore_x_h_distance_protein = False skip check on transfer of proton during QM optimisation
- ignore_lack_of_h_on_ligand = False skip check on protonation of ligand for entities such as MgF3
- capping_groups = True
- include_nearest_neighbours_in_optimisation = False include the side chains of protein in the QM optimisation
- include_inter_residue_restraints = False
- do_not_update_restraints = False For testing and maybe getting strain energy of standard restraints
- buffer_selection = None use this instead of distance from selection
- specific_atom_chargesatom_selection = None charge = None
- atom_selection = None
- charge = None
- specific_atom_multiplicitiesatom_selection = None multiplicity = None
- atom_selection = None
- multiplicity = None
- freeze_specific_atomsatom_selection = None
- atom_selection = None
- packageprogram = *mopac test charge = Auto multiplicity = Auto method = Auto basis_set = Auto solvent_model = None nproc = 1 read_output_to_skip_opt_if_available = False ignore_input_differences = False view_output = None
- program = *mopac test
- charge = Auto
- multiplicity = Auto
- method = Auto
- basis_set = Auto
- solvent_model = None
- nproc = 1
- read_output_to_skip_opt_if_available = False
- ignore_input_differences = False
- view_output = None
- outputsuffix = '.deposit' Suffix string added to automatically generated output filenames
- suffix = '.deposit' Suffix string added to automatically generated output filenames
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

QI Energies

qi_energies.html

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Authors

Nigel W. Moriarty

Introduction

Under Construction

Literature

In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

- In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. Acta Cryst. D79, 100-110 (2023).

QM Flipping (QMF)

qi_histidine.html

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Authors

Nigel W. Moriarty

Introduction

Histidine can be protonated on either or both of the two nitrogen atoms of the imidazole moiety. Each of the three possible forms occur as a result of the stereochemical environment of the histidine sidechain. In an atomic model, comparing the possible protonation states in situ, looking at possible H-bonding and metal coordination, it is possible to predict which is most likely correct. We describe a more direct method that uses quantum mechanical methods to calculate – also in situ – the minimum geometry and energy for comparison, and therefore more accurate identification of the most likely protonation state.

QMF is part of the Quantum Interface (QI) module that covers the interaction of Phenix with QM packages.

Running an example

QMF can only be run stand-alone in the `mmtbx.quantum_interface` procedure as it's not reasonable to do this during refinement. Using the same example as QMR of PDB:4w53, download from the PDB website using:

```
~> phenix.fetch_pdb 4w53 --mtz
```

Command to produce PHIL scope for QM minimisation using `mmtbx.quantum_interface` is:

```
~> mmtbx.quantum_interface 4w53.pdb iterate_NQH=HIS
```

which will fail because of the bulk test for hydrogen atoms.

```
Sorry: Model must have Hydrogen atoms
```

Also note that the `write_qmr_phil=True` option writes a PHIL file. In this example, the format of the PHIL file works for QI interface but the default works for `phenix.refine.Add Hydrogen atoms` to the model using:

```
~> phenix.ready_set 4w53.pdb
```

which produces a file with hydrogens atoms – `4w53.updated.pdb`. It also may produce a restraints file that needs to be used next but in this case because the restraints for BNZ (benzene) are shipped with Phenix it is not necessary to supply to the program unless you want a different list of restraints (not ideal values). This file can be created cheaply but is needed as the skeleton for the QMR values to be inserted in the bond, angle and torsion restraints. So, we can repeat the quantum interface command with the new files.:

```
~> mmtbx.quantum_interface 4w53.updated.pdb iterate_NQH=HIS
```

This will prompt for the selection of the ligand which in this case is “1”.

```
1 : "chain A and resid 31 and resname HIS"
2 : "chain A and resid 200 and resname MBN"
3 : "chain A and resid 201 and resname EPE"
```

Enter selection by choosing number or typing a new one ~>

The result is a PHIL scope for MBN QMR restraints:

```
4w53.updated_A_31_HIS.phil.
```

There is also a command given to run the six configurations.

```
~&gt; mmtbx.quantum_interface 4w53.updated.pdb iterate_NQH=HIS 4w53.updated_A_31_HIS.phil run_qmr=True qi.nproc
```

Note that you can edit the number of nproc to run all six configurations at once.

Once the jobs are done the important metrics are listed in the table:

```
Energies in units of kcal/mol
Proton energy used : 94.5100 kcal/mol

0. HD1 : rotamer "t70"
1. HD1, HE2 : -1272.239 kcal/mol ~&gt; 3.19 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
2. HD1 only : -1275.425 kcal/mol ~&gt; 0.00 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
3. HE2 only : -1249.172 kcal/mol ~&gt; 26.25 kcal/mol. H-Bonds : 13 rmsd : 0.10 rotamer "t70"
4. HD1, HE2 flipped : -1258.856 kcal/mol ~&gt; 16.57 kcal/mol. H-Bonds : 12 rmsd : 0.18 rotamer "t-90"
5. HD1 only flipped : -1265.615 kcal/mol ~&gt; 9.81 kcal/mol. H-Bonds : 13 rmsd : 0.22 rotamer "t-90"
6. HE2 only flipped : -1254.390 kcal/mol ~&gt; 21.03 kcal/mol. H-Bonds : 12 rmsd : 0.20 rotamer "t-90"
```

A Coot command is supplied:

```
phenix.start_coot qm_work_dir/iterate_HIS_01_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pdb qm_w
```

And a list of PyMOL commands:

```
phenix.pymol qm_work_dir/iterate_HIS_01_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_02_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_03_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_04_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_05_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
phenix.pymol qm_work_dir/iterate_HIS_06_cluster_final_A_31_HIS_3.5_C_D_T_PM6-D3H4_EPS_equals_78.4.pml &
```

Finally the information about the lowest (and near) energy configuration:

```
!!! 2. HD1 only : -1275.425 kcal/mol ~&gt; 0.00 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70" SAME

Close
~&gt;&lt; 1. HD1, HE2 : -1272.239 kcal/mol ~&gt; 3.19 kcal/mol. H-Bonds : 14 rmsd : 0.03 rotamer "t70"
```

Literature

Protonation of histidine rings using quantum mechanical methods. N.W. Moriarty, J. Moussa, and P.D. Adams. *Acta Cryst.* (2024).

- Protonation of histidine rings using quantum mechanical methods. N.W. Moriarty, J. Moussa, and P.D. Adams. *Acta Cryst.* (2024).

In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. *Acta Cryst.* D79, 100-110 (2023).

- In situ ligand restraints from quantum-mechanical methods. D. Liebschner, N.W. Moriarty, B.K. Poon, and P.D. Adams. *Acta Cryst.* D79, 100-110 (2023).

Model rebuilding keeping connectivity and allowing insertions and deletions

rebuild_model.html

Author(s)

- rebuild_model: Tom Terwilliger

Purpose

Rebuild a model, keeping all connectivity the same but allowing insertions and deletions.

How rebuild_model works:

Rebuild model rebuilds each chain in your model (optionally in segments of about 15-20 residues) and reassembles these into a new model with the same connectivity as the original.

Each segment is rebuilt with several different base methods (default base methods are refinement, fixing insertions and deletions, and rebuilding from CA positions).

For each base method, a sequence of optimization steps is carried out (typically refinement, the base method, replacement of side chains, regularization, and another replacement of side chains).

Rebuilt segments are scored based on map correlation. The highest-scoring segments are used to replace the corresponding segment in the original model.

If there are multiple chains in your model, each is rebuilt separately.

The quick option can be used to speed up the process. It sets the number of refinement cycles to 1 (otherwise there are 3 carried out) and skips all the optimization steps except the base method.

Examples

Standard run of rebuild_model:

Running rebuild_model is easy. From the command-line you can type:

```
phenix.rebuild_model my_model.pdb my_map.mrc resolution=3 nproc=8
```

This will rebuild my_model.pdb based on the map my_map.mrc at a resolution of 3 Å using 8 processors.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command line
input_filesseq_file = None Sequence file. Used to guess sequence if insertions and deletions are allowed. (Fasta format or sequences separated by blank lines)
selection = None If specified, use selected part of model (discard remainder)
map_modelfull_map = None Input full map file
half_map = None Input half map files
model = None Input model file
outputrebuilt_model = None Output file name
overwrite = True Overwrite files with same names
rebuild_modelfix_insertions_deletions = True Fix insertions/deletions
refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quickmacro_cycles = 1 Overall macro-cycles carrying out all steps
base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable
methodstake_anything_over_original = None True to not use original model
segment_length = None If set, break chains up into pieces about this size
crystal_inforesolution = None Nominal resolution of map
minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
sequence = None Sequences
controlnproc = 1 Number of processors (if None, use all available)
ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
verbose = False Verbose output
quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
guiGUI-specific parameter required for output directory
output_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesseq_file = None Sequence file. Used to guess sequence if insertions and deletions are allowed. (Fasta format or sequences separated by blank lines)
- selection = None If specified, use selected part of model (discard remainder)
- map_modelfull_map = None Input full map file
half_map = None Input half map files
model = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputrebuilt_model = None Output file name
overwrite = True Overwrite files with same names
- rebuilt_model = None Output file name
- overwrite = True Overwrite files with same names
- rebuild_modelfix_insertions_deletions = True Fix insertions/deletions
refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quickmacro_cycles = 1 Overall macro-cycles carrying out all steps
base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable

methodstake_anything_over_original = None True to not use original model
segment_length = None If set, break chains up into pieces about this size

- fix_insertions_deletions = True Fix insertions/deletions
- refine_cycles = None Refinement cycles (set to zero to not refine). Default is 3 for standard run and 1 for quick
- macro_cycles = 1 Overall macro-cycles carrying out all steps
- base_methods = *None fix_insertions_deletions ca_to_model refine Base methods to run. None will run all applicable methods
- optimization_sequence = *None refine base replace_side regularize_model replace_side Optimization sequence to run. None will run all applicable methods
- take_anything_over_original = None True to not use original model
- segment_length = None If set, break chains up into pieces about this size
- crystal_inforesolution = None Nominal resolution of map
minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
sequence = None Sequences
- resolution = None Nominal resolution of map
- minimum_resolution = 2.5 If resolution is finer than minimum_resolution, use minimum_resolution. This speeds up model-building.
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- sequence = None Sequences
- controlnproc = 1 Number of processors (if None, use all available)
ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
verbose = False Verbose output
quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
- nproc = 1 Number of processors (if None, use all available)
- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- verbose = False Verbose output
- quick = False Run quickly. Sets refinement_cycles to 1 and optimization_sequence to base_method
- guiGUI-specific parameter required for output directory
output_dir = None
- output_dir = None

<string>

reciprocal_space_arrays.html

phenix.reciprocal_space_arrays: compute various arrays such as Fcalc, Fmask, Fmodel, Fbulk, and more.

Inputs:

File with reflection data (Fobs or lobs), R-free flags, and optionally HL coefficients. It can be in most of known formats and spread across multiple files; label(s) selecting which reflection data arrays should be used (in case there are multiple choices in input file, there is no need to provide labels otherwise); Model file (PDB or mmCIF) with input model.

- File with reflection data (Fobs or lobs), R-free flags, and optionally HL coefficients. It can be in most of known formats and spread across multiple files;
- label(s) selecting which reflection data arrays should be used (in case there are multiple choices in input file, there is no need to provide labels otherwise);
- Model file (PDB or mmCIF) with input model.

Usage examples:

```
phenix.reciprocal_space_arrays model.pdb data.hkl f_obs_label="IOBS" phenix.reciprocal_space_arrays  
model.pdb data.hkl r_free_flags_label="FREE"
```

- phenix.reciprocal_space_arrays model.pdb data.hkl f_obs_label="IOBS"
- phenix.reciprocal_space_arrays model.pdb data.hkl r_free_flags_label="FREE"

Output:

MTZ file with data arrays. Depending on the input data, the following arrays may be present: FOBS : data from input reflection file SIGFOBS : corresponding sigmas R_FREE_FLAGS : R-free flags (0 - work, 1 - test) FMODEL : total model structure factor. See phenix.fmodel for details PHIFMODEL : corresponding phases FCALC : Fcalc from atomic model PHIFCALC : corresponding phases FMASK : Fmask from bulk-solvent mask. See phenix.fmodel for details PHIFMASK : corresponding phases FBULK : bulk-solvent contribution. See phenix.fmodel for details PHIFBULK : corresponding phases FB_CART : overall anisotropic scale factor FOM : figures of merit ALPHA : ML parameter used in m&D calculation for 2mFo-DFc maps BETA : ML parameter used in m&D calculation for 2mFo-DFc maps HLA : HL coefficients from from input reflection file HLB : HL coefficients from from input reflection file HLC : HL coefficients from from input reflection file HLD : HL coefficients from from input reflection file HLmodelA : HL coefficients from the model (C=D=0) HLmodelB : HL coefficients from the model (C=D=0) HLmodelC : HL coefficients from the model (C=D=0) HLmodelD : HL coefficients from the model (C=D=0) HLcombA : combined HL: model + input HLcombB : combined HL: model + input HLcombC : combined HL: model + input HLcombD : combined HL: model + input RESOLUTION : resolution per reflection

MTZ file with data arrays. Depending on the input data, the following arrays may be present:

- FOBS : data from input reflection file
- SIGFOBS : corresponding sigmas
- R_FREE_FLAGS : R-free flags (0 - work, 1 - test)
- FMODEL : total model structure factor. See phenix.fmodel for details
- PHIFMODEL : corresponding phases

- FCALC : Fcalc from atomic model
- PHIFCALC : corresponding phases
- FMASK : Fmask from bulk-solvent mask. See phenix.fmodel for details
- PHIFMASK : corresponding phases
- FBULK : bulk-solvent contribution. See phenix.fmodel for details
- PHIFBULK : corresponding phases
- FB_CART : overall anisotropic scale factor
- FOM : figures of merit
- ALPHA : ML parameter used in m&D calculation for 2mFo-DFc maps
- BETA : ML parameter used in m&D calculation for 2mFo-DFc maps
- HLA : HL coefficients from from input reflection file
- HLB : HL coefficients from from input reflection file
- HLC : HL coefficients from from input reflection file
- HLD : HL coefficients from from input reflection file
- HLmodelA : HL coefficients from the model (C=D=0)
- HLmodelB : HL coefficients from the model (C=D=0)
- HLmodelC : HL coefficients from the model (C=D=0)
- HLmodelD : HL coefficients from the model (C=D=0)
- HLcombA : combined HL: model + input
- HLcombB : combined HL: model + input
- HLcombC : combined HL: model + input
- HLcombD : combined HL: model + input
- RESOLUTION : resolution per reflection

Restraints Editor Especially Ligands (REEL)

reel.html

Contents

- Author
- Purpose
- Screen Shots
- General Procedure
- Input
- Editing
- Menus
- Examples
- References

Author

Nigel W. Moriarty

Purpose

Edit the geometry restraints of a ligand using a Graphical User Interface (GUI) including a 3D view of the ligand and a tabular view of the restraints.

Screen Shots

General Procedure

The general procedure is to load a restraints file (CIF) and manipulate the restraints via the table or molecule view. The geometry of the revised restraints can be tested using the Action->Geometry optimisation option. The final restraints can be saved to a CIF file for use with phenix.refine. The corresponding PDB file can also be saved.

Input

Restraints can be loaded into REEL using the command line, the pull-down menu to open a file and a pull-down menu to run eLBOW. Restraints files from eLBOW contain both the restraints and cartesian coordinates. For the purposes of REEL, the coordinates are generated by the restraints and can not be edited directly. The background colour is light steel blue to show that it is not used in the geometry editing actions.

REEL can load a molecule geometry from any format the eLBOW can read, including PDB, Mol2D and SDF. If the file does not contain the bonding information, the bonding is automatically determined using proximity. The limit on the size of molecule that the bonding is automatically determined is set at 200 atoms.

Molecule up to 2000 atoms can be loaded using the --view but only the bond connectivity is determined. The bond order is set to one but can be changed interactively.

Molecules can be loaded into REEL using the `--reel` option for eLBOW or using the eLBOW GUI dialog available in REEL.

The Chemical Components database can also be accessed using the `--chem` option so load the cartesian coordinates and atom names of a certain ligand.

The `--smiles` option will automatically run eLBOW on the SMILES string and load the CIF file in REEL.

It may be that a CIF without cartesian coordinates is the file to be loaded. REEL will load it and attempt but its better to provide the cartesian coordinates. A simple way is to use:

```
phenix.reel ligand.cif ligand.pdb
```

and to use the `--overlay` option to force it in the case of slightly different ligand details (ligand code).

Editing

The geometry restraints (bonds, angles, dihedrals, planes and chirals) are the driving coordinates in this editor. The cartesian coordinates are displayed in the atoms table only because there are generated for the viewer display. They are the driven coordinates and are therefore ignored if changed in the editor.

Many cells in the table view of the restraints can be changed but care must be taken to ensure that the changes are local or global. For example, changing an atom name in the bonds table view will only change it in that row. If you wish to change the name of the atom in all the restraints for should use the right mouse menu in the viewer window or change it in the atoms table view. The colour of the cell gives the impression as to whether the changes are propagated elsewhere. The cells, in some cases, have not been made read only to allow the user to make changes as desired.

Clicking an atom in the molecule view will highlight the various related topological elements in the table view and vice versa. Use the checkbox in the table view or a right-mouse pop-menu to remove a restraint from optimisation and output file. Chiral centres can be changed in the table view.

Right clicking in the viewer or in the table view provide pop-menu for various actions. Examples are below.

Double clicking the label headers will sort the rows.

Menus

File

REEL can open a large number of file formats but the best file is the CIF output from eLBOW. Using the menu or the shortcut will provide a dialog to choose an input file.

The template input allows for the transfer of atom names from a PDB to the molecule in the viewer

Two different cartesian options for overlaying onto the molecule in the viewer are in the "Cartesian coordinate inputs" sub-menu. The main difference is the adjustment of the restraints' ideal value from the input geometry.

The saving options include a data sheet which is a postscript file including a two-dimensional representation of the molecule.

The view defaults including the window size and position, the molecular representation (atom labels and bond view) and background can be managed using the "View defaults" menu items.

View

View options in include toggling the hydrogens, the background, the labels on the atoms ...

and showing the bond orders.

The listing of the geometry allows the ideal restraint values and the actual geometric values to be displayed. Any actual values more than three sigma the estimated standard deviations are highlighted.

The 2D options shows a 2-dimensional representation of the molecule.

Action

Geometry optimisation using a simple force field can be performed using the first option. This is also available in the toolbar as is the semi-empirical quantum method, AM1, optimisation using the eLBOW dialog.

The last three options will open the eLBOW dialog with either no defaults, the file initially loaded for the molecule or the current CIF restraints, respectively. Once the eLBOW job is finished the user is prompted to choose what to do with the results.

"Geometry transfer" will update the cartesian coordinates of the molecule while the "Geometry and restraints transfer" will update the ideal values of the restraints. The former can be combined with the View -> "List geometry values" menu option to see the differences.

Examples

To load a previously created restraints file

```
phenix.reel atp.cif
```

To load a restraints file into REEL from eLBOW

```
phenix.elbow --smiles 0 --reel
```

To load all the unknown ligands from a PDB file

```
phenix.reel model.pdb --do-all
```

or a single residue

```
phenix.reel model.pdb --residue ATP
```

References

An editor for the generation and customization of geometry restraints. N.W. Moriarty, E.J. Draizen, and P.D. Adams. Acta Crystallogr D Biol Crystallogr 73, 123-130 (2017).

- An editor for the generation and customization of geometry restraints. N.W. Moriarty, E.J. Draizen, and P.D. Adams. Acta Crystallogr D Biol Crystallogr 73, 123-130 (2017).

Running Phenix regression tests

regression_tests.html

Author(s)

- : All

Purpose

It is important to run regression tests any time a change is made in the Phenix code to make sure that the change does not break something that is already written.

Also regression tests can be used by users to make sure everything is working correctly.

Framework

Regression tests are located in many places in the Phenix and CCTBX codebase. They are typically either in subdirectories called regression or in the phenix_regression directory, or are specified in a run_tests.py file that can be anywhere in the CCTBX codebase.

Regression tests for GUI programs are created specifically for each program and can be run as a group in the GUI. These tests and their setup is described in Setting up GUI tests).

Running regression tests

You can run all the regression tests (usually on a machine with many processors) with a simple command:

```
phenix_regression.test_all_parallel nproc=96
```

There are some simple keywords that you might want to add to this:

```
sort: sort the tests by timings, run the longest ones first
git: identify all tests that use any files that are changed in the git
    database

tests_to_run="xx,yy": Run only tests that include the words "xx" or "yy"
tests_to_skip="aa bb": Skip any tests that include the words "aa" or "bb"
```

These keywords can be combined:

```
phenix_regression.test_all_parallel nproc=96 sort tests_to_run=refine
```

Remove test set reflections from a map file with `remove_free_from_map`

`remove_free_from_map.html`

Author(s)

- `remove_free_from_map`: Tom Terwilliger

Purpose

`remove_free_from_map` is a tool to remove test set reflections from a map coefficients file so that the map can be used in real-space refinement without including the test set reflections.

NOTE: Normally `remove_free_from_map` is called from `build_one_model` or `phase_and_build`, but you can run it interactively if you want.

Usage

How `remove_free_from_map` works:

`remove_free_from_map` uses `resolve` to set all test set reflections to zero.

Output files from `remove_free_from_map`

`map_coeffs_without_freer_set.mtz`: An MTZ file with test set reflections set to F=0

Examples

Standard run of `remove_free_from_map`:

Running `remove_free_from_map` is easy. From the command-line you can type:

```
phenix.remove_free_from_map map_coeffs.mtz free_r.mtz
```

where `free_r.mtz` has your `FreeR_flag` indicating which reflections are in the test set.

Removing protein chains built into RNA density with `remove_poor_fragments`

`remove_poor_fragments.html`

Author(s)

- `remove_poor_fragments`: Tom Terwilliger

Purpose

This routine takes a model that contains RNA and protein and tries to identify fragments of protein that were accidentally built into RNA density.

Usage

When carrying out model-building for complexes between protein and RNA it sometimes happens that protein can be built accidentally into regions that are really RNA. The `phenix.remove_poor_fragments` tool is used to try and identify such incorrectly-built regions.

How `remove_poor_fragments` works:

The key idea in this approach is that these incorrectly-built regions tend to be isolated (not in the middle of a protein domain) and to have lower map-model correlation than the correct parts of the protein chains. The `phenix.remove_poor_fragments` tool ranks all protein segments on these criteria and removes the worst ones. The threshold that is chosen based on the number of residues of RNA expected in the molecule that were not built, the fraction of protein that was built, and the quality of the model, using the formula:

$$\text{residues_to_remove} = A * \text{protein_built} * \text{rna_not_built} / (\text{cc} * \text{protein_present})$$

where `cc` is the map-model correlation for the protein part of the model, `protein_present` is the number of protein residues in the sequence file, `protein_built` are residues of protein built, and `rna_not_built` are residues of RNA in the sequence file minus the number built.

The logic of this formula is that there is a certain volume, proportional to `rna_not_built`, that is really RNA that might be accidentally built as protein. Furthermore the logic is that the number of residues of protein that might be built in that volume is higher if more residues of protein have been built and higher if the model-map correlation for the protein model is low. These relationships were seen in an analysis of ribosome structures. A scale factor `A` of 0.53 relating the optimal number of residues to remove to the other factors was found empirically.

Output files from `remove_poor_fragments`

`trimmed_pdb.pdb`: A PDB file with your trimmed model.

Examples

Standard run of `remove_poor_fragments`:

Running `remove_poor_fragments` is easy. From the command-line you can type:

```
phenix.remove_poor_fragments model_with_protein_and_rna.pdb \
seq.dat \
map.ccp4
```

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

map_file_name = None Map file name
map_coeffs_file = None File with map coefficients (alternative to map file)
map_coeffs_labels = None Optional label specifying which columns of map coefficients to use
model_file_name = None Model file names. Normally only one specified.
output_pdb_file_name = trimmed_pdb.pdb Model file name
seq_file_name = None Sequence file name. Used to calculate n_protein_in_complex, n_rna_in_complex and rna_ratio
select_good_models = True Select good models from input models and write to output PDB file
cutoff_method = *volume_of_rna_not_built sigma_cutoff Choice of method for deciding how many residues to remove. Volume of RNA not built means calculate the volume in the structure that should be RNA, subtract the volume of RNA actually built, weight by 1/mean map CC and the scale factor rna_volume_ratio. Sigma cutoff means use a cutoff based on resolution and ratio of RNA:protein residues in the model with parameters. Default is Volume of RNA not built.
rna_volume_ratio = 0.53 Scale factor on RNA volume used to decide how many residues to remove.
n_protein_in_complex = None Number of protein residues in complex. Normally obtained from sequence file.
n_rna_in_complex = None Number of RNA residues in complex. Normally obtained from sequence file.
n_protein_in_model = None Number of protein residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
n_rna_in_model = None Number of RNA residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
rna_ratio = None Used to estimate optimal cutoff in sigma_cutoff method. Calculated automatically from sequence file. Note that this whole procedure is optimized for cases where the rna_ratio is greater than 0.2.
toss_sigma_cutoff_a = 1.4 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$. You can just set the value by setting toss_sigma_cutoff_a and setting toss_sigma_cutoff_b and toss_sigma_cutoff_c to zero. To accept a few more chains, increase toss_sigma_cutoff_a by a little (try 0.1 higher).
toss_sigma_cutoff_b = -1.1 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
toss_sigma_cutoff_c = -3.3 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
resolution = None Resolution
metrics = *mean_delta_rama *map_value *cc fraction_with_neighbor n_residues fraction_favored Metrics for a chain (fragment) being likely to be correct. Mean_delta_rama is mean change in Ramachandran angles between neighboring residues. Map_value is maximum map value at coordinates of a CA atom in chain for low_resolution map (typically 30 Å) calculated from the model. Fraction_with_neighbor is fraction of residues in a chain near another chain.

N_residues is length of chain. Fraction_favored is fraction of residues with favored Ramachandran angles
 low_resolution_map_d_min = 30 Resolution for low-resolution map showing location of molecule
 distance_cutoff = 7 In Score for fraction_with_neighbor, a residue in the same chain has to be within distance_cutoff to count as being close to another chain
 any_in_series = 4 Score for fraction_with_neighbor, fraction of residues in a chain that are near another chain, counts a residue as close if any residue in a string of any_in_a_series is near another chain
 n_offset = 10 In Score for fraction_with_ne...

- map_file_name = None Map file name
- map_coeffs_file = None File with map coefficients (alternative to map file)
- map_coeffs_labels = None Optional label specifying which columns of map coefficients to use
- model_file_name = None Model file names. Normally only one specified.
- output_pdb_file_name = trimmed_pdb.pdb Model file name
- seq_file_name = None Sequence file name. Used to calculate n_protein_in_complex, n_rna_in_complex and rna_ratio
- select_good_models = True Select good models from input models and write to output PDB file
- cutoff_method = *volume_of_rna_not_built sigma_cutoff Choice of method for deciding how many residues to remove. Volume of RNA not built means calculate the volume in the structure that should be RNA, subtract the volume of RNA actually built, weight by 1/mean map CC and the scale factor rna_volume_ratio. Sigma cutoff means use a cutoff based on resolution and ratio of RNA:protein residues in the model with parameters. Default is Volume of RNA not built.
- rna_volume_ratio = 0.53 Scale factor on RNA volume used to decide how many residues to remove.
- n_protein_in_complex = None Number of protein residues in complex. Normally obtained from sequence file.
- n_rna_in_complex = None Number of RNA residues in complex. Normally obtained from sequence file.
- n_protein_in_model = None Number of protein residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
- n_rna_in_model = None Number of RNA residues in model. Normally obtained from the input model. Note distinction from n_protein_in_complex which is how many there are based on the sequence file.
- rna_ratio = None Used to estimate optimal cutoff in sigma_cutoff method. Calculated automatically from sequence file. Note that this whole procedure is optimized for cases where the rna_ratio is greater than 0.2.
- toss_sigma_cutoff_a = 1.4 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$. You can just set the value by setting toss_sigma_cutoff_a and setting toss_sigma_cutoff_b and toss_sigma_cutoff_c to zero. To accept a few more chains, increase toss_sigma_cutoff_a by a little (try 0.1 higher).
- toss_sigma_cutoff_b = -1.1 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
- toss_sigma_cutoff_c = -3.3 For sigma_cutoff method, At the conclusion of the procedure, the worst portion of chains, with low z-scores, is removed. The cutoff is $\text{toss_sigma_cutoff} = \text{toss_sigma_cutoff_a} + \text{toss_sigma_cutoff_b} * (\text{resolution} - 3.0) + \text{toss_sigma_cutoff_c} * (\text{rna_ratio} - 0.5)$
- resolution = None Resolution

- `metrics = *mean_delta_rama *map_value *cc fraction_with_neighbor n_residues fraction_favored`
Metrics for a chain (fragment) being likely to be correct. `Mean_delta_rama` is mean change in Ramachandran angles between neighboring residues. `Map_value` is maximum map value at coordinates of a CA atom in chain for low_resolution map (typically 30 Å) calculated from the model. `Fraction_with_neighbor` is fraction of residues in a chain near another chain. `N_residues` is length of chain. `Fraction_favored` is fraction of residues with favored Ramachandran angles
- `low_resolution_map_d_min = 30` Resolution for low-resolution map showing location of molecule
- `distance_cutoff = 7` In Score for `fraction_with_neighbor`, a residue in the same chain has to be within `distance_cutoff` to count as being close to another chain
- `any_in_series = 4` Score for `fraction_with_neighbor`, fraction of residues in a chain that are near another chain, counts a residue as close if any residue in a string of `any_in_a_series` is near another chain
- `n_offset = 10` In Score for `fraction_with_neighbor`, a residue in the same chain has to be separated by `n_offset` to count as being from another chain
- `atom_selection = "name ca and (not element Ca)"` Specification of atoms to use in `fraction_with_neighbor`. Normally CA only
- `top_half_value = 2`. The top set of chains (typically about one half) are used to estimate statistics for good chains. Value is $1/\text{top_half_value}$.
- `cross_validate = True` Cross-validate values of mean/sd for each metric.
- `optimize_weights = False` Optimize weights. Development only. Only applies if `model_file_name` specifies a model with good chains in a structure, `second_model_file_name` specifies one with poor chains in the same structure, and weights are to be optimized.
- `n_cycle = 2` Number of minor cycles of estimation of histograms.
- `n_big_cycle = 1` Number of big cycles of estimation of histograms.
- `weight_mean_delta_rama = 1` Weight on `mean_delta_rama` metric
- `weight_map_value = 1` Weight on `map_value` metric
- `weight_cc = 1` Weight on `cc` metric
- `weight_fraction_favored = 1` Weight on `fraction_favored` metric
- `weight_n_residues = 1` Weight on `n_residues` metric
- `weight_fraction_with_neighbor = 1` Weight on `fraction_with_neighbor` metric
- `minimum_number_of_fragments = 10` Minimum number of fragments evaluated with a metric to use that metric.
- `distance_cutoff_for_split_model = 5`. Distance cutoff for splitting chain into fragments.
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `guiGUI-specific parameter required for output directory`
`output_dir = None`

phenix.renumber_fab

renumber_fab.html

Description

The renumber_fab tool is designed to streamline the renumbering process for FAB chains. It performs a search among a set of supplied sequences to find the most similar sequence. Once identified, this sequence is sent to the server to retrieve the appropriate numbering. The tool supports various renumbering schemes, such as Kabat, Chothia, and Martin.

Requirements: A model and its corresponding sequence.

Ensure that the sequence encompasses the entire FAB structure and adequately covers the entire chains in the model.

Additionally, the tool will update HELIX/SHEET records describing secondary structure. However, it will neither renumber nor output SSBOND, LINK, and certain other records.

You can customize the similarity threshold by adjusting the 'identity' parameter in case the sequences are not closely matched. The tool generates detailed logs, giving insights into similarity and matched chains.

!!! WARNING !!! This program sends supplied sequence to AbNum server at <http://www.bioinf.org.uk/abs/abnum/> Please acknowledge the server authors: Abhinandan KR, Martin AC. Mol Immunol. 2008 Aug; 45(14):3832-9. <https://doi.org/10.1016/j.molimm.2008.05.022>

Command line usage examples

```
% phenix.renumber_fab model.pdb sequence.fa % phenix.renumber_fab 2jjj.cif sequence.fa scheme=chothia identity=0.4
```

```
% phenix.renumber_fab model.pdb sequence.fa
```

```
% phenix.renumber_fab 2jjj.cif sequence.fa scheme=chothia identity=0.4
```

GUI

This tool is available in the Phenix GUI in the "PDB deposition" section.

List of all available keywords

scheme = *kabat chothia martin which scheme to use
verbose = True More log output
identity_cutoff = 0.5 chains with less identity between model and sequence file will be skipped. If you have a chain with a lot of deletions in sequence, lower this number.
job_title = None Job title in PHENIX GUI, not used on command line
outputsuffix = None Suffix string added to automatically generated output filenames. If None, _fab_renumbered and scheme name will be used.
gui GUI-specific parameter required for output directory
output_dir = None

- scheme = *kabat chothia martin which scheme to use
- verbose = True More log output
- identity_cutoff = 0.5 chains with less identity between model and sequence file will be skipped. If you have a chain with a lot of deletions in sequence, lower this number.

- `job_title = None` Job title in PHENIX GUI, not used on command line
- `outputsuffix = None` Suffix string added to automatically generated output filenames. If `None`, `_fab_renumbered` and scheme name will be used.
- `suffix = None` Suffix string added to automatically generated output filenames. If `None`, `_fab_renumbered` and scheme name will be used.
- `guiGUI-specific parameter required for output directory``output_dir = None`
- `output_dir = None`

Replace side chains in a model to match the sequence with `replace_side_chains`

`replace_side_chains.html`

Author(s)

- `replace_side_chains`: Tom Terwilliger

Purpose

`replace_side_chains` is a tool to take a main-chain model and to add side-chains from a rotamer library based on an electron density map.

NOTE: Normally this method is called from `phenix.phase_and_build` but you can use it interactively if you want.

You can also use it to identify side chains (sequence) from a model without side chains (or with) and a map using `sequence_from_density=True`.

Usage

How `replace_side_chains` works:

`replace_side_chains` uses `resolve` to replace the side chains in a model. You can choose to use the default `resolve` rotamer library or the Lovell et al penultimate rotamer library.

Output files from `replace_side_chains`

`replace_side_chains.pdb`: A PDB file with side chains replaced.

Examples

Standard run of `replace_side_chains`:

Running `replace_side_chains` is easy. From the command-line you can type:

```
phenix.replace_side_chains model.pdb map_coeffs.mtz sequence.dat
```

You should supply a sequence file that matches the sequence of the model.

Using `replace_side_chains` to sequence a protein from density:

From the command-line you can type:

```
phenix.replace_side_chains model.pdb map_coeffs.mtz sequence_from_density=True
```

You don't need a sequence file but if you supply one the residues allowed are those in your sequence file.

Possible Problems

Specific limitations and problems:

Literature

Additional information

Rebuild a model using fragments from the PDB

`replace_with_fragments_from_pdb.html`

Author(s)

- `replace_with_fragments_from_pdb`: Tom Terwilliger

Purpose

Create a composite model similar to a supplied model using fragments from the PDB.

This tool can be used in early stages of model-building to try and improve parts of a preliminary model by replacing them with similar fragments from a deposited structure.

Note that the composite model may not actually have good geometry because the fragments from the PDB are morphed (distorted) to match the supplied model and also because the junctions between fragments taken from the PDB are not refined.

Usage

`replace_with_fragments_from_pdb` can create composite models for protein chains

You can adjust the maximum number of insertions/deletions in each replacement fragment and whether to morph structures obtained by SSM matching.

You can supply a map file for refinement or calculation of map-model correlations.

You can also choose to exclude or include specific PDB entries in SSM matching and fragment searching, or exclude all PDB entries with sequences similar to a particular PDB entry.

You can choose to create a replacement for just a selected part of your model (by supplying a selection string that specifies what to select).

How `replace_with_fragments_from_pdb` works:

Search structures in the PDB (using SSM matching and fragment searching) for fragments similar to segments in a supplied model. Then choose the best matching segments for each part of the supplied model and create a composite model using the fragments from the PDB.

`replace_with_fragments_from_pdb` uses SSM-based (secondary structure matching) to find structures in the PDB that are similar to a supplied model. The SSM matching is the same as in `phenix.superpose_models`.

`replace_with_fragments_from_pdb` also uses fragment searching to find shorter fragments matching each loop (segment between two helices or strands) in the supplied model.

The fragments obtained with fragment searching are always morphed (distorted) to make the distance between starting and ending residues match the distance in the supplied structure.

Fragments obtained from SSM matching are optionally morphed to match the target structure as well. The distance over which morphing occurs (the shift-field distance) is adjustable (typically 10 Å), where structure is largely preserved over shorter distances and may be changed over longer distances.

The supplied model is then used as a template and matching parts of structures from the PDB are superimposed on the template. The best-matching segments (smallest rmsd and longest length) are chosen

and connected. Gaps are filled if possible by a targeted fragment search of the PDB.

Examples

Standard run of `replace_with_fragments_from_pdb`:

Running `replace_with_fragments_from_pdb` is easy. From the command-line you can type:

```
phenix.replace_with_fragments_from_pdb model.pdb nproc=8
```

This will find fragments in the PDB similar to `model.pdb`, put them together and write out a new model using these fragments.

Possible Problems

Specific limitations and problems:

The final model may have poor geometry at connections between fragments, so further refinement is normally required. Additionally the final model may have poor geometry within segments because morphing is carried out to match the target structure.

Literature

Additional information

List of all available keywords

`job_title` = None Job title in PHENIX GUI, not used on command line
`input_filesselection` = None If specified, use selected part of model
`local_pdb_dir` = None Local PDB mirror (something like `/User/files/pdb_mirror/`)
`map_model` = None Input full map
`full_map` = None Input half map
`half_map` = None Input model file
`outputrebuilt_model` = None Output file name. Default is to append `_rebuilt` to input file
`overwrite` = True Overwrite files with same names
`replace_with_fragments_from_pdb` = 3 Refinement cycles (set to zero to not refine)
`maximum_insertions_or_deletions` = None Maximum insertions/deletions per fragment
`database` = *pdb100 pdb2018_90 Database to search. `pdb100` is unique set of PDB. `pdb2018_90` is quality-filtered PDB at 90% identity
`number_of_sequence_neighbors` = 100 Number of sequence neighbors to search if using `pdb100`
`max_models_to_fetch` = 100 Maximum models to fetch from PDB
`allow_reverse_connectivity` = False Allow connectivity to be backwards or mixed
`morph` = False Morph replacement segments (NOTE: morphing of loops is intrinsic)
`shift_field_distance` = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 Å.
`pdb_include_list` = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
`pdb_exclude_list` = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
`exclude_similar_to_this_pdb_id` = None Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to `exclude_similar_to_this_pdb_id`
`maximum_percent_identity_to_this_pdb_id` = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > `maximum_percent_identity_to_this_pdb_id` to `exclude_similar_to_this_pdb_id`
`maximum_residues_to_try_fragment_search` = 30 If `try_fragment_search_for_short_models` and model has no more than `maximum_residues_to_try_fragment_search`, try simple fragment search

firsttry_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search
 firstcrystal_inforesolution = None Nominal resolution of mapscattering_table = n_gaussian wk1995 it1992
 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequenceswrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).controlnproc = 1 Number of processors (if None, use all available)ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputquick = False Run quicklyguiGUI-specific parameter required for output directoryoutput_dir = None

- job_title = None Job title in PHENIX GUI, not used on command line
- input_filesselection = None If specified, use selected part of modellocal_pdb_dir = None Local PDB mirror (something like /User/files/pdb_mirror/)map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- selection = None If specified, use selected part of model
- local_pdb_dir = None Local PDB mirror (something like /User/files/pdb_mirror/)
- map_modelfull_map = None Input full map filehalf_map = None Input half map filesmodel = None Input model file
- full_map = None Input full map file
- half_map = None Input half map files
- model = None Input model file
- outputrebuilt_model = None Output file name. Default is to append _rebuilt to input fileoverwrite = True Overwrite files with same names
- rebuilt_model = None Output file name. Default is to append _rebuilt to input file
- overwrite = True Overwrite files with same names
- replace_with_fragments_from_pdbrefine_cycles = 3 Refinement cycles (set to zero to not refine)maximum_insertions_or_deletions = None Maximum insertions/deletions per fragmentdatabase = *pdb100 pdb2018_90 Database to search. pdb100 is unique set of PDB. pdb2018_90 is quality-filtered PDB at 90% identitynumber_of_sequence_neighbors = 100 Number of sequence neighbors to search if using pdb100max_models_to_fetch = 100 Maximum models to fetch from PDBallow_reverse_connectivity = False Allow connectivity to be backwards or mixedmorph = False Morph replacement segments (NOTE: morphing of loops is intrinsic)shift_field_distance = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 A.pdb_include_list = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)pdb_exclude_list = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)exclude_similar_to_this_pdb_id = None Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to
 exclude_similar_to_this_pdb_idmaximum_percent_identity_to_this_pdb_id = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to
 exclude_similar_to_this_pdb_idmaximum_residues_to_try_fragment_search = 30 If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search
 firsttry_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model

has no more than maximum_residues_to_try_fragment_search, try simple fragment search first

- refine_cycles = 3 Refinement cycles (set to zero to not refine)
- maximum_insertions_or_deletions = None Maximum insertions/deletions per fragment
- database = *pdb100 pdb2018_90 Database to search. pdb100 is unique set of PDB. pdb2018_90 is quality-filtered PDB at 90% identity
- number_of_sequence_neighbors = 100 Number of sequence neighbors to search if using pdb100
- max_models_to_fetch = 100 Maximum models to fetch from PDB
- allow_reverse_connectivity = False Allow connectivity to be backwards or mixed
- morph = False Morph replacement segments (NOTE: morphing of loops is intrinsic)
- shift_field_distance = None Shift field distance. Distance over which morphing leaves model mostly the same. Over longer distances morphing changes model. Default is 10 Å.
- pdb_include_list = None List of PDB ID or pdb ID and chain_id to include (2DY1 or 2DY1_A)
- pdb_exclude_list = None List of PDB ID or pdb ID and chain_id to exclude (2DY1 or 2DY1_A)
- exclude_similar_to_this_pdb_id = None Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to exclude_similar_to_this_pdb_id
- maximum_percent_identity_to_this_pdb_id = 30 Maximum percent identity to supplied PDB chain to include. Exclude all PDB chains with sequence identity > maximum_percent_identity_to_this_pdb_id to exclude_similar_to_this_pdb_id
- maximum_residues_to_try_fragment_search = 30 If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search first
- try_fragment_search_for_short_models = True If try_fragment_search_for_short_models and model has no more than maximum_residues_to_try_fragment_search, try simple fragment search first
- crystal_inforesolution = None Nominal resolution of mapscattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.sequence = None Sequenceswrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- resolution = None Nominal resolution of map
- scattering_table = n_gaussian wk1995 it1992 *electron neutron Choice of scattering table for structure factor calculations. Standard for X-ray is n_gaussian, for cryoEM is electron.
- sequence = None Sequences
- wrapping = None You can specify whether the map is wrapped (can map values outside bounds to inside with cell translations).
- controlnproc = 1 Number of processors (if None, use all available)ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.verbose = False Verbose outputquick = False Run quickly
- nproc = 1 Number of processors (if None, use all available)
- ignore_symmetry_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box_map for example.
- verbose = False Verbose output

- quick = False Run quickly
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Crystallographic refinement with Rosetta (phenix.rosetta_refine)

rosetta_refine.html

Contents

- Overview
- Installation
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- List of all available keywords

Overview

The program phenix.rosetta_refine provides a convenient wrapper for running Rosetta refinement of protein X-ray crystal structures (DiMaio et al. 2013), which integrates the Rosetta methods for conformational sampling with the X-ray targets, B-factor refinement, and map generation in phenix.refine (Afonine et al. 2012). This is intended for use in difficult cases, especially at low resolution where it combines a wide radius of convergence with excellent geometry, and for preparing crystal structures for further modelling in Rosetta.

Currently both the setup and execution of Rosetta refinement have some limitations; please read this entire document before attempting to use the program! Note that Rosetta does not currently run on Windows; you must have a Mac or Linux system to use this program.

Installation

First, you should install the appropriate version of Phenix. The binary installers should be sufficient; users of the Mac graphical package can set up the command-line environment with this command (replacing the version number as needed):

```
source /Applications/PHENIX-1.8.4-1496/Contents/phenix-1.8.4-1496/phenix_env.sh
```

To run phenix.rosetta_refine, you must have installed Rosetta, software developed from the Baker laboratory at the University of Washington.

See the central installation notes for Rosetta

Execution

Running phenix.rosetta_refine is superficially similar to running phenix.refine, but with a much more limited set of options. Required inputs are a PDB file, a data file (MTZ format), and CIF restraint files for any non-standard ligands. Other common parameters include the number of processors to use, the method for parallelizing Rosetta jobs, and the refinement protocol, for example:

```
phenix.rosetta_refine model.pdb data.mtz nproc=5 technology=sge protocol=hi-res
```

The default behavior is to run the full refinement protocol as described in the program citation, creating five unique models, and picking the model with the best free R-factor. This will then be passed through

phenix.refine with a null strategy to optimize hydrogen and bulk solvent parameters. If desired, the argument `postref=True` can be given to enable additional cycles of refinement in phenix.refine, including both coordinates and B-factors with weight optimization.

The main choice of refinement strategy is between the "full" and "hires" protocols, both of which are described using the RosettaScripts syntax (Fleishman et al. 2011). The "hires" protocol is intended both for high-resolution structures and in general any structure that is already near convergence. The XML scripts for these methods are included in the Rosetta distribution. You may also create your own XML script with a customized protocol and pass that to `phenix.rosetta_refine`.

You can ask the program to parallelize jobs across multiple CPUs or several different queuing systems; the `technology` parameter determines the method used, and `nproc` indicates the number of jobs to run at once. The actual number of processes will be limited by the `number_of_models` parameter, which defaults to 5. Requesting more models may provide incremental improvements in some cases but we have found it rarely makes a difference of more than 1-2% in R-free.

Output will be to a new directory with a name such as "rosetta_1", with the directory number increasing sequentially as additional runs are created. Because phenix.refine is used as the final step, the output files will be identical to those output by that program.

Limitations

The program has several restrictions compared to phenix.refine:

Molecules are limited to the standard amino acids and a small number of common ions and ligands. If you have additional molecules in your model, these will be removed prior to Rosetta refinement and replaced afterwards. You must still supply and necessary CIFs to the program, as these will be used later when running the final phenix.refine step. RNA is not supported. Alternate conformations are not supported. The B-factor refinement is limited to simply isotropic refinement at present; for more flexible parameterization you should run phenix.refine separately. Rosetta scales poorly above approximately 1000 residues; although larger structures may work, the performance tends to be worse. At present it is not possible to impose NCS restraints on structures in Rosetta.

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- Rosetta scales poorly above approximately 1000 residues; although larger structures may work, the performance tends to be worse.
- At present it is not possible to impose NCS restraints on structures in Rosetta.

Also note that the program is significantly more processor-intensive, although less so than MR-Rosetta (DiMaio et al. 2011). We recommend that you use a multiprocessor computer or queuing system for greatest efficiency.

Troubleshooting

This program is under development; please address any questions concerning its use to help@phenix-online.org. The following guidelines may be useful for best performance:

The default settings are optimized for difficult structures, where conventional refinement gets stuck quickly. The assumption is that fine details such as waters and ligand molecules have not yet been added. For now, the only alternative is the "high-resolution" protocol, which employs gentler optimization; additional refinement scripts will be provided in the future. For optimal results, especially when preparing a structure for further Rosetta modelling, you may wish to design your own Rosetta refinement protocol using RosettaScripts. Unlike methods such as simulated annealing with the default geometry restraints, Rosetta refinement will almost never improve the fit to the experimental data at the expense of model geometry. If your structure contains severe distortions as a result of aggressive optimization, Rosetta refinement may actually degrade the model. While Rosetta refinement can improve structures that are outside the radius of convergence of traditional crystallographic refinement, it still requires that the molecule(s) be correctly placed in the unit cell. If you are uncertain of your molecular replacement solution, we recommend trying MR-Rosetta instead. The tests performed in DiMaio et al. 2013 indicated that the best results may come from combining orthogonal refinement methods such as DEN (Schroder et al. 2010) with Rosetta.

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List of all available keywords

inputsequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False twin_law = Auto Enter twin law if known.xray_dataScope of X-ray data and free-R flagsfile_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution binsmin_bin_size = 40 Minimum number of reflections per binr_free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore_r_free_flags = False Use all reflections in refinement (work and test)disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n_shells = 20 pdbfile_name = None Model file(s) name (PDB)monomersfile_name = None Monomer file(s) name (CIF)mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZfilerosettaprotocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.script = None XML file in RosettaScripts syntax describing a custom refinement protocol.rosetta_path = None Path to

Rosetta distribution. Normally this is set by the PHENIX_ROSETTA_PATH environment variable.
 database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.
 score_weights = *score12_full standard score13_env_hb number_of_models = 5 density_sampling = *fast thorough ignore_unrecognized = False keep_ligands = False adp_type = *individual group map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.
 ligand_params_file = None phenixpost_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.
 strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.
 cycles = 3 tls = False phil_file = None waters = False Add water molecules in phenix.refine after Rosetta refinement.
 outputprefix = None directory_number = None verbose = False debug = True gui_output_dir = None Output directory (for PHENIX GUI only)
 job_title = None Job title in PHENIX GUI, not used on command line
 runtimeproc = 1 technology = *multiprocessing sge lsf pbs condor slurm threading qsub_command = None

- inputsequence = None scattering_table = wk1995 it1992 *n_gaussian neutron electron wavelength = None energy = None use_experimental_phases = False twin_law = Auto Enter twin law if known.
 xray_dataScope of X-ray data and free-R flags
 file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
 french_wilson_scale = True sigma_fobs_rejection_criterion = None sigma_iobs_rejection_criterion = None ignore_all_zeros = True force_anomalous_flag_to_be_equal_to = None
 convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60 Maximum number of resolution bins
 min_bin_size = 40 Minimum number of reflections per bin
 free_flagsfile_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
 label = None required = True Specify if free-r flags are must be present (or else generated)
 test_flag_value = None This value is usually selected automatically - do not change unless you really know what you're doing!
 ignore_r_free_flags = False Use all reflections in refinement (work and test)
 disable_suitability_test = False ignore_pdb_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
 generate = False Generate R-free flags (if not available in input files)
 fraction = 0.1 max_free = 2000 lattice_symmetry_max_delta = 5 use_lattice_symmetry = True use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
 n_shells = 20 pdbfile_name = None Model file(s) name (PDB)
 monomersfile_name = None Monomer file(s) name (CIF)
 mapsmap_file_name = None A CCP4-formatted map
 map_min = None Resolution of map
 map_coefficients_file_name = None MTZ file containing map
 map_coefficients_label = None Data label for complex map coefficients in MTZ file

- sequence = None

- scattering_table = wk1995 it1992 *n_gaussian neutron electron

- wavelength = None

- energy = None

- use_experimental_phases = False

- twin_law = Auto Enter twin law if known.

- xray_dataScope of X-ray data and free-R flags
 file_name = None labels = None high_resolution = None low_resolution = None twin_law = None outliers_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
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 convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5 french_wilsonmax_bins = 60

Maximum number of resolution bins `min_bin_size = 40` Minimum number of reflections per bin `r_free_flagsfile_name = None` This is normally the same as the file containing Fobs and is usually selected automatically. `label = None` `required = True` Specify if free-r flags are must be present (or else generated) `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing! `ignore_r_free_flags = False` Use all reflections in refinement (work and test) `disable_suitability_test = False` `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs. `generate = False` Generate R-free flags (if not available in input files) `fraction = 0.1` `max_free = 2000` `lattice_symmetry_max_delta = 5` `use_lattice_symmetry = True` `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry. `n_shells = 20`

- `file_name = None`
- `labels = None`
- `high_resolution = None`
- `low_resolution = None`
- `twin_law = None`
- `outliers_rejection = True` Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- `french_wilson_scale = True`
- `sigma_fobs_rejection_criterion = None`
- `sigma_iobs_rejection_criterion = None`
- `ignore_all_zeros = True`
- `force_anomalous_flag_to_be_equal_to = None`
- `convert_to_non_anomalous_if_ratio_pairs_lone_less_than_threshold = 0.5`
- `french_wilsonmax_bins = 60` Maximum number of resolution bins `min_bin_size = 40` Minimum number of reflections per bin
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- disable_suitability_test = False
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- fraction = 0.1
- max_free = 2000
- lattice_symmetry_max_delta = 5
- use_lattice_symmetry = True
- use_dataman_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- n_shells = 20
- pdbfile_name = None Model file(s) name (PDB)
- file_name = None Model file(s) name (PDB)
- monomersfile_name = None Monomer file(s) name (CIF)
- file_name = None Monomer file(s) name (CIF)
- mapsmap_file_name = None A CCP4-formatted mapd_min = None Resolution of mapmap_coefficients_file_name = None MTZ file containing mapmap_coefficients_label = None Data label for complex map coefficients in MTZ file
- map_file_name = None A CCP4-formatted map
- d_min = None Resolution of map
- map_coefficients_file_name = None MTZ file containing map
- map_coefficients_label = None Data label for complex map coefficients in MTZ file
- rosetta_protocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.script = None XML file in RosettaScripts syntax describing a custom refinement protocol.rosetta_path = None Path to Rosetta distribution. Normally this is set by the PHENIX_ROSETTA_PATH environment variable.database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.score_weights = *score12_full standard score13_env_hb number_of_models = 5 density_sampling = *fast thorough ignore_unrecognized = False keep_ligands = False adp_type = *individual group map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.ligand_params_file = None
- protocol = *full hires Refinement protocol to use, corresponding to XML scripts in the Rosetta distribution. The hires protocol is less aggressive and faster, but has a smaller radius of convergence.
- script = None XML file in RosettaScripts syntax describing a custom refinement protocol.
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- database = None Path to the Rosetta database. Normally this is set by the ROSETTA3_DB environment variable.

- score_weights = *score12_full standard score13_env_hb
- number_of_models = 5
- density_sampling = *fast thorough
- ignore_unrecognized = False
- keep_ligands = False
- adp_type = *individual group
- map_type = *Auto 2mFo-DFc mFo dm prime_and_switch feature_enhanced composite_omit This controls the procedure for generating a map for Rosetta real-space refinement. All map types other than mFo will essentially be similar to 2mFo-DFc, but with different processing steps.
- ligand_params_file = None
- phenixpost_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.cycles = 3
- tls = False phil_file = None waters = False Add water molecules in phenix.refine after Rosetta refinement.
- post_refine = False Perform additional refinement (coordinates and B-factors) in phenix.refine after Rosetta is done.
- strip_hydrogens = False Remove hydrogen atoms after Rosetta refinement.
- cycles = 3
- tls = False
- phil_file = None
- waters = False Add water molecules in phenix.refine after Rosetta refinement.
- outputprefix = None directory_number = None verbose = False debug = True gui_output_dir = None Output directory (for PHENIX GUI only)job_title = None Job title in PHENIX GUI, not used on command line
- prefix = None
- directory_number = None
- verbose = False
- debug = True
- gui_output_dir = None Output directory (for PHENIX GUI only)
- job_title = None Job title in PHENIX GUI, not used on command line
- runtimenproc = 1 technology = *multiprocessing sge lsf pbs condor slurm threading qsub_command = None
- nproc = 1
- technology = *multiprocessing sge lsf pbs condor slurm threading
- qsub_command = None

Running SOLVE/RESOLVE in PHENIX

running-solve-resolve.html

Author(s)

- SOLVE/RESOLVE: Tom Terwilliger

Purpose

SOLVE and RESOLVE can be run directly in the PHENIX environment. This feature is normally only for advanced SOLVE/RESOLVE users who want to access the keywords in SOLVE/RESOLVE directly.

Usage

Running SOLVE/RESOLVE from the command-line or in a script.

- You can run solve with the command:

```
phenix.solve
```

This command will set the environmental variables CCP4_OPEN, SYMOP, SYMLIB, and SOLVEDIR and will run solve. If you want to run a different size of solve, then you can specify:

```
phenix.solve --giant
```

For a bigger version still, choose --huge, for biggest, --extra_huge.

- Running resolve or resolve_pattern is similar:

```
phenix.resolve  
phenix.resolve_pattern
```

- Running solve/resolve from a command file is simple. Here is a command file to run resolve:

```
phenix.resolve<<<EOD  
hklin solve.mtz  
labin FP=FP SIGFP=SIGFP PHIB=PHIB FOM=FOM HLA=HLA HLB=HLB HLC=HLC HLD=HLD  
solvent_content 0.43  
database 5  
EOD
```

Literature

Additional information

All the solve/resolve keywords are available in the PHENIX versions of solve and resolve. See the full documentation for solve/resolve at <https://solve.lanl.gov/> . A PDF of this documentation is available at https://phenix-online.org/phenix_data/terwilliger/solve_resolve_archive_doc/SOLVE_RESOLVE_213.pdf .

Downloading data from PDB and setting up for SAD structure determination with phenix.sad_data_from_pdb

sad_data_from_pdb.html

Author(s)

- phenix.sad_data_from_pdb: Tom Terwilliger

Purpose

phenix.sad_data_from_pdb is a command line tool for downloading coordinates and anomalous structure factor data for a PDB deposition and setting up for running one or more wavelengths of data as SAD datasets.

Usage

How phenix.sad_data_from_pdb works:

phenix.sad_data_from_pdb first uses phenix.fetch_pdb to download the data from the PDB.

Then it converts the structure factor data to Scalepack unmerged original index format, calculates model map coefficients, and identifies the anomalously-scattering atoms in the PDB file, the wavelengths of data collection, and the resolution of the data. If these are not present in the PDB deposition then the procedure will not work.

Examples

Standard run of phenix.sad_data_from_pdb:

Running phenix.sad_data_from_pdb is easy. You can just type:

```
phenix.sad_data_from_pdb pdb_ids="2etd"
```

and phenix.sad_data_from_pdb will download the data for 2etd and set up a directory with this data and with subdirectories for each wavelength of data present (w1, w2, and w3 in this case). In each subdirectory a file run_autosol_std.csh will be created that is suitable for running phenix.autosol using that anomalous data.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

base_directory = None Directory where files are to be located
 pdb_ids = None PDB ID to fetch and use to create SAD datasets
 wavelength_id = None Select only the wavelength specified by wavelength_id
 ha_list = None List of anomalously-scattering atoms (used instead of list extracted from PDB file)
 steps = *fetch *cif_to_sca *fmodel *ha *autosol Steps to carry out: fetch downloads files from the PDB, cif_to_sca converts PDB CIF files to Scalepack unmerged original index, fmodel calculates model structure factors, ha identifies anomalously-scattering atoms in the PDB file, autosol sets up for a run of autosol. To select several steps put them in quotes with the stars: steps='*ha *autosol'
 initialize = None If set, remove any existing data in the PDB directories
 best_only = True Keep only atom with strongest anomalous scattering and ignore the rest
 merge = None If set, merge multiple observations
 replace_full_directory = None You can change the full directory name for a single dataset

- base_directory = None Directory where files are to be located
- pdb_ids = None PDB ID to fetch and use to create SAD datasets
- wavelength_id = None Select only the wavelength specified by wavelength_id
- ha_list = None List of anomalously-scattering atoms (used instead of list extracted from PDB file)
- steps = *fetch *cif_to_sca *fmodel *ha *autosol Steps to carry out: fetch downloads files from the PDB, cif_to_sca converts PDB CIF files to Scalepack unmerged original index, fmodel calculates model structure factors, ha identifies anomalously-scattering atoms in the PDB file, autosol sets up for a run of autosol. To select several steps put them in quotes with the stars: steps='*ha *autosol'
- initialize = None If set, remove any existing data in the PDB directories
- best_only = True Keep only atom with strongest anomalous scattering and ignore the rest
- merge = None If set, merge multiple observations
- replace_full_directory = None You can change the full directory name for a single dataset

Scaling unmerged anomalous data or multiple datasets with `scale_and_merge`

`scale_and_merge.html`

Author(s)

- `scale_and_merge`: Tom Terwilliger

Purpose

`scale_and_merge` is a tool for scaling unmerged anomalous data or multiple data files and creating a scaled dataset and two scaled half-datasets. This tool normally is used in combination with `anomalous_signal` to create a scaled dataset and analyze anomalous signal in a SAD dataset. It can also be used to scale any other group of datasets.

Usage

How `scale_and_merge` works:

- You supply `scale_and_merge` with a directory containing datafiles or the name of a single unmerged datafile.
- You can optionally also specify a pair of labels that identifies datasets that are to be kept together. For example if you collected your data as pairs of datafiles with inverse beam geometry, you might have called the members of a pair `data_1_0_w1.HKL` and `data_1_0_w2.HKL`, related by `w1` and `w2`.
- `scale_and_merge` will first check the cell dimensions of all the datasets. Normally it will choose the largest set of similar crystals (you can have it keep all datasets with `only_similar_datasets=False`). It will also check the anisotropy in all the datafiles and calculate the average anisotropy (to be applied by default to all datafiles before scaling).
- `scale_and_merge` will then scale all the data together. It will split your datafiles into smaller files if your datafiles contain duplicate measurements of the same indices. The intensities in each datafile will be adjusted for anisotropy to match the average anisotropy of all the datafiles. In this way all the datafiles are matched but the overall character of the data is not changed.
- `scale_and_merge` will then scale each individual file with local scaling, producing files in the temporary directory with names ending in `_scale.mtz`. and then merge them all together.
- Merging of the individual datasets is done twice, and then optionally two additional times to optimize anomalous differences.
- In the first merging the individual datasets are simply averaged with weights based on the sigmas for each reflection. Then the individual datasets are compared with the overall mean dataset and dataset variances (estimates of systematic differences between datasets and the mean) are calculated based on the rms differences from the mean and the individual sigmas. In essence the dataset variance plus the individual variances are then to be used as estimates of the total variance for each reflection. The second merging uses these total variances in weighting rather than the original sigmas. The correlations of each individual dataset with the merged dataset are calculated.

- If anomalous differences are optimized (default with `optimize_anomalous=True`), merging is carried out another time in order to optimize the weighting of anomalous differences in the merging step. For each unique reflection in the asymmetric unit of each dataset, the mean I^+ and mean I^- are used to calculate anomalous differences. The anomalous differences from each individual dataset are then compared with the anomalous differences from the merged dataset to estimate individual dataset anomalous difference variances. Then the anomalous differences from each individual dataset are averaged, with weights based on the original sigmas and the dataset variances. These merged anomalous differences are then used to replace the anomalous differences in the merged dataset above (for example, a reflection in the merged dataset above that has $I^+, \text{sig}I^+, I^-, \text{sig}I^-$ would get new values of I^+ and I^- that have a difference equal to the appropriate merged anomalous difference, but the same mean as before.) The correlation of the anomalous differences in the original merged dataset and after optimization is printed (this should be high, for example 0.80).
- The original datasets are then split into two parts for creation of two half-datasets. These half-datasets are useful for estimating the quality of the data and are used in `anomalous_signal` for this purpose.
- The splitting into half-datasets is done in one of three ways, and the method is chosen based on the number of anomalous differences available for comparison using each method. With each method the data in each half-dataset are scaled just as the entire dataset was scaled. The preferred method is to split by files. Half of the datafiles are used to create each half-dataset. The next preferred method is to split with the first half of each dataset in one half-dataset and the second half in the other. The third preferred method is splitting alternate reflections with each unique index (after mapping to the asymmetric unit) into the two half-datasets. The last method is to randomly assign reflections to the two half-datasets. The reason for this hierarchical approach is that reflections measured close in time, within the same dataset are better matched than those measured further in time or within different datasets. These approaches for splitting the datafiles attempt to pair anomalous differences measured close in time and in the same dataset.
- After half-datasets are obtained, the correlation of anomalous differences between half-datasets is estimated.

Output from `scale_and_merge`

`scale_and_merge` provides a summary half-dataset correlations in your dataset. Here is an example with very weak anomalous signal (but present):

```
Scale and merge...analysis of multi-dataset SAD data
input\_files {
  data = "dd"
  data\_labels = None
  paired\_group\_ids = None
}
output\_files {
  output\_file = "scaled\_data.mtz"
  output\_half\_dataset\_a = "half\_dataset\_a.mtz"
  output\_half\_dataset\_b = "half\_dataset\_b.mtz"
  output\_file\_format = *mtz sca
}
crystal\_info {
  resolution = None
  low\_resolution = None
  space\_group = None
  unit\_cell = None
}
data\_selection {
  minimum\_datafile\_fraction = 0.3
  require\_fpfm = True
}
```

```

only\_similar\_datasets = True
relative\_length\_tolerance = None
absolute\_angle\_tolerance = None
choose\_optimal\_datasets = False
sort\_datasets\_by\_anomalous\_cc = True
}
scaling {
  make\_anisotropy\_uniform = True
  overall\_scale = False
  skip\_scaling = False
  lowest\_resolution\_range = 6
}
merging {
  optimize\_anomalous = True
  use\_best\_group\_as\_target = False
  rescale\_sigmas = False
}
half\_dataset\_cc {
  get\_half\_dataset\_cc = True
  half\_dataset\_cc\_by\_files = True
  split\_as\_first\_second\_half = True
  split\_alternately = True
  split\_randomly = True
}
directories {
  temp\_dir = None
  output\_dir = ""
  gui\_output\_dir = None
}
control {
  verbose = False
  random\_seed = 714215
  comparison\_file = None
  clean\_up = True
}

```

Comparing crystal symmetries and noting anisotropy in data

Relative length tolerance: 0.010 Angle tolerance: 1.000 degrees

Taking initial symmetry from /Users/terwill/unix/misc/scale_and_merge/dd/WNV_NS1_205_w2_2.9_90.HKL
with 3 similar files

File SG A B C Alpha Beta Gamma

Symmetry group 1

```

WNV\_NS1\_202\_w1\_2.9\_90.HKL P 3 2 1 166.997 166.997 94.156 90.000 90.000 120.000
WNV\_NS1\_202\_w2\_2.9\_90.HKL P 3 2 1 166.704 166.704 94.113 90.000 90.000 120.000
WNV\_NS1\_205\_w1\_2.9\_90.HKL P 3 2 1 167.578 167.578 93.934 90.000 90.000 120.000
WNV\_NS1\_205\_w2\_2.9\_90.HKL P 3 2 1 167.639 167.639 93.859 90.000 90.000 120.000

```

Scaling 4 datasets

Scaling data files with local scaling

Files to scale:

```

/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_202\_w1\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_202\_w2\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_205\_w1\_2.9\_90.HKL
/Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\_NS1\_205\_w2\_2.9\_90.HKL

```

Selecting just reflections that have both F+ and F- in the same
dataset or are centric

Splitting datafiles into sub-files with one copy of each unique hkl

File Refl (rejected) B1 B2 B3 B-avg

```

TEMP0/WNV\_NS1\_202\_w1\_2.9\_90\_1.sca: 138180 ( 6378) 63.9 63.9 55.00 60.9
TEMP0/WNV\_NS1\_202\_w1\_2.9\_90\_2.sca: 229 ( 0) 63.9 63.9 55.00 60.9

```

```

TEMP0\WNV\NS1\202\w2\2.9\90\1.sca: 136254 ( 6096) 59.4 59.4 49.78 56.2
TEMP0\WNV\NS1\202\w2\2.9\90\2.sca: 206 ( 0) 59.4 59.4 49.78 56.2
TEMP0\WNV\NS1\205\w1\2.9\90\1.sca: 148714 ( 4464) 73.7 73.7 63.92 70.4
TEMP0\WNV\NS1\205\w1\2.9\90\2.sca: 239 ( 0) 73.7 73.7 63.92 70.4
TEMP0\WNV\NS1\205\w2\2.9\90\1.sca: 147256 ( 4560) 70.3 70.3 60.50 67.0
TEMP0\WNV\NS1\205\w2\2.9\90\2.sca: 241 ( 0) 70.3 70.3 60.50 67.0

```

Notes:

```

Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\202\w1\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\202\w2\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\205\w1\2.9\90.HKL into 2 files:
Splitting /Users/terwill/unix/misc/scale\_and\_merge/dd/WNV\NS1\205\w2\2.9\90.HKL into 2 files:
Keeping split datafiles with at least 44614 reflections

```

High-resolution limit: 2.89

Scaling data in batches from individual data files

List of scaled data files:

```

TEMP0\WNV\NS1\205\w1\2.9\90\1\_scale.mtz (30844 refl)
TEMP0\WNV\NS1\205\w2\2.9\90\1\_scale.mtz (30828 refl)
TEMP0\WNV\NS1\202\w1\2.9\90\1\_scale.mtz (29641 refl)
TEMP0\WNV\NS1\202\w2\2.9\90\1\_scale.mtz (29527 refl)

```

Scaling and merging data files with overall scale factor

Files to scale:

```
TEMP0\WNV\NS1\205\w1\2.9\90\1\_scale.mt...
```

Possible Problems

Literature

List of all available keywords

input_filesdata = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensitiespaired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this example you could also use paired_group_ids='w1.sca w2.sca' and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.check_file_type = False You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.reference_data = None Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.reference_data_labels = None Optional label specifying which columns of reference data to use.output_filesoutput_file = scaled_data.mtz Output data file name for merged scaled dataoutput_half_dataset_a = half_dataset_a.mtz Output data file name for half dataset A. Only written if get_half_datset_cc=True. This is the scaled merged data from half of the input data.output_half_dataset_b = half_dataset_b.mtz Output data file name for half dataset B. Only written if get_half_datset_cc=True. This is the scaled merged data from the other half of the input data.output_file_format = *mtz sca Output format for scaled data. Choices are MTZ and unmerged scalepack

formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.

crystal_inforesolution = None high-resolution limit
low_resolution = None Low-resolution limit
space_group = None Space group (normally read from the data file)
unit_cell = None Unit Cell (normally read from the data file)
data_selectionanomalous = True Consider data as anomalous (otherwise average Bijvoet mates).
minimum_datafile_fraction = 0.3 You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile.
require_fpfm = True If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
only_similar_datasets = True Group data by lattice similarity and take biggest group only. If space_group and unit_cell are set, then take those only that match this space_group and unit_cell. Use relative_length_tolerance and absolute_angle_tolerance to decide if two datasets are similar.
relative_length_tolerance = None Tolerance for considering two cells similar (fractional) You can allow anything with only_similar_datasets=False
resolution_length_tolerance = None Tolerance for considering two cells similar (fraction of resolution. For example normalized_length_tolerance=0.25 means the unit cells must match within 1/4 of the resolution. You can allow anything with only_similar_datasets=False
absolute_length_toler...

- input_filesdata = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.
data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.
assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensities
paired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this example you could also use paired_group_ids='w1.sca w2.sca' and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.
check_file_type = False You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.
reference_data = None Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.
reference_data_labels = None Optional label specifying which columns of reference data to use.
- data = None Anomalous data (I+ and I- or F+ and F-). Any standard format is fine. You can have any number of input files. If you have multiple files, specify the directory that contains them and nothing else.
- data_labels = None Optional label specifying which columns of anomalous data to use. Not necessary if your input files have only one set of anomalous data. If you have multiple input files they need to have the same labels.
- assume_shelx_observation_type_is = *amplitudes intensities For shelx files you need to specify if they are amplitudes or intensities
- paired_group_ids = None Optional pair of identifiers that specify datasets that should be grouped together. The paired group ids must be the only characters that differ between datasets that should be grouped together, and the characters must uniquely define what datasets to group. For example paired_group_ids='w1 w2' would be the pair for the two datasets se_w1.sca and se_w2.sca. In this

example you could also use `paired_group_ids='w1.sca w2.sca'` and that would have the same effect. Use this option when some of your datafiles are related to others by inverse-beam data collection for example. This matching of datasets can be very helpful.

- `check_file_type = False` You can request that the data type of input files be checked if a data directory is supplied. Default is to read everything in the data directory.
- `reference_data = None` Reference data for use in scaling. All datasets will be scaled to this reference dataset. Useful if you have very accurate native data.
- `reference_data_labels = None` Optional label specifying which columns of reference data to use.
- `output_filesoutput_file = scaled_data.mtz` Output data file name for merged scaled data
- `dataoutput_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Only written if `get_half_datset_cc=True`. This is the scaled merged data from half of the input data.
- `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Only written if `get_half_datset_cc=True`. This is the scaled merged data from the other half of the input data.
- `data.output_file_format = *mtz sca` Output format for scaled data. Choices are MTZ and unmerged scalepack formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.
- `output_file = scaled_data.mtz` Output data file name for merged scaled data
- `output_half_dataset_a = half_dataset_a.mtz` Output data file name for half dataset A. Only written if `get_half_datset_cc=True`. This is the scaled merged data from half of the input data.
- `output_half_dataset_b = half_dataset_b.mtz` Output data file name for half dataset B. Only written if `get_half_datset_cc=True`. This is the scaled merged data from the other half of the input data.
- `output_file_format = *mtz sca` Output format for scaled data. Choices are MTZ and unmerged scalepack formats. Note: unmerged scalepack format not normally used because indices may be changed from original values and that data from within one input file may be merged, so that this unmerged data is partially merged and not the same as the original data.
- `crystal_inforesolution = None` high-resolution limit `low_resolution = None` Low-resolution limit
`space_group = None` Space group (normally read from the data file)
`unit_cell = None` Unit Cell (normally read from the data file)
- `resolution = None` high-resolution limit
- `low_resolution = None` Low-resolution limit
- `space_group = None` Space group (normally read from the data file)
- `unit_cell = None` Unit Cell (normally read from the data file)
- `data_selectionanomalous = True` Consider data as anomalous (otherwise average Bijvoet mates).
- `minimum_datafile_fraction = 0.3` You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile
- `require_fpm = True` If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
- `only_similar_datasets = True` Group data by lattice similarity and take biggest group only. If `space_group` and `unit_cell` are set, then take those only that match this `space_group` and `unit_cell`. Use `relative_length_tolerance` and `absolute_angle_tolerance` to decide if two datasets are similar.
- `relative_length_tolerance = None` Tolerance for considering two cells similar (fractional) You can allow anything with `only_similar_datasets=False`
- `resolution_length_tolerance = None` Tolerance for

considering two cells similar (fraction of resolution. For example `normalized_length_tolerance=0.25` means the unit cells must match within 1/4 of the resolution. You can allow anything with `only_similar_datasets=False`
`absolute_length_tolerance = None` `absolute_angle_tolerance = None`
 Tolerance for considering two cell angles similar (degrees) You can allow anything with `only_similar_datasets=False`
`choose_optimal_datasets = False` After initial merging, sort datasets by correlation to merged dataset, remove datasets until overall $I/\sigma I$ is maximized, including dataset systematic errors in $I/\sigma I$.
`sort_datasets_by_anomalous_cc = True` For choosing optimal datasets, sort the datasets by anomalous CC. Alternative is sorting by non-anomalous CC.

- `anomalous = True` Consider data as anomalous (otherwise average Bijvoet mates).
- `minimum_datafile_fraction = 0.3` You can reject any datasets (after splitting) that have less than this fraction of the maximum number of reflections found in any datafile
- `require_fpfm = True` If set, then when files are split into unique sets of indices only centric reflections and reflections with both F+ and F- present are written out. Note that in most space groups there may be more than one F+ or F- symmetry mate present. If any one is present for F+ and any one for F- the reflection is written out.
- `only_similar_datasets = True` Group data by lattice similarity and take biggest group only. If `space_group` and `unit_cell` are set, then take those only that match this `space_group` and `unit_cell`. Use `relative_length_tolerance` and `absolute_angle_tolerance` to decide if two datasets are similar.
- `relative_length_tolerance = None` Tolerance for considering two cells similar (fractional) You can allow anything with `only_similar_datasets=False`
- `resolution_length_tolerance = None` Tolerance for considering two cells similar (fraction of resolution. For example `normalized_length_tolerance=0.25` means the unit cells must match within 1/4 of the resolution. You can allow anything with `only_similar_datasets=False`
- `absolute_length_tolerance = None`
- `absolute_angle_tolerance = None` Tolerance for considering two cell angles similar (degrees) You can allow anything with `only_similar_datasets=False`
- `choose_optimal_datasets = False` After initial merging, sort datasets by correlation to merged dataset, remove datasets until overall $I/\sigma I$ is maximized, including dataset systematic errors in $I/\sigma I$.
- `sort_datasets_by_anomalous_cc = True` For choosing optimal datasets, sort the datasets by anomalous CC. Alternative is sorting by non-anomalous CC.
- `scalingmake_anisotropy_uniform = True` Calculate anisotropy for all datasets and adjust all of them to the average anisotropy. This is designed to improve scaling without changing the overall character of the data (i.e., not sharpening the data or removing anisotropy).
`overallscale = False` If set, only use overall scale and B (no local scaling)
`skip_scaling = False` You can skip the scaling step and just get half-dataset CC for a pair of existing files that you have put in a directory by themselves.
`lowest_resolution_range = 6`. Lowest value of the high-resolution cutoff for CC as a function of high-resolution cutoff with constant low-resolution cutoff
`min_common_refl = 200` Minimum number of reflections in common to scale datasets
`sort_on_batch_number = True` You can sort XDS and MTZ reflections by frame number. Otherwise scale and merge assumes reflections are collected in the order in the data file.
`scale_reference_data = True` Scale reference data to maximum of 99999 if writing intermediate scalepack data files
- `make_anisotropy_uniform = True` Calculate anisotropy for all datasets and adjust all of them to the average anisotropy. This is designed to improve scaling without changing the overall character of the data (i.e., not sharpening the data or removing anisotropy).

- `overall_scale = False` If set, only use overall scale and B (no local scaling)
- `skip_scaling = False` You can skip the scaling step and just get half-dataset CC for a pair of existing files that you have put in a directory by themselves.
- `lowest_resolution_range = 6`. Lowest value of the high-resolution cutoff for CC as a function of high-resolution cutoff with constant low-resolution cutoff
- `min_common_refl = 200` Minimum number of reflections in common to scale datasets
- `sort_on_batch_number = True` You can sort XDS and MTZ reflections by frame number. Otherwise scale and merge assumes reflections are collected in the order in the data file.
- `scale_reference_data = True` Scale reference data to maximum of 99999 if writing intermediate scalepack data files
- `merging_include_dataset_variance = True` Scale datasets together including inter-dataset variance in in weighting. This takes into account the systematic differences among datasets and allows automatic down-weighting of datasets that are different than all the others. `optimize_anomalous = True` Scale datasets together including inter-dataset variance in anomalous differences in weighting. Involves separation of mean intensity of each reflection from anomalous differences and recombination after scaling to recreate I+ and I-. Alternative is to include inter-dataset variance in intensities. `use_best_group_as_target = False` Estimate inter-dataset variance by using the best dataset (highest I/sigma) as target for all datasets. This amounts to using all the other datasets to optimally estimate the data for this one crystal. `rescale_sigmas = False` Rescale sigmas assuming that at high resolution, there are no anomalous differences. This can sometimes be useful if the sigmas are all too high or low and the data are collected to sufficiently high resolution that there really are essentially no anomalous differences (only noise) at high resolution.
- `include_dataset_variance = True` Scale datasets together including inter-dataset variance in in weighting. This takes into account the systematic differences among datasets and allows automatic down-weighting of datasets that are different than all the others.
- `optimize_anomalous = True` Scale datasets together including inter-dataset variance in anomalous differences in weighting. Involves separation of mean intensity of each reflection from anomalous differences and recombination after scaling to recreate I+ and I-. Alternative is to include inter-dataset variance in intensities.
- `use_best_group_as_target = False` Estimate inter-dataset variance by using the best dataset (highest I/sigma) as target for all datasets. This amounts to using all the other datasets to optimally estimate the data for this one crystal.
- `rescale_sigmas = False` Rescale sigmas assuming that at high resolution, there are no anomalous differences. This can sometimes be useful if the sigmas are all too high or low and the data are collected to sufficiently high resolution that there really are essentially no anomalous differences (only noise) at high resolution.
- `half_dataset_cc_get_half_dataset_cc = True` Split datasets and get half-dataset CC if data are available. `half_dataset_cc_by_files = True` Split datasets into groups based on keeping all data in a file together (only applies if multiple data files are supplied). `half_dataset_cc_match_inverses = False` Split datasets into groups keeping (hkl, -h-k-l) together. Applies if only one file or `half_dataset_cc_by_files=False`. Does not apply if less than half of the reflections have (-h-k-l) pairs and other methods below have more matches. `split_as_first_second_half = True` split datasets into groups based on using the first or second half of each dataset in each half-dataset. Applies if `half_dataset_cc_by_files=False` or only one dataset is supplied and `half_dataset_cc_match_inverses=False`. Also applies if `half_dataset_cc_by_files` and

half_dataset_cc_match_inverses have failed to yield sufficient anomalous differences. split_alternately = True split datasets into groups based on sending observations of equivalent reflections alternately to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences. split_randomly = True split datasets into groups based on sending observations of equivalent reflections randomly to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and split_alternately is False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences

- get_half_dataset_cc = True Split datasets and get half-dataset CC if data are available
- half_dataset_cc_by_files = True Split datasets into groups based on keeping all data in a file together (only applies if multiple data files are supplied.)
- half_dataset_cc_match_inverses = False Split datasets into groups keeping (hkl, -h-k-l) together. Applies if only one file or half_dataset_cc_by_files=False. Does not apply if less than half of the reflections have (-h-k-l) pairs and other methods below have more matches.
- split_as_first_second_half = True split datasets into groups based on using the first or second half of each dataset in each half-dataset. Applies if half_dataset_cc_by_files=False or only one dataset is supplied and half_dataset_cc_match_inverses=False. Also applies if half_dataset_cc_by_files and half_dataset_cc_match_inverses have failed to yield sufficient anomalous differences.
- split_alternately = True split datasets into groups based on sending observations of equivalent reflections alternately to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences
- split_randomly = True split datasets into groups based on sending observations of equivalent reflections randomly to the first or second half-dataset. Applies if split_as_first_second_half=False and half_dataset_cc_by_files=False and split_alternately is False and only one dataset is supplied. Also applies if other methods have failed to yield sufficient anomalous differences
- directoriestemp_dir = None Optional temporary work directory. If specified, it will be used for temporary files. If clean_up=True and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
- output_dir = "" Output directory where files are to be written
- gui_output_dir = None GUI use only - does not apply to command line version
- temp_dir = None Optional temporary work directory. If specified, it will be used for temporary files. If clean_up=True and it did not exist before, it will be removed at the end of the run. If it did exist, it will be left.
- output_dir = "" Output directory where files are to be written
- gui_output_dir = None GUI use only - does not apply to command line version
- controljob_title = None Job title in PHENIX GUI, not used on command line
- verbose = False Verbose
- output_random_seed = 714215 random seed
- comparison_file = None Data file with comparison
- anomalous_differences = *multiprocessing sge lsq pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to use in scaling
- solve_size = None You can set the size of Solve. Normally done automatically.
- clean_up = True Clean up and remove temp directory after finishing
- job_title = None Job title in PHENIX GUI, not used on command line

- verbose = False Verbose output
- random_seed = 714215 random seed
- comparison_file = None Data file with comparison anomalous differences
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.
- nproc = 1 Number of processors to use in scaling
- solve_size = None You can set the size of Solve. Normally done automatically.
- clean_up = True Clean up and remove temp directory after finishing

Simplified Carbohydrate Line-Entry System (SCaLES)

scales.html

There are many formats for specifying a polysaccharides as seen in this diagram from Herget et al. 2008.

The article introduces a format known as GlycoCT.

A common and compact molecule specification format is known as SMILES (ref) which stands for. It relies on the specification of the elements and the bonds (both implicit and explicit) of a molecule. Branches and rings can also be specified.

Because protein crystallographers are conversant with the concept of residues and polysaccharides are polymers of various carbohydrate units, the step to equating an element to a sugar residue and a chemical bond to a glycosidic link is easy. Polysaccharides are simpler because there are no polymeric rings (only rings within the residue unit). However, the linking is more complicated. Firstly, there are more types of links compared to the basic three in molecules (single, double, triple). Secondly, the type of the link relies on the attaching carbohydrate (α , β).

Conceptually, it is easiest to think of one sugar unit linking to the rest of the polysaccharide. For instance, this is the portion that attached to the protein. So, the most common polysaccharide attached to a protein is an N-linked chain that has three units in a linear chain linked to an ASN. The three-letter code for each of the units in order is NAG, NAG, MAN [1]. It is not a leap to imagine that this could be represented using:

NAG-NAG-MAN [2]

As with SMILES, there is a default link. In SMILES, it's a single bound while in SCaLES the default is $\beta(1-4)$. This linkage means that the attaching unit is connecting its anomeric carbon (C1) to the forth carbon (C4) on the other unit. It also specifies that the attaching unit is a β -isomer. In the example, MAN is a mannose saccharide but it is the α -isomer so it should:

NAG-NAG-BMA

However, the SCaLES parser automatically detects that the final residue is the incorrect isomer and changes the code from MAN to BMA: The linkage takes precedence. The specify an alternative to the default linkage, it can be indicated thus:

NAG-NAG- α -(1-4)-MAN

The hand of the linkage can be specified with "a" or "alpha" and the carbon atoms being linked can be specified using "(1-4)" or "(1,4)".

Branching is specified using square brackets (compared to parenthesis in SMILES) thus:

NAG-[α -(1-6)-FUC]-NAG-BMA

The branching can contain more than one saccharide:

NAG-NAG-BMA-[α -(1-3)-MAN- α -(1-2)-MAN- α -(1-2)-MAN]- α -(1-6)-MAN-[α -(1-3)-MAN- α -(1-2)-MAN]- α -(1-6)-MAN- α -(1-2)-MAN

References

Weininger, David (February 1988). "SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules". *Journal of Chemical Information and Modeling* 28 (1): 31–6. doi:10.1021/ci00057a005.

phenix.sceds: SCEDS - identify domains with normal mode analysis

[sceds.html](#)

Description

SCEDS is useful for identifying domains in a model to be used by molecular replacement when it is anticipated that there may be conformational motion of one or more domains relative to the domains in the target structure. Typically in such cases MR underperforms as it is not possible to match both domains onto the target structure. By partitioning a multidomain structure into distinct domains it is then possible to do MR searches with each separate domain and achieve a clearer solution than would otherwise have been possible. Domains are identified with a normal mode analysis. See literature reference below.

To use SCEDS supply it with the MR model and execute it. Depending on the size of the model it may take some time before it completes.

Output is one or more domains each saved into separate pdb files. These can then be used in molecular replacement. It is good practice to inspect the identified domains in a graphical viewer such as Coot or Pymol.

The structure with PDB id 1XHB before running SCEDS. Two distinct domains can be discerned in the structure.

After running SCEDS the two domains have been identified and saved into separate pdb files.

Literature

SCEDS: protein fragments for molecular replacement in Phaser McCoy AJ, Nicholls RA, Schneider TR Acta Cryst. D69, 2216-2225 (2013)

List of all available keywords

file_name = None root = None gui_output_dir = None job_title = None Job title in PHENIX GUI, not used on command line
validate_domains = True keywordsnmamode = 5 combinations = 2 enmoscillators = *rtb all
rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1 perturbationincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro scedsenable = None ndom = 2 weightequality = 1
sphericity = 4 density = 1 continuity = 0 ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12

- file_name = None
- root = None
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- keywordsnmamode = 5 combinations = 2 enmoscillators = *rtb all rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1 perturbationincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro scedsenable = None ndom = 2 weightequality = 1 sphericity = 4 density = 1 continuity = 0 ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12

- nmamode = 5 combinations = 2
- mode = 5
- combinations = 2
- enmoscillators = *rtb all rtb_maxblocks = 125 rtb_nresidues = 0 radius = 5 force = 1
- oscillators = *rtb all
- rtb_maxblocks = 125
- rtb_nresidues = 0
- radius = 5
- force = 1
- perturbincrement = *rms dq dq = None rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro
- increment = *rms dq
- dq = None
- rmsstep = 0.2 maximum = 0.3 direction = forward backward *tofro
- step = 0.2
- maximum = 0.3
- direction = forward backward *tofro
- scedsenable = None ndom = 2 weightequality = 1 sphericity = 4 density = 1 continuity = 0
- enable = None
- ndom = 2
- weightequality = 1 sphericity = 4 density = 1 continuity = 0
- equality = 1
- sphericity = 4
- density = 1
- continuity = 0
- ddmslider = 0 cluster = *max first last distancestep = 50 min = 1 max = 5 separationmin = 7 max = 14 sequencemin = 0 max = 1 joinmin = 2 max = 12
- slider = 0
- cluster = *max first last
- distancestep = 50 min = 1 max = 5
- step = 50
- min = 1
- max = 5
- separationmin = 7 max = 14
- min = 7
- max = 14
- sequencemin = 0 max = 1
- min = 0

- $\max = 1$
- $\text{joinmin} = 2 \quad \max = 12$
- $\min = 2$
- $\max = 12$

Segmenting cryo-EM maps with `segment_and_split_map`

`segment_and_split_map.html`

Author(s)

- `segment_and_split_map`: Tom Terwilliger

Purpose

The routine `segment_and_split_map` will identify the asymmetric unit of a map (typically cryo-EM) and contiguous regions of density within the asymmetric unit of the map.

Usage

How `segment_and_split_map` works:

If you have a CCP4-style (mrc, etc) map and a sequence file, you can use `segment_and_split_map` to split the map into smaller pieces suitable for model-building or viewing.

The `segment_and_split_map` tool works best at resolutions of about 4.5 Å or better.

The tool `segment_and_split_map` will find where your molecule is in the map and cut out and work with just that part of the density.

If your map has been averaged based on NCS symmetry and you supply a file with that NCS information (`.ncs_spec`, `biomtr.dat`, etc), `segment_and_split_map` will find the asymmetric unit of NCS and work with that.

Finally, the `segment_and_split_map` tool will cut the density in the asymmetric unit of your map into small pieces of connected density and write out a map for each one.

All the maps that are written by `segment_and_split_map` are superimposable on each other. They typically are all shifted from the original map to place the origin of the maps on the grid point (0,0,0).

Output files from `segment_and_split_map`

`shifted_map.ccp4`: Original map, shifted to place the origin on grid point (0,0,0)

`shifted_ncs.ncs_spec`: NCS operators (if any), shifted to match `shifted_map.ccp4`.

`shifted_pdb.pdb`: Input PDB file (if any), shifted to match `shifted_map.ccp4`.

`box_map_au.ccp4`: Same as `shifted_map.ccp4`, except that everything except the asymmetric unit of NCS is zeroed out (map shows the asymmetric unit only).

`box_mask_au.ccp4`: Mask showing location of NCS asymmetric unit. Superimposes on `box_map_au.ccp4` and `shifted_map.ccp4`.

`segment_and_split_map_info.pkl`: Pickled file with information about the segmentation. Used in `phenix.map_to_model` and to restore a shifted PDB file to original location.

Shifting the map to the origin

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the origin in the middle of the map. To make a consistent map, any maps with an origin not at the corner are shifted to put the origin at grid point [0,0,0]. This map is the shifted map that is used for further steps in model-building. At the conclusion of model-building, the model is shifted back to superimpose on the original map.

Finding the region containing the molecule

By default (`density_select=True`), the region of the map containing density is cut out of the entire map. This is particularly useful if the original map is very large and the molecule only takes up a small part of the map. This portion of the map is then shifted to place the origin at grid point [0,0,0]. (At the conclusion of model-building, the final model is shifted back to superimpose on the original map.) The region containing density is chosen as a box containing all the points above a threshold, typically 5% of the maximum in the map.

Finding the NCS asymmetric unit of the map

If you supply NCS matrices describing the NCS used to average the map (if any), then `segment_and_split_map` will try to define a region of the map that represents the NCS asymmetric unit. Application of the NCS operators to the NCS asymmetric unit will generate the entire map, and application to a model built into the asymmetric unit will generate the entire model. Normally identification of the NCS asymmetric unit and segmentation of the map (below) are done as a single step, yielding an asymmetric unit and a set of contiguous regions of density within that asymmetric unit. The asymmetric unit of NCS will be written out as a map to the `segmentation_dir` directory, superimposed on the shifted map (so that they can be viewed together in Coot).

Segmentation of the map

By default (`segment=True`) the map or NCS asymmetric unit of the map will be segmented (cut into small pieces) into regions of connected density. This is done by choosing a threshold of density and identifying contiguous regions where all grid points are above this threshold. The threshold is chosen to yield regions that have a size corresponding to about 50 residues. The regions of density are written out to the `segmentation_dir` directory and are superimposed on the shifted map (if you load the shifted map in Coot and a region map in Coot, they should superimpose.)

Special treatment of structures with very high symmetry

If your structure has very high symmetry (by default, 20 or more symmetry operators), then `segment_and_split_map` will try to cut out a piece of your map and work with that instead of working with the whole map. You can control this with the keyword `select_au_box=True` (or `None`, which will give the default behavior). If you use `select_au_box` then a box that contains about `n_au_box` asymmetric units of the map (default of 5) will be cut out of your map. That map will be worked with along with the symmetry operators that apply to it for the remainder of the analysis. placed in the original reference frame. This process can greatly save on memory.

Examples

Standard run of `segment_and_split_map`:

Running `segment_and_split_map` is easy. From the command-line you can type:

```
phenix.segment_and_split_map my_map.map seq.fa ncs_file=find_ncs.ncs_spec
```

where my_map.map is a CCP4, mrc or other related map format, seq.fa is a sequence file, and find_ncs.ncs_spec is an optional file specifying any NCS operators used in averaging the map. This can be in the form of BIOMTR records from a PDB file as well.

Possible Problems

Specific limitations and problems:

Literature

Additional information

List of all available keywords

input_filesseq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.map_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_mapncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.pdb_file = None Optional PDB file matching map_file to be offsetpdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdbinfo_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.input_weight_map_pickle_file = None Weight map picklefileoutput_files magnification_map_file = magnification_map.ccp4 Input map file with magnification applied. Only written if magnification is applied.magnification_ncs_file = magnification_ncs.ncs_spec Input NCS with magnification applied. Only written if magnification is applied.shifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.shifted_sharpened_map_file = shifted_sharpened_map.ccp4 Input map file shifted to new origin and sharpened.sharpened_map_file = sharpened_map.ccp4 Output sharpened map file, superimposed on the original map.shifted_pdb_file = shifted_pdb.pdb Input pdb file shifted to new origin.shifted_ncs_file = shifted_ncs.ncs_spec NCS information shifted to new origin.shifted_used_ncs_file = shifted_used_ncs.ncs_spec NCS information (just the part that is used) shifted to new origin.output_directory = segmented_maps Directory where output files are to be written applied.box_map_file = box_map_au.ccp4 Output map file with one NCS asymmetric unit, cut out boxbox_mask_file = box_mask_au.ccp4 Output mask file with one NCS asymmetric unit, cut out boxbox_buffer = 5 Buffer (grid units) around NCS asymmetric unit in box_mask and mapau_output_file_stem = shifted_au File stem for output map files with one NCS asymmetric unitwrite_intermediate_maps = False Write out intermediate maps and masks for visualizationwrite_output_maps = True Write out mapsremainder_map_file = remainder_map.ccp4 output map file with remainder after initial regions identifiedoutput_info_file = segment_and_split_map_info.pkl Output pickle file with information about map and masksrestored_pdb = None Output name of PDB restored to position matching original map_file. Used in combination with info_file = xxx.pkl and pdb_to_restore = xxxx.pdboutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle

filecrystal_infochain_type = *None PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so).sequence = None Sequence as stringis_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry = True and False otherwise.use_sg_symmetry = False If you set use_sg_symmetry = True then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of is_crysta...

- input_filesseq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.map_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_mapncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.pdb_file = None Optional PDB file matching map_file to be offsetpdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdbinfo_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.input_weight_map_pickle_file = None Weight map pickle file
- seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file. If not supplied, must supply molecular mass or solvent content.
- map_file = None File with CCP4-style map
- half_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_file
- external_map_file = None External map to be used to scale map_file (power vs resolution will be matched). Not used in segment_and_split_map
- ncs_file = None File with symmetry information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.
- pdb_file = None Optional PDB file matching map_file to be offset
- pdb_to_restore = None Optional PDB file to restore to position matching original map_file. Used in combination with info_file = xxx.pkl and restored_pdb = yyyy.pdb
- info_file = None Optional pickle file with information from a previous run. Can be used with pdb_to_restore to restore a PDB file to to position matching original map_file.
- target_ncs_au_file = None Optional PDB file to partially define the ncs asymmetric unit of the map. The coordinates in this file will be used to mark part of the ncs au and all points nearby that are not part of another ncs au will be added.
- input_weight_map_pickle_file = None Weight map pickle file
- output_files magnification_map_file = magnification_map.ccp4 Input map file with magnification applied. Only written if magnification is applied.magnification_ncs_file = magnification_ncs.ncs_spec Input NCS

with magnification applied. Only written if magnification is applied. `shifted_map_file = shifted_map.ccp4` Input map file shifted to new origin. `shifted_sharpened_map_file = shifted_sharpened_map.ccp4` Input map file shifted to new origin and sharpened. `sharpened_map_file = sharpened_map.ccp4` Output sharpened map file, superimposed on the original map. `shifted_pdb_file = shifted_pdb.pdb` Input pdb file shifted to new origin. `shifted_ncs_file = shifted_ncs.ncs_spec` NCS information shifted to new origin. `shifted_used_ncs_file = shifted_used_ncs.ncs_spec` NCS information (just the part that is used) shifted to new origin. `output_directory = segmented_maps` Directory where output files are to be written applied. `box_map_file = box_map_au.ccp4` Output map file with one NCS asymmetric unit, cut out box. `box_mask_file = box_mask_au.ccp4` Output mask file with one NCS asymmetric unit, cut out box. `box_buffer = 5` Buffer (grid units) around NCS asymmetric unit in `box_mask` and `map`. `au_output_file_stem = shifted_au` File stem for output map files with one NCS asymmetric unit. `write_intermediate_maps = False` Write out intermediate maps and masks for visualization. `write_output_maps = True` Write out maps. `remainder_map_file = remainder_map.ccp4` output map file with remainder after initial regions identified. `output_info_file = segment_and_split_map_info.pkl` Output pickle file with information about map and masks. `restored_pdb = None` Output name of PDB restored to position matching original `map_file`. Used in combination with `info_file = xxx.pkl` and `pdb_to_restore = xxxx.pdb`. `output_weight_map_pickle_file = weight_map_pickle_file.pkl` Output weight map pickle file

- `magnification_map_file = magnification_map.ccp4` Input map file with magnification applied. Only written if magnification is applied.
- `magnification_ncs_file = magnification_ncs.ncs_spec` Input NCS with magnification applied. Only written if magnification is applied.
- `shifted_map_file = shifted_map.ccp4` Input map file shifted to new origin.
- `shifted_sharpened_map_file = shifted_sharpened_map.ccp4` Input map file shifted to new origin and sharpened.
- `sharpened_map_file = sharpened_map.ccp4` Output sharpened map file, superimposed on the original map.
- `shifted_pdb_file = shifted_pdb.pdb` Input pdb file shifted to new origin.
- `shifted_ncs_file = shifted_ncs.ncs_spec` NCS information shifted to new origin.
- `shifted_used_ncs_file = shifted_used_ncs.ncs_spec` NCS information (just the part that is used) shifted to new origin.
- `output_directory = segmented_maps` Directory where output files are to be written applied.
- `box_map_file = box_map_au.ccp4` Output map file with one NCS asymmetric unit, cut out box
- `box_mask_file = box_mask_au.ccp4` Output mask file with one NCS asymmetric unit, cut out box
- `box_buffer = 5` Buffer (grid units) around NCS asymmetric unit in `box_mask` and `map`
- `au_output_file_stem = shifted_au` File stem for output map files with one NCS asymmetric unit
- `write_intermediate_maps = False` Write out intermediate maps and masks for visualization
- `write_output_maps = True` Write out maps
- `remainder_map_file = remainder_map.ccp4` output map file with remainder after initial regions identified
- `output_info_file = segment_and_split_map_info.pkl` Output pickle file with information about map and masks
- `restored_pdb = None` Output name of PDB restored to position matching original `map_file`. Used in combination with `info_file = xxx.pkl` and `pdb_to_restore = xxxx.pdb`

- `output_weight_map_pickle_file = weight_map_pickle_file.pkl` Output weight map pickle file
- `crystal_infchain_type = *None` PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so). `sequence = None` Sequence as string `is_crystal = None` Defines whether this is a crystal (or cryo-EM). Default is True if `use_sg_symmetry = True` and False otherwise. `use_sg_symmetry = False` If you set `use_sg_symmetry = True` then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of `is_crystal` (same value as `use_sg_symmetry`) and `restrict_map_size` (False if `use_sg_symmetry = True`). `resolution = None` Nominal resolution of the map. This is used later to decide on resolution cutoffs for Fourier inversion of the map. Note: the resolution is not cut at this value, it is cut at $\text{resolution} * d_min_ratio$ if at all. `space_group = None` Space group (used for boxed maps) `unit_cell = None` Unit Cell (used for boxed maps) `original_unit_cell = None` Original unit cell (of input map). Used internally `original_unit_cell_grid = None` Original unit cell grid (of input map). Used internally `molecular_mass = None` Molecular mass of molecule in Da. Used as alternative method of specifying solvent content. `solvent_content = None` Solvent fraction of the cell. Used for ID of solvent content in boxed maps. `solvent_content_iterations = 3` Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps. `wang_radius = None` Wang radius for solvent identification. Default is $1.5 * \text{resolution}$ `buffer_radius = None` Buffer radius for mask smoothing. Default is `resolution` `pseudo_likelihood = None` Use pseudo-likelihood method for half-map sharpening. (In development)
- `chain_type = *None` PROTEIN RNA DNA Chain type. Determined automatically from sequence file if not given. Mixed chain types are fine (leave blank if so).
- `sequence = None` Sequence as string
- `is_crystal = None` Defines whether this is a crystal (or cryo-EM). Default is True if `use_sg_symmetry = True` and False otherwise.
- `use_sg_symmetry = False` If you set `use_sg_symmetry = True` then the symmetry of the space group will be used. For example in P1 a point at one end of the unit cell is next to a point on the other end. Normally for cryo-EM data this should be set to False and for crystal data it should be set to True. This will normally also set the value of `is_crystal` (same value as `use_sg_symmetry`) and `restrict_map_size` (False if `use_sg_symmetry = True`).
- `resolution = None` Nominal resolution of the map. This is used later to decide on resolution cutoffs for Fourier inversion of the map. Note: the resolution is not cut at this value, it is cut at $\text{resolution} * d_min_ratio$ if at all.
- `space_group = None` Space group (used for boxed maps)
- `unit_cell = None` Unit Cell (used for boxed maps)
- `original_unit_cell = None` Original unit cell (of input map). Used internally
- `original_unit_cell_grid = None` Original unit cell grid (of input map). Used internally
- `molecular_mass = None` Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.
- `solvent_content = None` Solvent fraction of the cell. Used for ID of solvent content in boxed maps.
- `solvent_content_iterations = 3` Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.
- `wang_radius = None` Wang radius for solvent identification. Default is $1.5 * \text{resolution}$
- `buffer_radius = None` Buffer radius for mask smoothing. Default is `resolution`

- `pseudo_likelihood = None` Use pseudo-likelihood method for half-map sharpening. (In development)
- `reconstruction_symmetry` = None Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral), T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if `ncs_file` is supplied.
- `include_helical_symmetry = True` You can include or exclude searches for helical symmetry
- `must_be_consistent_with_space_group_number = None` Searches for symmetry must be compatible with this space group number.
- `symmetry_center = None` Center (in Å) for symmetry operators (if symmetry is found automatically). If set to None, first guess is the center of the cell and then if that fails, found automatically as the center of the density in the map.
- `optimize_center = False` Optimize position of symmetry center. Also checks for center at (0, 0, 0) vs center of map
- `helical_rot_deg = None` helical rotation about z in degrees
- `helical_trans_z_angstrom = None` helical translation along z in Angstrom units
- `max_helical_optimizations = 2` Number of optimizations of helical parameters when finding symmetry
- `max_helical_ops_to_check = 5` Number of helical operations in each direction to check when finding symmetry
- `max_helical_rotations_to_check = None` Number of helical rotations to check when finding symmetry
- `two_fold_along_x = None` Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.
- `smallest_object = None` Dimension of smallest object to consider when finding symmetry. Default is 5 * resolution
- `score_basis = ncs_score cc` Symmetry score basis. Normally `ncs_score (sqrt(n) * cc)` is used except for identification of helical symmetry
- `scale_weight_fractional_translation = 1.05` Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.
- `random_points = 100` Number of random points in map to examine in finding symmetry
- `identify_ncs_id = True` If symmetry is not point-group symmetry, try each possible operator when evaluating symmetry and choose the one that results in the most uniform density at symmetry-related points.
- `min_ncs_cc = 0.75` Minimum symmetry CC to keep operators when identifying automatically
- `n_rescore = 5` Number of symmetry operators to rescore
- `op_max = 14` If symmetry is ANY, try up to op_max-fold symmetry
- `tol_r = 0.02` tolerance in rotations for point group or helical symmetry
- `tol_t = 2` tolerance in translations (Å) for point group or helical symmetry
- `max_helical_operators = None` Maximum helical operators (if extending existing helical operators)
- `rel_tol_t = .05` tolerance in translations (fractional) for point group or helical symmetry
- `require_helical_or_point_group_symmetry = False` normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.
- `symmetry = None` Symmetry used in reconstruction. For example D7, C3, C2 I (icosahedral), T (tetrahedral), or ANY (try everything and use the highest symmetry found). Not needed if `ncs_file` is supplied.

- `max_helical_ops_to_check` = 5 Number of helical operations in each direction to check when finding symmetry
- `max_helical_rotations_to_check` = None Number of helical rotations to check when finding symmetry
- `two_fold_along_x` = None Specifies if D or I two-fold is along x (True) or y (False). If None, both are tried.
- `smallest_object` = None Dimension of smallest object to consider when finding symmetry. Default is 5 * resolution
- `score_basis` = `ncs_score cc` *None Symmetry score basis. Normally `ncs_score (sqrt(n)* cc)` is used except for identification of helical symmetry
- `scale_weight_fractional_translation` = 1.05 Give slight increase in weighting in helical symmetry search to translations that are a fraction (1/2, 1/3) of the d-spacing of the peak of intensity in the fourier transform of the density.
- `random_points` = 100 Number of random points in map to examine in finding symmetry
- `identify_ncs_id` = True If symmetry is not point-group symmetry, try each possible operator when evaluating symmetry and choose the one that results in the most uniform density at symmetry-related points.
- `min_ncs_cc` = 0.75 Minimum symmetry CC to keep operators when identifying automatically
- `n_rescore` = 5 Number of symmetry operators to rescore
- `op_max` = 14 If symmetry is ANY, try up to `op_max`-fold symmetries
- `tol_r` = 0.02 tolerance in rotations for point group or helical symmetry
- `abs_tol_t` = 2 tolerance in translations (A) for point group or helical symmetry
- `max_helical_operators` = None Maximum helical operators (if extending existing helical operators)
- `rel_tol_t` = .05 tolerance in translations (fractional) for point group or helical symmetry
- `require_helical_or_point_group_symmetry` = False normally helical or point-group symmetry (or none) is expected. However in some cases (helical + rotational symmetry for example) this is not needed and is not the case.
- `map_modificationmagnification` = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.
`b_iso` = None Target B-value for map (sharpening will be applied to yield this value of `b_iso`). If sharpening method is not supplied, default is to use `b_iso_to_d_cut` sharpening.
`b_sharpen` = None Sharpen with this b-value. Contrast with `b_iso` that yield a targeted value of `b_iso`. B_sharpen greater than zero is sharpening. Less than zero is blurring.
`b_blur_hires` = 200 Blur high_resolution data (higher than `d_cut`) with this b-value. Contrast with `b_sharpen` applied to data up to `d_cut`. Note on defaults: If None and `b_sharpen` is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and `b_sharpen` is negative (blurring) high-resolution data is also blurred.
`resolution_dependent_b` = None If set, apply resolution_dependent_b (`b0 b1 b2`). Log10(amplitudes) will start at 1, change to `b0` at half of resolution specified, changing linearly, change to `b1/2` at resolution specified, and change to `b1/2+b2` at `d_min_ratio*resolution`
`normalize_amplitudes_in_resdep` = False Normalize amplitudes in resolution-dependent sharpening
`d_min_ratio` = 0.833 Sharpening will be applied using `d_min` equal to `d_min_ratio` times resolution. Default is 0.833
`scale_max` = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
`input_d_cut` = None High-resolution limit for sharpening
`rmsd` = None RMSD of model to true model (if supplied). Used to estimate expected fall-off with resolution of correct part of model-based map. If None, assumed to be resolution times

`rmsd_resolution_factor`.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution
`factorfraction_complete` = None Completeness of model (if supplied). Used to estimate correct part of
 model-based map. If None, estimated from max(FSC).regions_to_keep = None You can specify a limit to
 the number of regions to keep when generating the asymmetric unit of density.auto_sharpen = True
 Automatically determine sharpening using kurtosis maximization or adjusted surface
`areaauto_sharpen_methods` = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent
`model_sharpening` half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening.
`b_iso` searches for `b_iso` to maximize sharpening target (kurtosis or adjusted_sa). `b_iso_to_d_cut` applies
`b_iso` only up to resolution specified, with fall-over of `k_sharpen`. Resolution dependent adjusts 3
 parameters to sharpen variably over resolution range. Default is `b_iso_to_d_cut` . `target_b_iso_to_d_cut`
 uses `target_b_iso_ratio` to set `b_iso`.`box_in_auto_sharpen` = False Use a representative box of density
 for initial auto-sharpening instead of the entire map.`density_select_in_auto_sharpen` = True Choose
 representative box of density for initial auto-sharpening with `density_select` method (choose region where
 there is high density). Normally use this as well as `density_select` = True which carries out `density_select`
 at start of segmentation.`density_select_threshold_in_auto_sharpen` = None Threshold for density select
 choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger
 such as 0.2.`allow_box_if_b_iso_set` = False Allow `box_in_auto_sharpen` (if set to True) even if `b_iso` is
 set. Default is to set `box_n_auto_sharpen` = False if `b_iso` is set.`soft_mask` = True Use soft mask (smooth
 change from inside to outside with radius based on resolution of map). Required if you use half-map
 sharpening without a model, otherwise optional.`use_weak_density` = False When choosing box of
 representative density, use poor density (to get optimized map for weaker density)`discard_if_worse` =
 None Discard sharpening if worse`local_sharpening` = None Sharpen locally using overlapping regions.
 NOTE: Best to turn off `local_aniso_in_local_sharpening` if symmetry is present. If
`local_aniso_in_local_sharpening` is True and symmetry is present this can distort the map for s...

- `magnification` = None Magnification to apply to input map. Input map grid will be scaled by magnification factor before anything else is done.
- `b_iso` = None Target B-value for map (sharpening will be applied to yield this value of `b_iso`). If sharpening method is not supplied, default is to use `b_iso_to_d_cut` sharpening.
- `b_sharpen` = None Sharpen with this b-value. Contrast with `b_iso` that yield a targeted value of `b_iso`. `B_sharpen` greater than zero is sharpening. Less than zero is blurring.
- `b_blur_hires` = 200 Blur high_resolution data (higher than `d_cut`) with this b-value. Contrast with `b_sharpen` applied to data up to `d_cut`. Note on defaults: If None and `b_sharpen` is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and `b_sharpen` is negative (blurring) high-resolution data is also blurred.
- `resolution_dependent_b` = None If set, apply `resolution_dependent_b` (`b0` `b1` `b2`). $\log_{10}(\text{amplitudes})$ will start at 1, change to `b0` at half of resolution specified, changing linearly, change to `b1/2` at resolution specified, and change to `b1/2+b2` at `d_min_ratio*resolution`
- `normalize_amplitudes_in_resdep` = False Normalize amplitudes in resolution-dependent sharpening
- `d_min_ratio` = 0.833 Sharpening will be applied using `d_min` equal to `d_min_ratio` times resolution. Default is 0.833
- `scale_max` = 100000 Scale amplitudes from inverse FFT to yield maximum of this value
- `input_d_cut` = None High-resolution limit for sharpening
- `rmsd` = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times `rmsd_resolution_factor`.

- `rmsd_resolution_factor` = 0.25 default RMSD is resolution times resolution factor
- `fraction_complete` = None Completeness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).
- `regions_to_keep` = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.
- `auto_sharpen` = True Automatically determine sharpening using kurtosis maximization or adjusted surface area
- `auto_sharpen_methods` = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent
`model_sharpening` half_map_sharpening target_b_iso_to_d_cut None Methods to use in sharpening.
`b_iso` searches for `b_iso` to maximize sharpening target (kurtosis or adjusted_sa). `b_iso_to_d_cut` applies `b_iso` only up to resolution specified, with fall-over of `k_sharpen`. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is `b_iso_to_d_cut`. `target_b_iso_to_d_cut` uses `target_b_iso_ratio` to set `b_iso`.
- `box_in_auto_sharpen` = False Use a representative box of density for initial auto-sharpening instead of the entire map.
- `density_select_in_auto_sharpen` = True Choose representative box of density for initial auto-sharpening with `density_select` method (choose region where there is high density). Normally use this as well as `density_select` = True which carries out `density_select` at start of segmentation.
- `density_select_threshold_in_auto_sharpen` = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.
- `allow_box_if_b_iso_set` = False Allow `box_in_auto_sharpen` (if set to True) even if `b_iso` is set. Default is to set `box_in_auto_sharpen` = False if `b_iso` is set.
- `soft_mask` = True Use soft mask (smooth change from inside to outside with radius based on resolution of map). Required if you use half-map sharpening without a model, otherwise optional.
- `use_weak_density` = False When choosing box of representative density, use poor density (to get optimized map for weaker density)
- `discard_if_worse` = None Discard sharpening if worse
- `local_sharpening` = None Sharpen locally using overlapping regions. NOTE: Best to turn off `local_aniso_in_local_sharpening` if symmetry is present. If `local_aniso_in_local_sharpening` is True and symmetry is present this can distort the map for some symmetry copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.
- `local_aniso_in_local_sharpening` = None Use local anisotropy in local sharpening. Default is True unless symmetry is present.
- `overall_before_local` = True Apply overall scaling before local scaling
- `select_sharpened_map` = None Select a single sharpened map to use
- `read_sharpened_maps` = None Read in previously-calculated sharpened maps
- `write_sharpened_maps` = None Write out local sharpened maps
- `smoothing_radius` = None Sharpen locally using `smoothing_radius`. Default is 2/3 of mean distance between centers for sharpening
- `box_center` = None You can specify the center of the box (A units)
- `box_size` = 40 40 40 You can specify the size of the boxes to use (grid units)
- `target_n_overlap` = 10 You can specify the targeted overlap of boxes in local sharpening

- `restrict_map_size = None` Restrict box map to be inside full map (required for cryo-EM data). Default is True if `use_sg_symmetry = False`.
- `restrict_z_turns_for_helical_symmetry = 1` Restrict Z turns for helical symmetry. Number of turns of helix going each direction in Z is specified.
- `restrict_z_distance_for_helical_symmetry = None` Restrict Z distance (+/- this distance from center) for helical symmetry.
- `remove_aniso = True` You can remove anisotropy (overall and locally) during sharpening
- `cc_cut = 0.2` Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values.
- `max_cc_for_rescale = 0.2` Min reliable CC in half-maps. Used along with `cc_cut` and `scale_using_last` to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above `max_cc_for_rescale`, assume these values are correct and do not correct them. To keep all original values use `max_cc_for_rescale = 1`
- `scale_using_last = 3` If set, assume that the last `scale_using_last` bins in the FSC for half-map or model sharpening are about zero (corrects for errors in the half-map process).
- `max_box_fraction = 0.5` If box is greater than this fraction of entire map, use entire map.
- `density_select_max_box_fraction = 0.95` If box is greater than this fraction of entire map, use entire map for `density_select`. Default is 0.95
- `mask_atoms = True` Mask atoms when using model sharpening
- `mask_atoms_atom_radius = 3` Mask for `mask_atoms` will have `mask_atoms_atom_radius`
- `value_outside_atoms = None` Value of map outside atoms (set to mean to have mean value inside and outside mask be equal)
- `k_sharpen = 10` Steepness of transition between sharpening (up to resolution) and not sharpening ($d < \text{resolution}$). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if `k_sharpen` is zero, then no transition is applied and all data is sharpened or blurred. Note 3: only used if `b_iso` is set.
- `iterate = False` You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.
- `optimize_b_blur_hires = False` Optimize value of `b_blur_hires`. Only applies for `auto_sharpen_methods` `b_iso_to_d_cut` and `b_iso`. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.
- `optimize_d_cut = None` Optimize value of `d_cut`. Only applies for `auto_sharpen_methods` `b_iso_to_d_cut` and `b_iso`. Not normally carried out.
- `adjust_region_weight = True` Adjust `region_weight` to make overall change in surface area equal to overall change in normalized regions over the range of `search_b_min` to `search_b_max` using `b_iso_to_d_cut`.
- `region_weight_method = initial_ratio * delta_ratio b_iso` Method for choosing `region_weights`. `Initial_ratio` uses ratio of surface area to regions at low B value. `Delta ratio` uses change in this ratio from low to high B. `B_iso` uses resolution-dependent `b_iso` (not weights) with the formula $b_iso = 5.9 * d_min^{**2}$
- `region_weight_factor = 1.0` Multiplies `region_weight` after calculation with `region_weight_method` above
- `region_weight_buffer = 0.1` `Region_weight` adjusted to be `region_weight_buffer` away from minimum or maximum values

- `region_weight_default = 30`. Region_weight adjusted to be `region_weight_default` if no information available
- `target_b_iso_ratio = 5.9` Target b_iso ratio : b_iso is estimated as `target_b_iso_ratio * resolution**2`
- `signal_min = 3.0` Minimum signal in estimation of optimal b_iso. If not achieved, use any other method chosen.
- `target_b_iso_model_scale = 0`. For model sharpening, the target_biso is scaled (normally zero).
- `search_b_min = -100` Low bound for b_iso search.
- `search_b_max = 300` High bound for b_iso search.
- `search_b_n = 21` Number of b_iso values to search.
- `residual_target = 'adjusted_sa'` Target for maximization steps in sharpening. Can be kurtosis or `adjusted_sa` (adjusted surface area)
- `sharpening_target = 'adjusted_sa'` Overall target for sharpening. Can be kurtosis or `adjusted_sa` (adjusted surface area) or `adjusted_path_length`. Used to decide which sharpening approach is used. Note that during optimization, `residual_target` is used (they can be the same.)
- `region_weight = 40` Region weighting in adjusted surface area calculation. Score is surface area minus `region_weight` times number of regions. Default is 40. A smaller value will give more sharpening.
- `sa_percent = 30`. Percent of target regions used in calculation of adjusted surface area. Default is 30.
- `fraction_occupied = 0.20` Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20
- `n_bins = 20` Number of resolution bins for sharpening. Default is 20.
- `max_regions_to_test = 30` Number of regions to test for surface area in `adjusted_sa` scoring of sharpening
- `eps = None`
- `k_sol = 0.35` k_sol value for model map calculation. IGNORED (Not applied)
- `b_sol = 50` b_sol value for model map calculation. IGNORED (Not applied)
- `segmentationselect_au_box = None` Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least `n_ops_to_use_au_box`
`n_ops_to_use_au_box = 25` If number of operators is this big or more and `select_au_box` is None, set it to True.
`n_au_box = 5` Number of symmetry copies to try and get inside `au_box`
`lower_bounds = None` You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
`upper_bounds = None` You can select a part of your map for analysis with `lower_bounds` and `upper_bounds`.
`density_select = True` Run `map_box` with `density_select = True` to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Done before segmentation is carried out.
`density_select_threshold = 0.05` Choose region where density is this fraction of maximum or greater
`get_half_height_width = None` Use 4 times half-width at half-height as estimate of max size
`box_ncs_au = True` Box the map containing just the au of the map
`cell_cutoff_for_solvent_from_mask = 150` For cells with average edge over this cutoff, use the low resolution mask (backup) method for solvent estimation
`mask_padding_fraction = 0.025` Adjust threshold of standard deviation map in low resolution mask identification of solvent content to give this much more inside mask than would be obtained with the value of `fraction_of_max_mask_threshold`.
`fraction_of_max_mask_threshold = .05` threshold of standard deviation map in low resolution mask identification of solvent content.
`mask_threshold = None` threshold in identification of overall mask. If None, guess volume of molecule from sequence and symmetry

copies.grid_spacing_for_au = 3 Grid spacing for asymmetric unit when constructing asymmetric unit.radius = None Radius for constructing asymmetric unit.value_outside_mask = 0.0 Value to assign to density outside maskdensity_threshold = None Threshold density for identifying regions of density. Applied after normalizing the density in the region of the molecule to an rms of 1 and mean of zero.starting_density_threshold = None Optional guess of threshold densityiteration_fraction = 0.2 On iteration of finding regions, assume target volume is this fraction of the value on previous iterationmax_overlap_fraction = 0.05 Maximum fractional overlap allowed to density in another asymmetric unit. Definition of a bad region.remove_bad_regions_percent = 1 Remove the worst regions that are part of more than one NCS asymmetric unit, up to remove_bad_regions_percent of the totalrequire_complete = True Require all symmetry copies to be represented for a regionsplit_if_possible = True Split regions that are split in some symmetry copies. If None, split if most copies are split.write_all_regions = False Write all regions to ccp4 map files.max_per_au = None Maximum number of regions to be kept in the NCS asymmetric unitmax_per_au_ratio = 5. Maximum ratio of number of regions to be kept in the NCS asymmetric unit to those expectedmin_ratio_of_ncs_copy_to_first = 0.5 Minimum ratio of the last ncs_copy region size to maximummin_ratio = 0.1 Minimum ratio of region size to maximum to keep itmax_ratio_to_target = 3 Maximum ratio of grid points in top region to targetmin_ratio_to_target = 0.3 Minimum ratio of grid points in top region to targetmin_volume = 10 Minimum region size to consider (in grid points)residues_per_region = 50 Target number of residues per regionseeds_to_try = 10 Number of regions to try as centersiterate_with_remainder = True Iterate looking for regions based on remainder from first analysisweight_rad_gyr = 0.1 Weight on radius of gyration of group of regions in NCS AU relative to weight on closeness to neighbors. Normalized to largest cell dimension with weight = weight_rad_gyr*300/cell_maxexpand_size = None Grid points to expand size of regions when excluding for next round. If None, set to approx number of grid points to get expa...

- select_au_box = None Select box containing at least one representative region of the map. Also select just symmetry operators relevant to that box. Default is true if number of operators is at least n_ops_to_use_au_box
- n_ops_to_use_au_box = 25 If number of operators is this big or more and select_au_box is None, set it to True.
- n_au_box = 5 Number of symmetry copies to try and get inside au_box
- lower_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- upper_bounds = None You can select a part of your map for analysis with lower_bounds and upper_bounds.
- density_select = True Run map_box with density_select = True to cut out the region in the input map that contains density. Useful if the input map is much larger than the structure. Done before segmentation is carried out.
- density_select_threshold = 0.05 Choose region where density is this fraction of maximum or greater
- get_half_height_width = None Use 4 times half-width at half-height as estimate of max size
- box_ncs_au = True Box the map containing just the au of the map
- cell_cutoff_for_solvent_from_mask = 150 For cells with average edge over this cutoff, use the low resolution mask (backup) method for solvent estimation
- mask_padding_fraction = 0.025 Adjust threshold of standard deviation map in low resolution mask identification of solvent content to give this much more inside mask than would be obtained with the value of fraction_of_max_mask_threshold.

- `fraction_of_max_mask_threshold` = .05 threshold of standard deviation map in low resolution mask identification of solvent content.
- `mask_threshold` = None threshold in identification of overall mask. If None, guess volume of molecule from sequence and symmetry copies.
- `grid_spacing_for_au` = 3 Grid spacing for asymmetric unit when constructing asymmetric unit.
- `radius` = None Radius for constructing asymmetric unit.
- `value_outside_mask` = 0.0 Value to assign to density outside masks
- `density_threshold` = None Threshold density for identifying regions of density. Applied after normalizing the density in the region of the molecule to an rms of 1 and mean of zero.
- `starting_density_threshold` = None Optional guess of threshold density
- `iteration_fraction` = 0.2 On iteration of finding regions, assume target volume is this fraction of the value on previous iteration
- `max_overlap_fraction` = 0.05 Maximum fractional overlap allowed to density in another asymmetric unit. Definition of a bad region.
- `remove_bad_regions_percent` = 1 Remove the worst regions that are part of more than one NCS asymmetric unit, up to `remove_bad_regions_percent` of the total
- `require_complete` = True Require all symmetry copies to be represented for a region
- `split_if_possible` = True Split regions that are split in some symmetry copies. If None, split if most copies are split.
- `write_all_regions` = False Write all regions to ccp4 map files.
- `max_per_au` = None Maximum number of regions to be kept in the NCS asymmetric unit
- `max_per_au_ratio` = 5. Maximum ratio of number of regions to be kept in the NCS asymmetric unit to those expected
- `min_ratio_of_ncs_copy_to_first` = 0.5 Minimum ratio of the last `ncs_copy` region size to maximum
- `min_ratio` = 0.1 Minimum ratio of region size to maximum to keep it
- `max_ratio_to_target` = 3 Maximum ratio of grid points in top region to target
- `min_ratio_to_target` = 0.3 Minimum ratio of grid points in top region to target
- `min_volume` = 10 Minimum region size to consider (in grid points)
- `residues_per_region` = 50 Target number of residues per region
- `seeds_to_try` = 10 Number of regions to try as centers
- `iterate_with_remainder` = True Iterate looking for regions based on remainder from first analysis
- `weight_rad_gyr` = 0.1 Weight on radius of gyration of group of regions in NCS AU relative to weight on closeness to neighbors. Normalized to largest cell dimension with `weight = weight_rad_gyr*300/cell_max`
- `expand_size` = None Grid points to expand size of regions when excluding for next round. If None, set to approx number of grid points to get `expand_target` below
- `expand_target` = 1.5 Target expansion of regions (A)
- `mask_additional_expand_size` = 1 Mask expansion in addition to `expand_size` for final map
- `mask_expand_ratio` = 1 Mask expansion relative to resolution for `save_box_map_ncs_au`
- `exclude_points_in_ncs_copies` = True Exclude points that are in symmetry copies when creating NCS au. Does not apply if `add_neighbors` = True

- `add_neighbors = True` Add neighboring regions around the au. Turns off `exclude_points_in_ncs_copies` also.
- `add_neighbors_dist = 1`. Max increase in radius of gyration by adding region to keep it.
- `controlverbose = False` Verbose output
- `shift_only = None` Shift map and half_maps and stop
- `sharpen_only = None` Sharpen map and stop
- `check_ncs = None` Check the NCS symmetry by estimating NCS correlation and stop
- `resolve_size = None` Size of resolve to use.
- `quick = True` Run quickly if possible
- `memory_check = True` Map-to-model checks to make sure you have enough memory on your machine to run. You can disable this by setting this keyword to False. The estimates are approximate so it is possible your job could run even if the check fails. Note the check does not take any other uses of the memory on your machine into account.
- `save_box_map_ncs_au = False` Controls whether the map_box ncs_au is saved. Internal use only
- `write_files = True` Controls whether files are written
- `multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm` Choices are multiprocessing (single machine) or queuing systems
- `queue_run_command = None` run command for queue jobs. For example qsub.
- `nproc = 1` Number of processors to use